Role of business formation and exit for unemployment and vacancy dynamics

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Abstract

Vacancies comove positively with establishment entry and negatively with exit; states with higher exit rates accumulate larger labor market gaps during recessions. We develop a model of endogenous business formation and exit with labor market frictions to explain these patterns. Monopolistically competitive firms rent labor from specialized recruiters and face exit risk from product obsolescence and stochastic fixed costs. Both business creation and vacancy posting require sunk costs, making them partially predetermined. Three results follow. First, the business formation margin affects labor markets through love of variety and the endogenous exit rate. Second, because vacancy posting is a sunk cost, exiting firms simultaneously eliminate jobs and posted vacancies, so destruction shocks shift the Beveridge curve outward–even if vacancies can respond elastically. Third, match separation shocks destroy only filled positions; firms optimally repost, generating positive unemployment-vacancy comovement. Both technology and destruction shocks produce second-round amplification through further exit and reduced product variety. We estimate by Bayesian methods targeting conditional moments—impulse responses to a Bartik-identified destruction shock—and unconditional labor market moments. Technology and destruction shocks account for the predominant share of labor market volatility.

1. Introduction

Labor market variables do not recover at equal speed, and the disparity does not just depend on the severity of the initial shock. The 2020 recession produced the highest peak unemployment rate of any postwar episode at 13.0 %, yet labor markets had recovered within two years,

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accumulating only 26.6 percentage-point quarters of total unemployment excess over twenty quarters from recession onset. The 2008–09 recession generated a smaller peak unemployment rate of 9.9 percent yet produced 76.6 percentage-point quarters of cumulative damage and a recovery stretching across six years.

The standard Diamond–Mortensen–Pissarides (DMP) framework, in which all separations are identical and firms repost vacancies freely after any job loss, offers no structural account of why two recessions of comparable peak unemployment can generate such different cumulative damage. The reason is that the composition of job loss has no influence on persistence. We argue that a key margin to close the gap is firm destruction — the permanent elimination of business opportunities and their associated vacancy slots — which ran at 7.8 percent of base employment in 2008–09 but only 2.9 percent in 2020.[2] When a firm exits, its vacancies vanish permanently and must be rebuilt from scratch at sunk cost. Recovery requires not merely a rebound in profitability but a costly reconstruction of the vacancy stock, generating a persistent outward shift of the Beveridge curve. By contrast, after a separation at a surviving firm, the position is reposted provided reposting is cheap relative to initial creation, so the aggregate vacancy stock partially self-corrects even while unemployment remains elevated. This produces a transitory displacement along the Beveridge curve rather than a persistent outward shift.

Coles and Moghaddasi Kelishomi (2018) show that extending the DMP model with finitely elastic vacancy creation enables separation shocks to generate a downward-sloping Beveridge curve and, jointly with technology shocks, fit a broad set of labor market moments well. But CK assume a single separation shock, conflating firm destruction and match dissolution into one margin, and separations are entirely exogenous. Gabrovski and Silva (2025) build on CK by distinguishing between destruction of a product line δ and breakdown of a match s, and show that, for empirically reasonable values of δ, the model generates mild unemployment-labor productivity comovement under technology shocks alone, and also matches other key labor market moments, including the Beveridge curve. In both frameworks, sunk entry costs make the vacancy stock predetermined: v_{t-1} enters the equilibrium alongside u_t, and surviving firms repost vacancies freely following match dissolution, substantially weakening the link between

[2]Financial frictions, nominal rigidities, and other margins absent from the benchmark model have each received substantial attention in the literature; we focus on firm destruction and its interaction with endogenous business formation.

unemployment and contemporaneous productivity.

This paper opens additional channels related to business formation and exit. First, we introduce love-of-variety preferences over differentiated product lines, so the mass of active product lines N_t becomes a third state variable. Employment and variety feed each other, an aggregate demand externality in the sense of Schaal and Taschereau-Dumouchel (2019). A new product line raises the welfare-based relative price $\rho(N_t)$, and with it recruiter compensation $w_t^{int} = \rho(N_t)z_t/\mu$ and the return to vacancy creation, while greater employment expands the resources available for product-line formation.

We segment the product and labor markets so that a product line employs many workers. Employment and the number of product lines therefore do not move one for one, and u_t, v_{t-1}, and N_t are jointly determined without an accounting restriction.[3]

Second, we endogenize firm exit. The realized destruction rate $\delta_{e,t}$ combines an exogenous component with a margin that responds to profitability through a continuation-cost cutoff, so a recession that lowers profits or raises separations induces additional exit. The establishment death rate measured in Section 4 is this composite. The same cost distribution supplies the draws that govern which vacated positions are reposted, so costly reposting adds no separate distributional assumption.

These extensions sharpen the asymmetry between δ and s shocks. A destruction shock damages all three state variables at once: it raises u_t, depletes the vacancy stock, and eliminates product lines that would otherwise sustain vacancy creation. A separation shock raises u_t, but under costless reposting each separation returns its position to the vacancy pool. Section 3.9 establishes the asymmetry analytically, and Section 5 assesses how much of the cross-recession gap in cumulative labor market damage it accounts for.

One might ask whether the composition of job loss could be studied by building on Gabrovski and Silva (2025), which already separates destruction from match dissolution, and leaving preferences aside. Our measurement strategy favors the business formation block. We identify δ

[3]In Schaal and Taschereau-Dumouchel (2019), each matched worker–firm pair produces one variety of an intermediate good, so employment and the range of goods fluctuate one-for-one. The demand externality therefore operates simultaneously through both matching and variety. In our setting, firms employ multiple workers per product line, so employment adds output within existing varieties while the variety externality operates separately through firm entry, governed by the business-formation Euler equation.

from establishment deaths, so a business opportunity must correspond to an establishment, which holds many positions; where each opportunity carries a single vacancy, δ has to be calibrated to a rate of job destruction instead. A finite and endogenous mass of multi-position lines sustained by a sunk entry cost requires positive profit per line: free entry equates the value of a line to $\rho(N_t)f_e/\mu(N_t)$, which vanishes at $\mu = 1$, leaving no finite mass of lines. Under Dixit–Stiglitz preferences the markup and the taste for variety are then the same parameter, $\zeta(N) = \mu(N) - 1 = 1/(\varepsilon - 1)$. The monopolistic competition block is therefore the minimal structure in which an establishment-level destruction margin is coherent, with the variety externality an implication rather than a separate assumption. The structure also makes the weighting of the measured exit rate substantive. Because a line holds many positions, the employment-weighted and count-weighted establishment death rates are distinct objects in the data, and it is the former that maps to the mass of positions destroyed.

1.1. Motivating reduced form evidence

Three empirical facts motivate treating firm destruction as a distinct and quantitatively important margin. *First, establishment entry and exit constitute a large and cyclically sensitive component of gross job flows.* Table 1 extends Table 2 of Jaimovich and Floetotto (2008) through 2019Q4 using BLS Business Employment Dynamics (BED) data.[4] We stop the sample in 2019Q4. Extending it leaves the mean shares almost unchanged but collapses every cyclicality ratio, because 2020 inflates total job losses without a corresponding rise in permanent exit.

The entry and exit margins are of comparable size. Opening establishments account for 20.0 percent of gross job gains and closing establishments for 19.5 percent of gross job losses, a share that is stable across industries, from construction (20.6 percent) to financial activities (24.7 percent). Dying establishments, the permanent subset, account for 12.0 percent. The difference, close to two-fifths of the closings margin, is the temporary-shutdown component that the δ instrument must remove. Permanent exit is a minority of job destruction but far from a rounding error: about one job loss in eight occurs at an establishment that never reopens. The two margins are separately measurable, which is what makes the δ-versus-s decomposition

[4]The closings columns use BED *closings*, for comparability with Jaimovich and Floetotto (2008), whose question concerns the number of firms actively competing in a period; BED deaths were not published until May 2009. The model's δ shock is the permanent elimination of a product line, so the rest of the paper uses BED *deaths*, defined in Section 4.2.

empirically tractable.

All three margins move with the cycle. For total private, the opening flow is the most variable, at 0.269 times the cyclical variation of total job gains under the Hamilton filter and 0.275 under the HP filter. The closing flow moves at 0.239 times the cyclical variation of total job losses, against 0.187 for deaths (0.223 and 0.180 under the HP filter). Closings are more cyclical than deaths in every supersector, but the gap is wide only in natural resources and mining (0.205 against 0.104), where project-based and seasonal establishments suspend operations without exiting. The cyclical sensitivity that Jaimovich and Floetotto (2008) document is therefore not an artifact of temporary shutdowns. Permanent exit is itself cyclical, which is what the δ shock requires.

Entry matters too. The damage a destruction shock does is undone only as new establishments replace the lines that closed, so the cost and speed of replacement govern how long the damage lasts. Openings are procyclical: their correlation with cyclical net job change is 0.47 under the Hamilton filter and 0.31 under the HP filter, while permanent exit runs the other way, at -0.08 and -0.49.[5] Replacement is therefore slowest when destruction is heaviest, though the exit side of this comparison rests mainly on the HP estimate. Where a business opportunity is a single vacancy slot, replacing the opportunity and posting the vacancy are the same act, so the entry margin in the table has no separate counterpart.

[5] Openings and deaths are themselves positively correlated (0.56 and 0.46), as two gross flows that share a common reallocation component, so we compare each flow with net job change.

Table 1: Job Gains and Losses from Opening, Closing, and Dying Establishments (*Extended to the COVID cutoff, preferred; estimated on 1992Q3–2019Q4*)

Source: BLS Business Employment Dynamics (BED), quarterly, 12 Bartik supersectors. Replication and extension of Jaimovich & Floetotto (2008, JME) Table 2.

	Mean share			Rel. std dev (Ham)			Rel. std dev (HP)		
Industry	$\bar{g}^O$	$\bar{g}^C$	$\bar{g}^D$	σ^O/σ	σ^C/σ	σ^D/σ	σ^O/σ	σ^C/σ	σ^D/σ
Total private	0.200	0.195	0.120	0.269	0.239	0.187	0.275	0.223	0.180
Natural resources & mining	0.164	0.158	0.058	0.295	0.205	0.104	0.261	0.215	0.118
Construction	0.208	0.206	0.108	0.188	0.198	0.139	0.212	0.192	0.133
Manufacturing	0.125	0.139	0.102	0.234	0.123	0.092	0.260	0.119	0.091
Wholesale trade	0.204	0.227	0.155	0.260	0.249	0.206	0.296	0.250	0.213
Retail trade	0.172	0.154	0.109	0.239	0.223	0.185	0.311	0.257	0.232
Transportation & utilities	0.165	0.175	0.119	0.208	0.223	0.187	0.147	0.174	0.149
Information	0.206	0.210	0.150	0.292	0.367	0.318	0.327	0.399	0.361
Financial activities	0.235	0.247	0.170	0.352	0.387	0.331	0.481	0.455	0.410
Prof. & business services	0.205	0.212	0.127	0.288	0.239	0.215	0.290	0.246	0.193
Education & health services	0.202	0.213	0.128	0.341	0.336	0.275	0.455	0.429	0.313
Leisure & hospitality	0.245	0.198	0.115	0.271	0.332	0.250	0.315	0.350	0.271
Other services	0.221	0.219	0.130	0.291	0.352	0.287	0.344	0.402	0.301

Notes. $\bar{g}^O$ ($\bar{g}^C$, $\bar{g}^D$): mean share of gross job gains (losses) from opening (closing, dying) establishments, e.g. $\sum L_D / \sum L$. A death is a closing that reports zero employment for four consecutive quarters, so $\bar{g}^C - \bar{g}^D$ is the temporary-shutdown component. Closings columns replicate Jaimovich & Floetotto (2008, JME) Table 2, whose sample predates the publication of BED deaths; deaths columns are the permanent-exit margin used in the model. σ^k/σ: relative std dev of the k flow against total flows, raw levels, via the Hamilton (2018) filter (h=8, p=4, robust to GFC/Covid outliers) and the HP filter (λ=1600). HP values for total private in the original sample (≈ 0.33) closely match Jaimovich & Floetotto Table 2. Values < 1: extensive margin less volatile than total flows. * Noisy estimate (|ratio| > 5).

Second, cumulative establishment exit predicts the severity and persistence of labor market disruption across recessions. Figure 1 plots the permanence-adjusted establishment exit rate against cumulative unemployment and vacancy gaps across the three post-1992 NBER recessions. The 2008–09 recession generated the largest exit rate (7.8% of base employment) and the

largest cumulative unemployment gap (76.6 pp-quarters), reflecting both a deep trough and a protracted recovery. The 2020 recession, with the smallest exit rate (2.9%), recovered rapidly on both margins, with a negligible cumulative vacancy shortfall (1.7 pp-quarters). The 2001 recession (exit rate 5.6%) is the exception on vacancies: its shortfall (25.7 pp-quarters) exceeds the GFC's (18.8), likely reflecting the post-dot-com collapse in investment demand rather than exit. Across the three episodes, larger establishment exit goes with greater and more persistent labor market damage, particularly for unemployment.

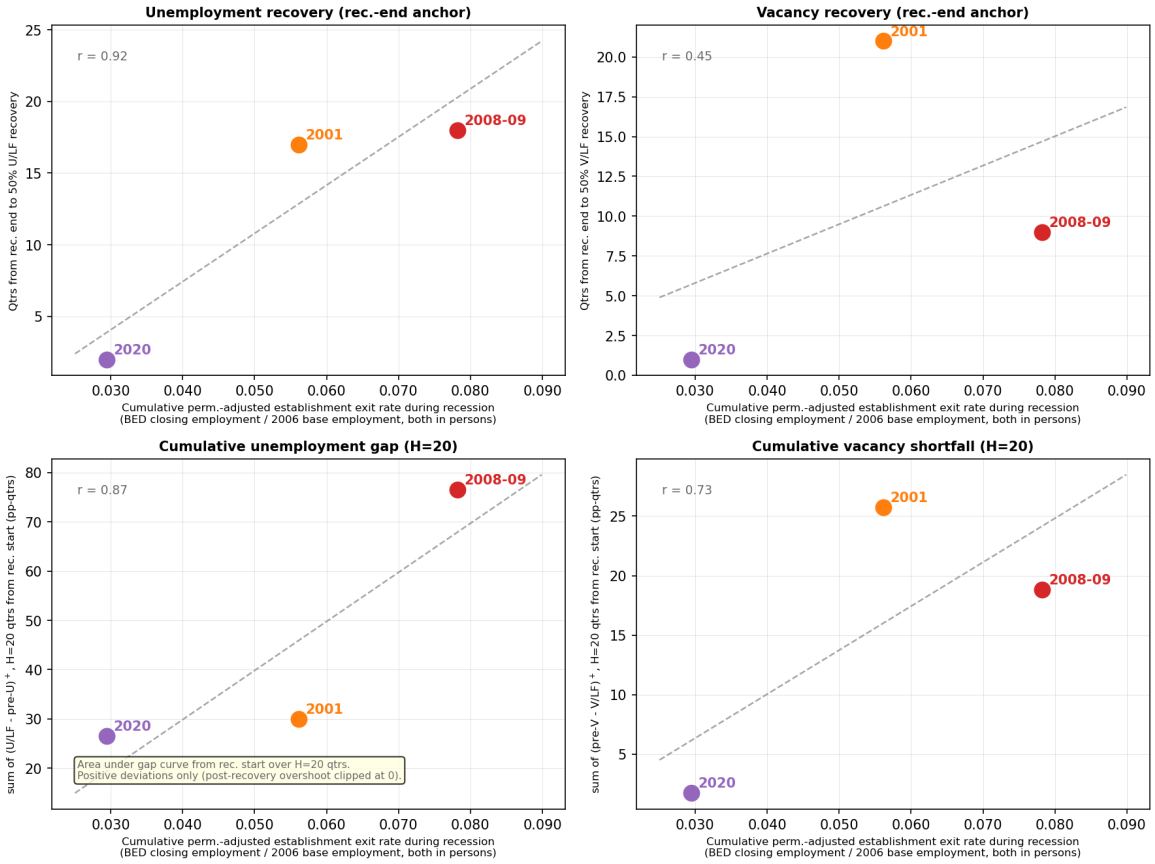


Figure 1: Relationship between cumulative establishment exit rates (BED Deaths) during recessions and labor market outcome measures. The top row shows quarters from the NBER recession end date until unemployment (left) and vacancy (right) rates recover 50% of the trough gap. The bottom row shows the cumulative unemployment gap (CUG, left) and cumulative vacancy shortfall (CVS, right), defined as the sum of positive deviations from the pre-recession baseline over $H = 20$ quarters from recession onset. Exit rates are constructed from BLS Business Employment Dynamics (BED) establishment deaths (establishments absent for four or more consecutive quarters), normalized by 2006 base-year employment.

Third, states with structurally higher establishment exit rates accumulate larger labor market gaps during recessions. Figure 2 plots each state's full-sample average Bartik-weighted estab-

lishment exit rate against its cumulative unemployment gap (CUG) and cumulative vacancy shortfall (CVS), both summed over 20 quarters from recession onset and averaged across the 2001, 2007–09, and 2020 recessions. The labor-force-weighted cross-sectional correlations are 0.65 (CUG) and 0.28 (CVS): states with structurally higher exit propensity accumulate substantially more total unemployment damage during downturns, with the vacancy pattern weaker but directionally consistent. The stronger CUG correlation is itself informative — it suggests that exit intensity shapes the *duration* of labor market distress more than the *depth* of the vacancy shortfall, consistent with the model's prediction that firm destruction generates persistent Beveridge curve dislocations rather than transitory displacements.[6] However, the cross-sectional pattern conflates structural differences in industrial composition with the causal effect of exit shocks: high-exit states may differ from low-exit states along dimensions — industry mix, institutional environment, average firm age — that independently predict labor market performance. To isolate causal transmission, we construct Bartik shift-share instruments that exploit within-state variation in exit exposure driven by national industry-level closing rates rather than local conditions, building separate instruments for δ from BED establishment deaths and for s from JOLTS layoff and discharge rates. We then estimate local projection impulse response functions to externally identify the dynamic transmission of each shock to state-level unemployment and vacancies, providing a quasi-experimental foundation for the structural parameters prior to Bayesian estimation.

[6]The weaker CVS correlation is also consistent with the model: the vacancy stock partially self-corrects through reposting by surviving firms, attenuating the cross-state vacancy signal relative to the unemployment signal.

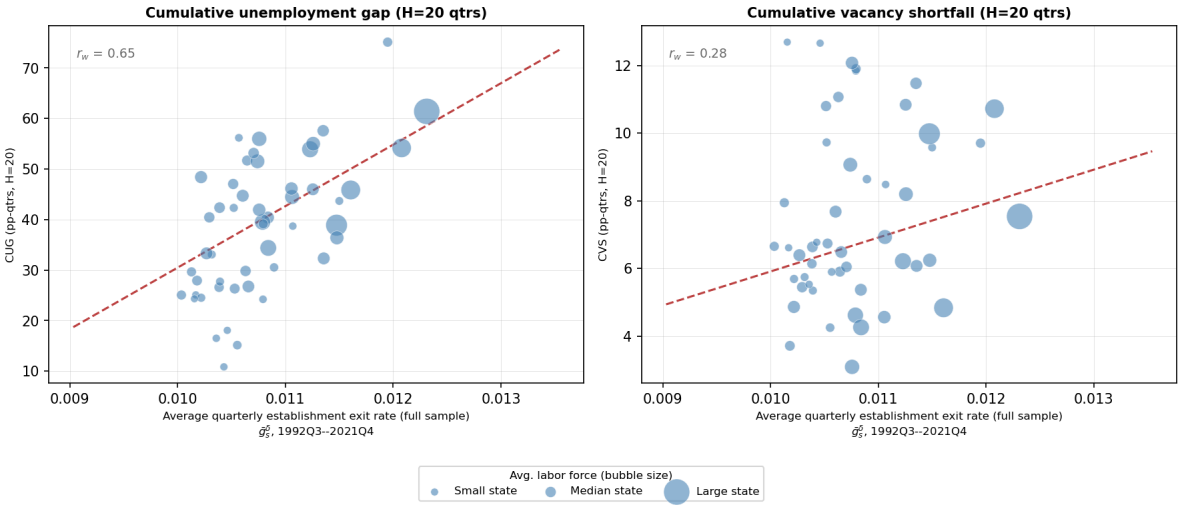


Figure 2: Cross-state relationship between average establishment exit rates and cumulative recession labor market damage. Each point is one of the 50 U.S. states; bubble size is proportional to average labor force. The horizontal axis is the full-sample time average of the state-level Bartik-weighted establishment exit rate ($\bar{g}_s^\delta$, 1992Q3–2021Q4), constructed from BLS Business Employment Dynamics (BED) data using 2006 base-year employment shares. The left panel shows the cumulative unemployment gap (CUG): the sum of quarterly deviations of the unemployment rate above its pre-recession average over $H = 20$ quarters from recession onset, in percentage-point-quarters. The right panel shows the analogous cumulative vacancy shortfall (CVS) for the vacancy rate. Both outcomes are averaged across the three post-1992 NBER recessions (2001, 2007–2009, 2020), weighted by state labor force at recession onset. The regression line is estimated by WLS with state labor force weights; r_w denotes the labor-force-weighted correlation.

Following Goldsmith-Pinkham, Sorkin, and Swift (2020), we bring quasi-experimental discipline to the δ transmission channel using Bartik shift-share instruments constructed from BED establishment deaths — the subset of closings absent from the UI payroll base for four or more consecutive quarters, aligning the instrument directly with the model's concept of irreversible product-line destruction. The instrument weights leave-one-out national industry-level death rates by each state's base-year (2006) industry employment shares. The identifying variation is geographic: a state with high pre-recession exposure to industries experiencing national waves of establishment closings — manufacturing in 2008–09, for instance — faces a large exit shock driven by national industry conditions rather than local demand, and can be compared to low-exposure states in the same quarter to isolate the causal labor market effect of firm destruction. The instrument is residualized on aggregate productivity growth to address cross-instrument correlation arising mechanically during aggregate downturns.

We estimate panel local projections (Jordà, 2005) with state and time fixed effects, standard

errors clustered at the state level, and outcomes capped at 2019Q4 to exclude the Covid episode. A one-standard-deviation rise in the δ instrument raises unemployment and lowers vacancies persistently. The unemployment rate rises by 0.22 percentage points on impact, significant from $h = 0$, and keeps rising, to about 1.71 points after four to five years. The vacancy rate falls, significantly from $h = 2$, to -0.58 points at $h = 18$. Both patterns are consistent with the model's prediction that firm destruction generates a persistent outward shift of the Beveridge curve rather than a transitory displacement along it.[7]

We estimate the model by Bayesian simulated method of moments with two sets of moments. The first follows Coles and Moghaddasi Kelishomi (2018): unconditional moments of unemployment, vacancies, labor productivity, and the job-finding and separation rates, augmented with the mean and relative volatility of the establishment death rate, which disciplines the destruction process directly rather than inferring it from labor market outcomes. The second is the $\delta \to u$ and $\delta \to v$ responses from the Bartik local projections, matched with inverse-variance weights. The unconditional moments pin down the destruction process and the steady state. The conditional moments discipline how costly and slow vacancy replacement is after a destruction shock, which the vacancy creation cost parameters ξ and x_m govern. Section 5 reports the results, together with a comparison with the clone-replacement benchmark of Section 3, which abstracts from variety and endogenous exit.

This paper contributes to four strands of literature. The first is search-and-matching models with finitely elastic vacancy creation, where our model builds most directly on Coles and Moghaddasi Kelishomi (2018) and Gabrovski and Silva (2025); Broer, Druedahl, Harmenberg, and Öberg (2025) independently combine finitely elastic vacancy creation with endogenous separation but abstract from product variety and the separate destruction margin. The second is endogenous product variety and firm dynamics in business cycles, where we build on Bilbiie, Ghironi, and Melitz (2012) and depart from the one-firm-one-worker structure of Shao and Silos (2013); Colciago, Fasani, and Rossi (2021) combine endogenous entry with labor market

[7] The δ unemployment response concentrates almost entirely in states experiencing tight financial conditions: at average financial conditions the point estimate is near zero and insignificant at all horizons beyond $h = 0$. This state-dependence in the unemployment channel, absent from the baseline model, is consistent with financial amplification of firm destruction and is left for future work; the vacancy channel, which enters the structural estimation as a moment condition, does not exhibit the same state-dependence.

frictions in a related framework but maintain infinitely elastic vacancy creation and abstract from the destruction margin. The third is Bartik shift-share identification in local labor markets (Goldsmith-Pinkham, Sorkin, and Swift, 2020; Nakamura and Steinsson, 2014), discussed alongside the instrument construction in Section 4. The fourth is Bayesian structural estimation of search models, extending the simulated method of moments approach of Coles and Moghaddasi Kelishomi (2018) to incorporate impulse response matching as an additional set of moment conditions (Christiano, Eichenbaum, and Trabandt, 2016).

Sections 2 and 3 present the model, its equilibrium, and the analytical results on the δ–s asymmetry. Section 4 presents the Bartik local projections, Section 5 the quantitative analysis, and Section 6 concludes.

2. Environment

Time is discrete and infinite. The economy is populated by households, intermediaries (recruiters), and retailers. Households and recruiters each have unit mass; the mass of retailers is endogenous, each supplying one product line. A household consists of a large family of workers who are either employed or unemployed, as in Merz (1995). Employed members supply one unit of labor inelastically. The household consumes and purchases equity. A unit measure of recruiters can each create one vacancy at a time, at a sunk cost drawn from a known distribution, with each vacancy specialized to a single product line. Retailers rent labor services from matched recruiters and produce the specialized variety.

Each period a number $m(u_t, v_t)$ of recruiter–worker pairs form, given unemployment u_t and vacancies v_t. We adopt the Cobb-Douglas matching function $m(u, v) = Au^{\eta_L} v^{1-\eta_L}$, where η_L is the elasticity of matches with respect to unemployment (Petrongolo and Pissarides, 2001). Constant returns to scale implies that the job-finding rate $f(\theta) = A\theta^{1-\eta_L}$ and the vacancy-filling rate $q(\theta) = A\theta^{-\eta_L}$ depend only on market tightness $\theta \equiv v/u$.

2.1. Preferences

We develop the model with general homothetic preferences as in Bilbiie, Ghironi, and Melitz (2012), then apply specific functional forms in the quantitative analysis. The household consumes a basket c_t, a homothetic aggregate over a continuum of goods Ω. The welfare-based price index P_t takes only individual prices as arguments, and demand for variety j satisfies

$$c_{jt} = c_t \partial P_t / \partial p_{jt}. \tag{1}$$

The symmetric price elasticity of demand is in general a function of the measure N_t of goods: $\varepsilon(N_t) = -\frac{\partial \log c_j}{\partial \log p_j}$. The welfare gain from an additional product variety is captured by the relative price and its elasticity:

$$\rho_j = \rho(N) \equiv \frac{p_j}{P}, \quad \zeta(N) = \frac{\rho'(N)}{\rho(N)} N.$$

The relative price rises with variety: with more products available at the same nominal price, the welfare-based cost of living falls, so ρ increases. Equivalently, spreading consumption across more goods benefits the household. The two statistics $\varepsilon(N_t)$ and $\rho(N_t)$ completely characterize symmetric homothetic preferences in the model.

Under standard Dixit–Stiglitz preferences (DS-CES), ε measures both the taste for diversity and the elasticity of substitution, so both the markup and the variety benefit are independent of the number of producers: $\zeta(N) = \mu(N) - 1 = 1/(\varepsilon - 1)$. The second specification is generalized CES following Benassy (1996), with consumption basket $N^{\zeta - 1/(\varepsilon-1)} \left(\int c_j^{(\varepsilon-1)/\varepsilon} dj \right)^{\varepsilon/(\varepsilon-1)}$; this disentangles the taste for variety ζ from the elasticity of substitution ε. The third specification uses the translog expenditure function of Feenstra (2003), under which $\varepsilon(N) = 1 + \eta N$, so goods become more substitutable as variety increases. Table 2 summarizes key statistics for all three specifications.

	DS-CES	General CES	Translog
Consumption index	$\left(\int c_j^{(\varepsilon-1)/\varepsilon} dj \right)^{\varepsilon/(\varepsilon-1)}$	$N^{\zeta - 1/(\varepsilon-1)} \left(\int c_j^{(\varepsilon-1)/\varepsilon} dj \right)^{\varepsilon/(\varepsilon-1)}$	–
Price index	$\left(\int p_j^{1-\varepsilon} dj \right)^{1/(1-\varepsilon)}$	$N^{-\zeta + 1/(\varepsilon-1)} \left(\int p_j^{1-\varepsilon} dj \right)^{1/(1-\varepsilon)}$	–
Markup $\mu(N)$	$\frac{\varepsilon}{\varepsilon-1}$	$\frac{\varepsilon}{\varepsilon-1}$	$1 + \frac{1}{\eta N}$
Relative price $\rho(N)$	$N^{\frac{1}{\varepsilon-1}}$	N^{ζ}	$\exp\left\{ -\frac{1}{2} \frac{\tilde{N} - N}{\eta \tilde{N} N} \right\}$
Benefit of variety $\zeta(N)$	$\mu - 1$	ζ	$\frac{1}{2\eta N} = \frac{\mu(N)-1}{2}$

Table 2: Preference specifications: indices, markup, relative price, and benefit of variety

2.2. Laws of motion for firms, unemployment, and vacancies

A product line survives into period t if it was not destroyed exogenously at the end of $t-1$ (probability $1 - \delta_{t-1}$) *and* its owner draws a continuation cost below the equilibrium cutoff χ_t^c at the start of t (probability $F(\chi_t^c)$). Both events occur before production and matching, so u_t is unemployment *after* firm exit but *before* matching. The law of motion for active product

lines is

$$N_t = (1 - \delta_{t-1})F(\chi_t^c)(N_{t-1} + N_{t-1}^e) \tag{2}$$

where N_{t-1}^e denotes last-period entrants. The entry lag means a fraction of entrants are destroyed before they produce. The total exit rate combining exogenous destruction and endogenous exit is

$$\delta_{e,t} \equiv 1 - (1 - \delta_{t-1})F(\chi_t^c), \tag{3}$$

where δ_t denotes the exogenous AR(1) destruction shock and $\delta_{e,t}$ denotes the total exit rate; we maintain this distinction throughout. The aggregate separation rate flowing into pre-matching unemployment u_t is

$$\tau_t = \delta_{e,t} + s_{t-1}(1 - \delta_{e,t}), \tag{4}$$

where s_{t-1} is the match-separation shock that realized at end of $t-1$. The unemployment law of motion is

$$u_t = \big[1 - (1 - \delta_{e,t})f(\theta_{t-1})\big]u_{t-1} + \tau_t(1 - u_{t-1}) \tag{5}$$

The first term is the stock of last period's unemployed who did not enter employment: those who did not find a job, plus those who matched in $t-1$ with a product line that then exited. The second is the fraction τ_t of the workers employed at the start of $t-1$ who separate before production in t. Matches formed in $t-1$ are exposed to product-line exit but not to match separation before they first produce. This is the standard DMP timing, also used by Gabrovski and Silva (2025), with the total separation rate τ_t replacing the constant separation rate. The separation base $1 - u_{t-1}$ also feeds the reposting inflow in (6): each separated match at a surviving product line returns a vacated position to its recruiter.

Sunk vacancy creation costs render vacancies partially predetermined: current vacancies evolve as

$$v_t = (1 - \delta_{e,t})[(1 - q(\theta_{t-1}))v_{t-1} + \Lambda_{r,t}s_{t-1}(1 - u_{t-1})] + e_t \tag{6}$$

Current vacancies are the sum of three components: (i) unmatched vacancies from last period that survived product destruction; (ii) filled jobs in place at the start of $t-1$ that experienced a match separation at end of $t-1$, whose firm survived both margins, and whose recruiter chose

to reactivate the vacancy; and (iii) vacancies created by new entrants e_t, who enter at the start of t before matching. Note that $(1-\delta_{e,t})$ multiplies both (i) and (ii): only firms surviving both exogenous destruction and the continuation-cost draw contribute to either term, so a shock to δ_t collapses the entire bracket. A vacated position is reactivated with probability $\Lambda_{r,t} = F(Q_t/\alpha)$, derived in (17), which carries the same date as the survival term. The remaining fraction $1-\Lambda_{r,t}$ is retired permanently.

2.3. Stochastic processes

Technology, product destruction, and idiosyncratic separation each follow an AR(1) process in logs:

$$\log z_t = \rho_z \log z_{t-1} + (1-\rho_z)\log \bar{z} + e_{zt} \tag{7}$$

$$\log \delta_t = \rho_\delta \log \delta_{t-1} + (1-\rho_\delta)\log \bar{\delta} + e_{\delta t} \tag{8}$$

$$\log s_t = \rho_s \log s_{t-1} + (1-\rho_s)\log \bar{s} + e_{st} \tag{9}$$

where $\bar{z}$, $\bar{\delta}$, and $\bar{s}$ are the long-run means of technology, exogenous product destruction, and idiosyncratic separation, respectively, and innovations e_{zt}, $e_{\delta t}$, e_{st} are mutually uncorrelated. As illustrated in Figure 3, the aggregate state, including s_t and δ_t, is realized at the start of period t, so all date-t decisions condition on it. The rates s_t and δ_t govern separation and destruction at the end of t, after production, bargaining, and matching; only which matches and product lines they hit is revealed then. The stocks entering t were therefore shaped by the rates of $t-1$, through $\delta_{e,t}$ and τ_t. This lag, combined with sunk vacancy creation costs, is the source of predetermination: vacancies and employment are partially predetermined with respect to current shocks. The two shocks also reach different stocks. Match separation s_t hits only the $1-u_t$ matches in place at the start of t, while product destruction removes every position attached to the line, including new matches and unfilled vacancies.

At the start of t, alongside the exit decision, recruiters choose whether to reactivate positions vacated when matches dissolved at the end of $t-1$. Conditional on the product line surviving, the recruiter draws a cost from F for each vacated position and resumes it whenever $\alpha\chi \leq Q_t$. Because the recruiter conditions on the realized Q_t rather than on its expectation, $\Lambda_{r,t}$ is procyclical and position destruction is countercyclical.

Because positions are held by recruiters and labor is rented competitively, how vacated positions are spread across recruiters does not matter to retailers. Only the aggregate vacancy

stock enters matching, and the law of large numbers delivers $\Lambda_{r,t}$.

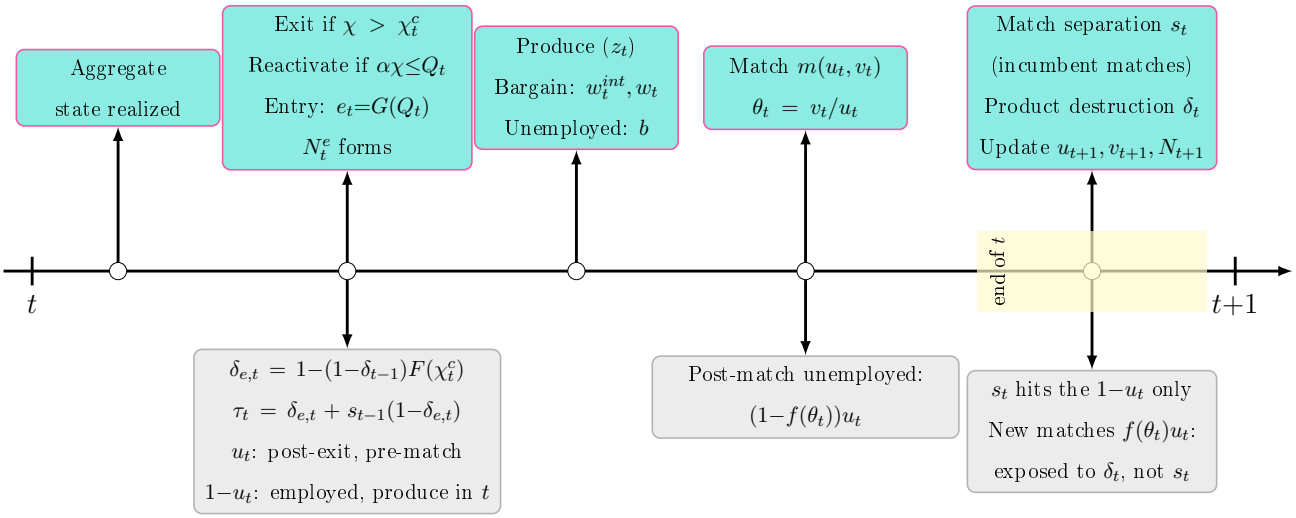


Figure 3: Timing. Turquoise: agent decisions. Grey: implied states. Yellow band: shocks realizing end of t, governing t+1. Match separation applies only to matches in place at the start of t; matches formed in t first face it at the end of t+1.

3. Equilibrium

This section derives equilibrium given the environment of Section 2. We then show that equilibrium comprises three blocks: a business formation block governing the evolution of product lines and firm values, a labor market block governing vacancy creation, matching, and wage determination, and stochastic processes for the three structural shocks. Table A.6 positions the paper relative to the closest antecedents across these dimensions. Section 3.9 derives formally the asymmetric transmission of δ and s shocks that motivates the paper's empirical strategy.

3.1. Household problem

Let ν_t denote the ex-dividend equity price of the aggregate firm portfolio and ν_t^f the post-dividend value of an individual firm, following Bilbiie, Ghironi, and Melitz (2012). Given risk-aversion parameter σ and discount factor β, the household chooses consumption c_t and equity shares a_t to maximize expected discounted utility:

$$\max_{c_t, a_t} \sum_{t=0}^{\infty} \mathbb{E}_0 \, \beta^t c_t^{1-\sigma}/(1-\sigma)$$

subject to

$$c_t + \nu_t a_{t+1} = w_t(1-u_t) + bu_t + (\nu_t + d_t)a_t + D_t^{int} - T_t \tag{10}$$

where b is the value of unemployment, d_t is the per-share dividend, D_t^{int} are intermediary dividends, and T_t are lump-sum taxes financing unemployment benefits. Since households have unit mass, $c_t = C_t$ in equilibrium. The first-order condition for equity shares yields the standard asset-pricing equation:

$$\lambda_t \nu_t = \beta \, \mathbb{E}_t \lambda_{t+1} [\nu_{t+1} + d_{t+1}] \tag{11}$$

where $\lambda_t = c_t^{-\sigma}$ is the marginal utility of consumption and $m_{t+1} \equiv \beta(\lambda_{t+1}/\lambda_t)$ is the stochastic discount factor. To express (11) at the firm level, use the mappings $\nu_t = \nu_t^f (N_t + N_t^e)$ and $\nu_t + d_t = (\nu_t^f + d_t^f) N_t$, together with the firm law of motion (2). This yields the firm-level Euler equation, familiar from BGM and extended here to incorporate endogenous exit:

$$\nu_t^f = \mathbb{E}_t m_{t+1} (1 - \delta_t) F(\chi_{t+1}^c)(\nu_{t+1}^f + d_{t+1}^f) \tag{12}$$

Equation (12) equates the current firm value to the expected discounted sum of next-period firm value and dividends, adjusted for the stochastic discount factor m_{t+1} and the survival probability $(1 - \delta_t)F(\chi_{t+1}^c)$, which combines the probability of surviving exogenous destruction $(1 - \delta_t)$ and the probability of drawing a continuation cost below the equilibrium cutoff χ_{t+1}^c, with probability $F(\chi_{t+1}^c)$.

3.2. Recruiters

There is a fixed unit measure of recruiters who can create vacancies subject to a sunk cost x drawn from cdf $G(\cdot)$. Additionally, if a vacancy is filled the recruiter pays a fixed real cost κ before bargaining with the newly matched worker.[8] After matching, the recruiter rents labor to an incumbent retailer with product line j. We introduce recruiters as a separate agent type to avoid the complications of Stole-Zwiebel intra-firm bargaining that arise when workers negotiate directly with retailers facing a downward- sloping demand curve: with a separate recruiter, the bilateral bargaining problem reduces to the standard DMP surplus-splitting equation regardless of the retail market structure.[9] Let Q_t^j and J_t^j denote the values of an unfilled and filled vacancy

[8] Pissarides (2009) first suggested this matching cost to make vacancy posting more responsive; Christiano, Eichenbaum, and Trabandt (2016) estimate it accounts for the dominant share of recruiting costs.

[9] Under Stole-Zwiebel bargaining, retailers would strategically over-hire to improve their bargaining position, generating a fixed-point problem in employment that significantly complicates aggregation. The recruiter segmentation is isomorphic to the retailer– wholesaler structure of Christiano, Eichenbaum, and Trabandt (2016), with recruiters replacing intermediate goods.

associated with product line j.

The value of an unfilled vacancy for product line j satisfies

$$\begin{aligned} Q_t^j &= \mathbb{E}_t m_{t+1}(1-\delta_t)F(\chi_{t+1}^c)\big[(1-q(\theta_t))Q_{t+1}^j + q(\theta_t)J_{t+1}^j\big] - q(\theta_t)\kappa \\ &= \mathbb{E}_t m_{t+1}(1-\delta_t)F(\chi_{t+1}^c)\big[Q_{t+1}^j + q(\theta_t)(J_{t+1}^j - Q_{t+1}^j)\big] - q(\theta_t)\kappa \end{aligned} \tag{13}$$

An unmatched recruiter survives to next period only if the product line survives both exogenous destruction and the continuation-cost draw, with probability $(1-\delta_t)F(\chi_{t+1}^c)$. In the next period, the vacancy fills with probability $q(\theta_t)$, generating value J_{t+1}^j, or remains unfilled with probability $1-q(\theta_t)$. The matching cost κ is paid upon filling. The recruiter chooses the best product line: $Q_t \equiv \max_j Q_t^j$. In equilibrium, competition equalizes values across active product lines: $Q_t^j = Q_t$ for all j with $v_t^j > 0$.

Define the flow value of a vacancy as the net benefit of holding a vacancy this period relative to next:

$$K_t \equiv Q_t - \mathbb{E}_t m_{t+1}(1-\delta_t)F(\chi_{t+1}^c)Q_{t+1} \tag{14}$$

Inserting (14) into (13) and rearranging places the average hiring cost on the left-hand side:

$$\kappa + \frac{K_t}{q(\theta_t)} = \mathbb{E}_t m_{t+1}(1-\delta_t)F(\chi_{t+1}^c)(J_{t+1}^j - Q_{t+1}) \tag{15}$$

A recruiter opens a vacancy whenever $x \le Q_t$, so total newly opened vacancies are $e_t = G(Q_t)$. Following CK, we adopt a power law distribution:

$$G(x) = \left(\frac{x}{x_m}\right)^{\xi}, \quad x \in [0,\, x_m]$$

with density $g(x) = \xi x^{\xi-1}/x_m^{\xi}$.[10] Provided $Q < x_m$, inverting $e = G(Q)$ gives

$$Q_t = e_t^{1/\xi}\, x_m$$

so the elasticity of the vacancy value with respect to entry is $1/\xi$. Free entry of vacancies ($\xi \to \infty$) nests the DMP baseline.

By symmetry of the retail problem (established in Section 3.4), recruiter compensation $w_t^{int,j}$ and the value of a filled vacancy J_t^j are identical across active product lines; we henceforth drop

[10] If $\xi > 1$ the density is rising (relatively more high-cost recruiters); if $\xi < 1$ it is falling; if $\xi = 1$ costs are uniform on $[0, x_m]$.

the superscript j. Recruiters supply labor in a competitive market, taking w_t^{int} as given. The value of a matched recruiter is[11]

$$J_t = w_t^{int} - w_t + \mathbb{E}_t m_{t+1}(1-\delta_t)F(\chi_{t+1}^c)\big(s_t\, Q_{t+1}^{rep} + (1-s_t)J_{t+1}\big) \tag{16}$$

The recruiter receives w_t^{int} from the retailer and pays the worker w_t. Conditional on product-line survival, the match continues with probability $1-s_t$. Otherwise the worker separates and the recruiter may reactivate the vacancy at cost $\alpha\chi$, with $\chi \sim F$ drawn from the same distribution as the continuation cost, scaled by α. Reactivation is optimal when the recovered vacancy covers the cost, $Q_{t+1} \geq \alpha\chi$, which occurs with probability $\Lambda_{r,t+1}$, where the reactivation rate is

$$\Lambda_{r,t} \equiv F\left(\frac{Q_t}{\alpha}\right) = \min\left\{(1-p_0) + p_0\left(\frac{Q_t}{\alpha\chi_m}\right)^{\psi},\, 1\right\} \tag{17}$$

using the form of F in (24). Like the survival probability (25), $\Lambda_{r,t}$ evaluates F at a threshold. The zero-cost atom always reactivates, so $\Lambda_{r,t} \in [1-p_0, 1]$, and every vacated position is reactivated once $Q_t \geq \alpha\chi_m$. Because Q_t is procyclical, so is the reactivation rate. The expected value of a separated position is therefore the reactivation option value

$$Q_t^{rep} \equiv \mathbb{E}\big[\max\{Q_t - \alpha\chi,\, 0\}\big] = Q_t\Lambda_{r,t} - \alpha\int_0^{Q_t/\alpha} \chi\, dF(\chi), \tag{18}$$

the mass of reactivated vacancies valued at Q_t net of the reactivation costs paid. Costless full reposting ($Q_t^{rep} = Q_t$) is nested as $\alpha \to 0$.

Job creation condition. Subtracting (13) from (16) and using the definition of K_t yields recruiter surplus:

$$J_t - Q_t = w_t^{int} - w_t - K_t + (1-s_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right) - s_t\mathcal{R}_t \tag{19}$$

where

$$\mathcal{R}_t \equiv \mathbb{E}_t m_{t+1}(1-\delta_t)F(\chi_{t+1}^c)\left(Q_{t+1} - Q_{t+1}^{rep}\right) \tag{20}$$

[11] The asset-pricing equations (13) and (16) can equivalently be obtained by having the household purchase shares of recruiter firms directly; the stochastic discount factor ensures equivalence. Writing it via the household problem makes the derivation more compact and facilitates comparison to BGM and Mortensen and Pissarides (1994).

is the expected discounted reposting shortfall: the value a separated position loses relative to costless reposting, through vacancies that are not reactivated and through the reactivation costs paid on those that are. It vanishes as $\alpha \to 0$. Imposing symmetry in (15) gives the job creation condition:

$$\kappa + \frac{K_t}{q(\theta_t)} = \mathbb{E}_t m_{t+1}(1-\delta_t)F(\chi^c_{t+1})\,(J_{t+1} - Q_{t+1}) \tag{21}$$

The left-hand side is the average hiring cost: the fixed matching cost κ plus the flow cost $K_t/q(\theta_t)$, equal to the opportunity cost of holding the vacancy this period multiplied by its expected duration $1/q(\theta_t)$. The right-hand side is the expected surplus of a match that begins producing next period, discounted by the stochastic discount factor and the product-line survival probability. By (19), that surplus is the recruiter's net revenue less the flow value of the vacancy, plus the hiring cost saved if the match survives separation, less the reposting shortfall if it does not.

3.3. Wage determination

Nash bargaining between the recruiter and worker yields

$$w_t = \phi\left(w_t^{int} - K_t + f(\theta_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right) - s_t\mathcal{R}_t\right) + (1-\phi)b \tag{22}$$

where ϕ is the worker's bargaining power.[12] Recruiter compensation $w_t^{int} = \rho(N_t)z_t/\mu(N_t)$ — derived in Section 3.4 — enters negatively through the markup. The flow entry cost K_t has opposing effects: it reduces the match surplus (dampening wages by ϕK_t) but improves the worker's threat point, raising the tightness slope of the wage curve by $\phi\theta K_t$; the second effect dominates when $\theta > 1$. The fixed cost κ raises wages by $\phi f(\theta_t)\kappa$ because a failed negotiation forces the firm to pay κ again upon next filling the vacancy. The reposting shortfall lowers the wage by $\phi s_t\mathcal{R}_t$: the expected loss of position value on separation works as an extra flow cost of holding a filled position, alongside K_t, and the worker bears share ϕ of it. The household's surplus carries no such term, because a separated worker becomes unemployed whether or not the position is reactivated (Appendix E.4). Setting $w_t^{int} = z_t$, $\kappa = 0$, and $\alpha \to 0$ recovers the CK wage equation.

[12] The wage equation (22) has the same general form as Shao and Silos (2013), their equations 21 and 22. Their setting includes capital affecting w_t^{int} and writes the wage using the vacancy value Q_t rather than the flow value K_t.

Substituting (22) into (19) to eliminate w_t yields the wage-substituted surplus:

$$J_t - Q_t = (1-\phi)\left(w_t^{int} - K_t - b\right) - \phi\theta_t\left(K_t + q(\theta_t)\kappa\right) + (1-s_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right) - (1-\phi)s_t\mathcal{R}_t \tag{23}$$

The recruiter bears share $1-\phi$ of the reposting shortfall; the worker bears the rest through the wage.

3.4. Retailers

A retailer selling variety j rents labor from recruiters at wage $w_t^{int,j}$. At the beginning of each period, before production, the firm draws a continuation cost $\chi \sim F$ in units of the consumption basket, where F is a mixture between a mass point at zero with probability $1-p_0$ and a bounded power law:

$$F(\chi) = (1-p_0) + p_0\left(\frac{\chi}{\chi_m}\right)^{\psi}, \quad \chi \in [0,\, \chi_m] \tag{24}$$

A firm with draw $\chi > \chi_t^c$ exits; one with $\chi \leq \chi_t^c$ pays the cost and continues. Define the survival probability:

$$\Lambda_t \equiv F(\chi_t^c) = (1-p_0) + p_0\left(\frac{\chi_t^c}{\chi_m}\right)^{\psi} \tag{25}$$

so that the endogenous exit rate is $1-\Lambda_t = p_0\left(1-(\chi_t^c/\chi_m)^{\psi}\right)$, consistent with the total exit rate $\delta_{e,t} = 1-(1-\delta_{t-1})\Lambda_t$ from Section 2. The elasticity of the endogenous exit rate with respect to the cutoff is $\psi p_0(\chi_t^c/\chi_m)^{\psi}/(1-\Lambda_t)$, which increases in ψ and falls as the mass point $1-p_0$ rises.

The value of a firm after its cost realization is:

$$\nu_t^f(\chi) = \max\left\{0,\, \max_{y_t}\left[\rho^d(y_t)y_t - w_t^{int}\frac{y_t}{z_t}\right] - \chi + (1-\delta_t)\mathbb{E}_t m_{t+1}\Lambda_{t+1}\,\nu_{t+1}^{ef}\right\} \tag{26}$$

where $\nu_t^{ef} \equiv (1/\Lambda_t)\int_0^{\chi_t^c}\nu_t^f(\tilde{\chi})\,dF(\tilde{\chi})$ is the expected firm value conditional on survival, and $\rho^d(y_t)$ is the inverse demand curve from (1), equal in symmetric equilibrium to $\rho(N_t)$. The profit maximization problem is static: letting $MR \equiv \rho'(y)y + \rho(y)$ denote marginal revenue, the first-order condition is

$$MR_t \cdot z_t = w_t^{int,j}$$

The left-hand side is the marginal revenue product; the right-hand side is recruiter compensation, which exceeds the worker's wage due to matching frictions. Equivalently, $MR_t = MC_t = w_t^{int}/z_t$. The standard markup condition $\rho_t = \mu(N_t)MC_t$ gives recruiter compensation:

$$w_t^{int} = \frac{\rho(N_t)}{\mu(N_t)} z_t \tag{27}$$

Recruiter compensation is increasing in the relative price $\rho(N_t)$ and technology z_t and decreasing in the markup $\mu(N_t)$. Since $\mu'(N_t) \leq 0$, w_t^{int} is increasing in N_t overall.[13] Retail profits gross of the continuation cost are a share $(\mu(N_t) - 1)/\mu(N_t)$ of sales:

$$R_t^f = \frac{\mu(N_t) - 1}{\mu(N_t)} \rho(N_t) y_t \tag{28}$$

3.5. Firm entry and exit

By definition $\nu_t^f(\chi_t^c) = 0$. Evaluating the firm Bellman (26) at $\chi = \chi_t^c$ yields the cutoff condition:

$$\chi_t^c = R_t^f + \nu_t^f \tag{29}$$

where $\nu_t^f \equiv \nu_t^{ef} \cdot \Lambda_t$ is the unconditional expected firm value. From the Bellman, the firm value conditional on the cost realization is $\nu_t^f(\chi) = \chi_t^c - \chi$ for all $\chi \leq \chi_t^c$, so that $\nu_t^{ef} = \chi_t^c - E(\chi \,|\, \chi \leq \chi_t^c)$. For the power law component, the conditional mean of a positive draw satisfies $E(\chi \,|\, 0 < \chi \leq \chi_t^c) = \psi_c \chi_t^c$ where $\psi_c \equiv \psi/(\psi + 1)$, and positive draws are a share $p_0(\chi_t^c/\chi_m)^\psi/\Lambda_t$ of the draws below the cutoff, so the overall conditional mean (including the mass point at zero) is $E(\chi \,|\, \chi \leq \chi_t^c) = p_0 \psi_c \chi_t^c (\chi_t^c/\chi_m)^\psi/\Lambda_t$. Hence:

$$\nu_t^f = (1 - \delta_t)\mathbb{E}_t m_{t+1} \Lambda_{t+1} \, \nu_{t+1}^{ef} \tag{30}$$

New firms are created using technology $N_t^e = z_t l_t^e / f_e$, where l_t^e is labor hired from recruiters at wage w_t^{int} and f_e is the sunk entry cost in labor units, equivalent to $w_t^{int} f_e / z_t = \rho(N_t) f_e / \mu(N_t)$ units of the consumption basket. Firms enter until the post-dividend firm value equals the sunk entry cost:

$$\nu_t^f = \frac{\rho(N_t) f_e}{\mu(N_t)} \tag{31}$$

[13] Following BGM, marginal cost is w_t^{int}/z_t. Under DS-CES, $\mu = \varepsilon/(\varepsilon - 1)$ is constant, recovering the standard formula.

Combining the free entry condition (31) with the cutoff condition (29) yields:

$$\chi_t^c = R_t^f + \frac{\rho(N_t) f_e}{\mu(N_t)} \tag{32}$$

The cutoff rises with current profitability R_t^f and the value of a new product line $\rho(N_t) f_e / \mu(N_t)$: when conditions improve, marginal firms that would otherwise exit find it worthwhile to continue.

3.6. Aggregation

Labor market clearing implies $L_t = 1 - u_t$. Define gross consumption output $Y_t^c \equiv \rho(N_t) N_t y_t$ and retail employment $L_t^c \equiv N_t l_t^c$. Aggregate continuation costs paid by surviving firms are

$$X_t^c \equiv (1-\delta_{t-1})(N_{t-1} + N_{t-1}^e) \int_0^{\chi_t^c} \chi \, dF(\chi) = N_t \, E(\chi \,|\, \chi \le \chi_t^c) = N_t \, \psi_c \, \chi_t^c \, \frac{p_0}{\Lambda_t} \left(\frac{\chi_t^c}{\chi_m} \right)^{\psi} \tag{33}$$

where $\psi_c \equiv \psi/(\psi+1)$. Every line that survives exogenous destruction draws a cost, and those drawing $\chi > \chi_t^c$ exit and pay nothing, so the pre-exit mass multiplies the partial expectation. Since $N_t = \Lambda_t (1-\delta_{t-1})(N_{t-1} + N_{t-1}^e)$, this is the surviving mass times the conditional mean. The factor $p_0 (\chi_t^c / \chi_m)^{\psi} / \Lambda_t$ is the share of survivors that drew a positive cost.

Aggregate reactivation costs are paid by surviving firms that resume a position vacated at the end of $t-1$:

$$X_t^r \equiv (1 - \delta_{e,t}) \, s_{t-1} (1 - u_{t-1}) \int_0^{Q_t/\alpha} \alpha \chi \, dF(\chi) \tag{34}$$

Only the power-law part of F contributes, since the atom at zero carries no cost, so the per-position integral is $\int_0^{Q_t/\alpha} \alpha\chi \, dF(\chi) = \psi_c Q_t \, p_0 \big(Q_t/(\alpha\chi_m)\big)^{\psi} = \psi_c Q_t (\Lambda_{r,t} - 1 + p_0)$, using the interior branch of (17). The cost scale α survives only through $\Lambda_{r,t}$. The count in (34) omits $\Lambda_{r,t}$ because the integral is a partial expectation, which already embeds the reactivation probability. As $\Lambda_{r,t} \to 1 - p_0$, only the zero-cost atom reactivates and $X_t^r \to 0$.

Aggregate retail profits gross of continuation costs are $\Pi_t^f = Y_t^c / \varepsilon(N_t) - X_t^c$, and recruiter profits, net of the vacancy creation and matching costs X_t and the reactivation costs X_t^r that recruiters pay, are $\Pi_t^{int} = (w_t^{int} - w_t) L_t - X_t - X_t^r$. The average per-firm dividend is $d_t^f = \Pi_t^f / N_t$.

Total vacancy creation costs are the sum of sunk posting costs and fixed matching costs:

$$X_t = X_v + X_f = e_t \frac{\xi}{\xi + 1} Q_t + \kappa q(\theta_t) v_t \tag{35}$$

where $X_v = x_m^{-\xi}\frac{\xi}{\xi+1}Q_t^{\xi+1}$ follows from integrating $xg(x)$ over $[0, Q_t]$. Given e_t and Q_t, the parameter x_m has no additional effect on entry costs provided $Q_t < x_m$.

Aggregating (10) with asset market clearing $a_{t+1} = a_t$ and $T_t = bu_t$, and using $Y_t^{Gross} = Y_t^c + \nu_t^f N_t^e$, real GDP net of intermediate inputs is:

$$Y_t \equiv Y_t^{Gross} - X_t - X_t^c - X_t^r = C_t + \nu_t^f N_t^e = w_t L_t + \Pi_t^f + \Pi_t^{int} \tag{36}$$

GDP comprises consumption and investment in new product lines. Moreover, decomposing output into income confirms the national accounts identity.

Using free entry (31) and $d_t^f = \Pi_t^f / N_t$, the firm-level Euler equation (12) becomes the business formation equation:

$$f_e\rho(N_t) = (1-\delta_t)\mathbb{E}_t m_{t+1}\Lambda_{t+1}\frac{\mu(N_t)}{\mu(N_{t+1})}\left\{f_e\rho(N_{t+1}) + \frac{Y_{t+1}^c}{N_{t+1}}\left(\mu(N_{t+1}) - 1 - \mu(N_{t+1})\frac{X_{t+1}^c}{Y_{t+1}^c}\right)\right\} \tag{37}$$

The left-hand side is the current cost of creating a product line. The right-hand side is the expected discounted return, adjusted for survival probability $(1-\delta_t)\Lambda_{t+1}$ and the current-to-future markup ratio $\mu(N_t)/\mu(N_{t+1})$. The term in braces is the next-period firm value plus the private marginal product of a new product line net of continuation costs; the net markup $\mu(N_{t+1}) - 1$ governs the return to entry.

3.7. Definition of equilibrium

Definition 1. *Given $\rho(N)$, $\mu(N)$, F, initial exogenous states $\{z_0, \delta_{-1}, s_{-1}\}$, and initial endogenous states $\{v_{-1}, u_{-1}, B_{-1}\}$, where $B_{-1} \equiv N_{-1} + N_{-1}^e$ is the stock of product lines entering period 0 before exit, an equilibrium is a sequence $\{N_t, C_t, \chi_t^c, Y_t^c, \theta_t, K_t, Q_t, e_t, v_t, u_t\}_{t=0}^{\infty}$ satisfying the business formation block:*

$$N_t = (1-\delta_{t-1})\Lambda_t\left(N_{t-1} + \frac{z_{t-1}(1-u_{t-1})}{f_e} - \frac{Y_{t-1}^c}{f_e\rho(N_{t-1})}\right) \tag{38}$$

$$f_e\rho(N_t) = (1-\delta_t)\mathbb{E}_t m_{t+1}\Lambda_{t+1}\frac{\mu(N_t)}{\mu(N_{t+1})}\left\{f_e\rho(N_{t+1}) + \frac{Y_{t+1}^c}{N_{t+1}}\left(\mu(N_{t+1}) - 1 - \mu(N_{t+1})\frac{X_{t+1}^c}{Y_{t+1}^c}\right)\right\} \tag{39}$$

$$Y_t^c = C_t + X_t + X_t^c + X_t^r \tag{40}$$

$$\chi_t^c = \frac{Y_t^c(\mu(N_t) - 1)}{\mu(N_t)N_t} + \frac{f_e\rho(N_t)}{\mu(N_t)} \tag{41}$$

and the labor market block:

$$\kappa + \frac{K_t}{q(\theta_t)} = \mathbb{E}_t m_{t+1}(1-\delta_t)\Lambda_{t+1}\left[(1-\phi)\left(\frac{\rho(N_{t+1})z_{t+1}}{\mu(N_{t+1})} - K_{t+1} - b\right) - \phi\theta_{t+1}\left(K_{t+1} + q(\theta_{t+1})\kappa\right)\right.$$

$$+ (1 - s_{t+1}) \left(\kappa + \frac{K_{t+1}}{q(\theta_{t+1})} \right) - (1 - \phi) s_{t+1} \mathcal{R}_{t+1} \Bigg] \tag{42}$$

$$K_t = Q_t - (1 - \delta_t) \mathbb{E}_t m_{t+1} \Lambda_{t+1} Q_{t+1} \tag{43}$$

$$e_t = \left(\frac{Q_t}{x_m} \right)^{\xi} \tag{44}$$

$$\theta_t = \frac{v_t}{u_t} \tag{45}$$

together with laws of motion (5)–(6)*, auxiliary definitions* $\delta_{e,t}$ (3)*,* τ_t (4)*,* Λ_t (25)*,* $\Lambda_{r,t}$ (17)*, matching rates* $f(\theta_t) = A\theta_t^{1-\eta_L}$ *and* $q(\theta_t) = A\theta_t^{-\eta_L}$*,* Q_t^{rep} (18)*,* $\mathcal{R}_t$ (20)*,* X_t (35)*,* X_t^c (33)*,* X_t^r (34)*,* m_t (11)*, and stochastic processes* (7)–(9)*.*

The job creation condition substitutes the wage-substituted surplus (23), dated $t + 1$, into (21), so the matched value J_t and the wage drop out. Carrying the reposting shortfall $\mathcal{R}_t$ as a variable with its own one-period equation (20) keeps the system first order, without a nested expectation. The two blocks interact through $\rho(N_t)$ and $\mu(N_t)$ in the job creation condition and through δ_t in the survival-adjusted discount factor $(1 - \delta_t)\Lambda_{t+1}$, which appears in both.

3.8. Limiting cases

Like Coles and Moghaddasi Kelishomi (2018), we study a destruction shock that removes jobs and vacancies together, and, like Gabrovski and Silva (2025), we consider separations that pertain to matches as well as business opportunities. However, those articles abstract from product variety and entry/exit decisions. This subsection connects the two settings. It first states when the labor market block can be solved on its own. It then constructs the benchmark economy against which the baseline is compared, and shows that this benchmark is a limit of the baseline. That limit is where the analytical results of Section 3.9 are proved. Proofs are in Appendix G.

Proposition 1 (Independence of labor market block)**.** *Consider general CES preferences with elasticity of substitution* ε*. The labor market block* $\{u_{t+1}, v_t, e_t, Q_t, K_t, J_t, w_t\}_{t=0}^{\infty}$ *can be solved independently of the business formation block for all values of the remaining parameters if and only if* $\sigma = 0$*,* $\zeta = 0$*, and* $p_0 = 0$*.*

When $\sigma = 0$, risk neutrality fixes $m_{t+1} = \beta$, eliminating the consumption dependence of the stochastic discount factor. When $\zeta = 0$, the taste for variety vanishes and $\rho(N_t) = 1$, removing

N_t from recruiter compensation $w_t^{int} = \rho(N_t)z_t/\mu(N_t)$ and from the job creation condition while leaving μ and hence the business formation block otherwise intact. Setting $p_0 = 0$ makes F a point mass at zero. This removes both endogenous exit ($\Lambda_t = 1$ and $\delta_{e,t} = \delta_{t-1}$) and partial reposting ($\Lambda_{r,t} = 1$, $Q_t^{rep} = Q_t$, and $X_t^r = 0$). That is, every vacated position is resumed at no cost and the reposting margin disappears along with the exit margin.[14] Under all three conditions, the labor market block closes on its own; the business formation block then determines C_t and N_t taking u_t and X_t as given from the labor market block.

Clone replacement. The benchmark must keep the destruction flow, the channel that Coles and Moghaddasi Kelishomi (2018) and Gabrovski and Silva (2025) study, while abstracting from the decisions to enter and to exit. We operationalize this with *clone replacement*. Product lines are destroyed at the rate δ_t, as in the baseline, and destruction still eliminates every position attached to the line. Each destroyed line is immediately replaced by an identical one, at no resource cost, so the mass of lines stays at $N_t = 1$ at all dates. Two things follow. There is no business formation margin, because entry is not a decision. And there is no variety effect, because N_t never moves and $\rho = \mu = 1$. What survives is the destruction channel itself. Since $\Lambda_t = 1$ implies $\delta_{e,t} = \delta_{t-1}$, δ_t is purely exogenous, and it still operates as a shock to jobs and vacancies.

Because replacement is costless, $\nu_t^f N_t^e$ drops out of the resource constraint. The construction also sets $p_0 = 0$, so it removes the reactivation margin along with endogenous exit.

Proposition 2 (Clone-replacement equilibrium)**.** *Under clone replacement with $N_t = 1$ and $\mu = 1$, the business formation block drops out and equilibrium labor market variables $\{u_{t+1}, v_t, e_t, Q_t, K_t\}_{t=0}^{\infty}$ satisfy:*

$$\kappa + \frac{K_t}{q(\theta_t)} = \mathbb{E}_t m_{t+1}(1-\delta_t)\left((1-\phi)(z_{t+1} - K_{t+1} - b) - \phi\theta_{t+1}(K_{t+1} + q(\theta_{t+1})\kappa) + (1-s_{t+1})\left(\kappa + \frac{K_{t+1}}{q(\theta_{t+1})}\right)\right) \tag{46}$$

$$K_t = Q_t - (1-\delta_t)\mathbb{E}_t m_{t+1} Q_{t+1} \tag{47}$$

$$v_t = (1-\delta_{t-1})\left[(1-q(\theta_{t-1}))v_{t-1} + s_{t-1}(1-u_{t-1})\right] + e_t \tag{48}$$

$$u_t = \left[1 - (1-\delta_{t-1})f(\theta_{t-1})\right]u_{t-1} + \tau_t(1-u_{t-1}) \tag{49}$$

[14] Had reactivation costs been drawn from a separate distribution, a fourth condition would have been required to shut down both margins.

together with $e_t = G(Q_t)$, $m_t = \beta(C_t/C_{t-1})^{-\sigma}$,

$$C_t = z_t(1-u_t) - X_t, \tag{50}$$

and stochastic processes (7)–(9), *with* $\delta_{e,t} = \delta_{t-1}$ *since endogenous exit is absent by construction, and hence* $\tau_t = \delta_{t-1} + s_{t-1}(1-\delta_{t-1})$.

We dub this the *augmented Gabrovski-Silva* (AGS) model. Remark 1 makes the connection to the two papers precise.

Remark 1 (Nested special cases of AGS)**.** *The AGS model nests two benchmark frameworks as limiting cases. Setting* $\sigma = \kappa = 0$ *renders the AGS system isomorphic to Gabrovski and Silva (2025), extended here with aggregate AR(1) shocks to* δ_t *and* s_t*. Additionally setting* $s_t = 0$ *recovers Coles and Moghaddasi Kelishomi (2018) exactly, with* $w_t^{int} = z_t$*. Both papers allow a flow cost of posting an unfilled vacancy and set it to zero in their quantitative work, as AGS does. Consumption is determined residually from* $z_t(1-u_t) = C_t + \frac{\xi}{\xi+1}Q_t^{\xi+1}$.

AGS as a limit of the baseline. Clone replacement is a heuristic. It supplies new product lines by fiat, so on its own it does not say whether AGS is a special case of our economy or a different model. The question matters because the free-entry results of Section 3.9 are proved in AGS. The natural route is the limit in which variety stops mattering. Under DS-CES the taste for variety vanishes as $\varepsilon \to \infty$, which delivers the condition $\zeta = 0$ of Proposition 1. That single limit fails here, and the reason is economic: business formation requires profits. Free entry equates the value of a line to $\rho(N_t)f_e/\mu(N_t)$, and under DS-CES the markup that generates profit per line is the taste for variety, $\mu - 1 = \zeta = 1/(\varepsilon - 1)$ (Section 1). With f_e fixed, profits vanish as $\varepsilon \to \infty$ while entry costs do not, so entry stops and $N \to 0$. Gross output $Y_t^c = \rho(N_t)N_t y_t$ then goes to zero with positive employment, and the limit is degenerate.[15] Entry costs must vanish with profits. Proposition 3 shows that scaling $f_e = \bar{f}_e/\varepsilon$ as $\varepsilon \to \infty$ keeps the mass of lines finite and positive while making business formation irrelevant for allocations.

Proposition 3 (Double limit)**.** *Consider DS-CES preferences. Set* $p_0 = 0$*, and let* $\varepsilon \to \infty$ *jointly with* $f_e = \bar{f}_e/\varepsilon$ *for fixed* $\bar{f}_e > 0$*. Then:* (i) *the product-line mass has the unique steady-state value* $\bar{N} = \bar{z}\bar{L}(1-\bar{\delta})/[\bar{f}_e(r+\bar{\delta})] > 0$*, and along any equilibrium path with entry interior*

[15] Letting $\delta \to 0$ as well would restore a positive mass of lines, but it removes the destruction margin that the benchmark is meant to preserve.

($N_t^e > 0$) it is set period by period,

$$N_{t+1} = \frac{(1-\delta_t)\,\mathbb{E}_t[m_{t+1}z_{t+1}]\,(1-u_{t+1})}{\bar{f}_e\big[1-(1-\delta_t)\,\mathbb{E}_t m_{t+1}\big]}; \tag{51}$$

(ii) *firm value ν_t^f, investment $\nu_t^f N_t^e$, and labor devoted to entry l_t^e all vanish at rate $1/\varepsilon$;* (iii) *recruiter compensation $w_t^{int} \to z_t$; and* (iv) *aggregate allocations $\{\theta_t, u_t, v_t, C_t, Y_t^c\}$ are independent of $\bar{f}_e$ and hence of $\bar{N}$, so the normalization $\bar{N} = 1$, which corresponds to $\bar{f}_e = \bar{z}\bar{L}(1-\bar{\delta})/(r+\bar{\delta})$, is without loss of generality for aggregate allocations. In the limit, the business formation block is irrelevant to real allocations.*

Every input to (51) is a labor market object or a shock, so causation runs one way: the labor market sets the mass of product lines, and the mass of product lines feeds nothing back. Product variety still moves in the limit. A destruction shock raises the user cost $r + \delta_t$ of a product line and lowers the target N_{t+1}. It simply has no effect on allocations. This is the sense in which the clone-replacement economy of Proposition 2, which fixes $N_t = 1$, implements the limit: it is the limit for allocations, not for N_t.

The limit is stronger than the separability of Proposition 1. The labor market block does not merely close on its own: business formation leaves no trace on allocations. Two consequences follow. First, the free-entry results of Section 3.9, Lemma 1 and Proposition 5, are statements about the baseline economy in this limit, with $\sigma = \kappa = 0$, not about a different model. The proposition concerns the equilibrium conditions. It does not show that impulse responses converge, so those results describe the limit economy, not the rate at which the baseline approaches it. Second, the limit identifies what the comparison with AGS measures. Away from it, entry costs do not vanish with profits, and business formation works through the channels below.

The baseline differs from AGS in two respects, and only the first is a matter of transmission channels. First, four objects move in the baseline labor market block and are constant in AGS. Variety effects enter the job creation condition through $\rho(N_t)/\mu(N_t)$. Profit dilution enters the return to entry through Y_t^c/N_t in (37): a larger mass of lines lowers profit per line, which regulates entry and feeds the survival cutoff χ_t^c through current profitability R_t^f. Endogenous exit operates through variation in that cutoff. Partial reactivation operates through $\Lambda_{r,t}$, which AGS loses because clone replacement sets $p_0 = 0$. Dilution is distinct from the variety effect. It works through the level of per-line profit rather than through the welfare-based relative price, so it survives when ρ is held at unity. The four are developed formally in Section 3.9 and evaluated

quantitatively in Section 5.

Second, and more basic, AGS does not price replacement. Clone replacement supplies a new line for every line destroyed, at no resource cost, within the period, and in every state of the world. The baseline makes each of these an equilibrium outcome of (37). Entry absorbs labor at rate f_e and so enters the resource constraint through $\nu_t^f N_t^e$ in (36), the mass of lines is a slow-moving state, and the rate of replacement is an equilibrium object rather than a constant. In the data replacement is slowest when destruction is heaviest (Section 1). AGS holds the replacement margin at its frictionless benchmark, so it cannot generate the joint behavior of the two margins that motivates the paper.

AGS also collapses measured and structural destruction. Since $\Lambda_t = 1$ implies $\delta_{e,t} = \delta_{t-1}$, the realized destruction rate is the shock. In the baseline the two differ by the endogenous exit margin, and it is the sum that the establishment death rate of Section 4 measures.

Because a single comparison with AGS moves all of this at once, the quantitative work of Section 5.3 isolates the channels one at a time. Comparison C removes variety effects by forcing $\rho \equiv 1$ while retaining $\mu > 1$, so entry, dilution, exit, and the entry component of the resource constraint all remain active. The gap between that specification and AGS therefore measures what business formation contributes net of the variety effect. Comparison B removes the endogenous exit margin by setting the endogenous share of destruction ω_δ to zero while holding the steady-state destruction rate fixed. A further comparison sets $\alpha \to 0$ to remove costly reactivation while leaving endogenous exit intact. The exit and reactivation margins must be switched separately rather than jointly through p_0, which removes both at once.

Strictly speaking, the baseline model does not nest Bilbiie, Ghironi, and Melitz (2012) since there is no intensive labor margin. However, it does nest a restricted version. Define *labor-restricted BGM* (LR-BGM) as the Bilbiie, Ghironi, and Melitz (2012) model with labor dropped from preferences: the household maximizes $\sum_t \beta^t c_t^{1-\sigma}/(1-\sigma)$ subject to $c_t = w_t L_t + \pi_t - X_t$, where $\{L_t\}$ and $\{X_t\}$ are exogenous sequences and X_t is the vacancy-creation cost absorbed as an intermediate input. The resource constraint of LR-BGM is therefore $Y_t = C_t + \nu_t^f N_t^e$, matching the GDP identity (36), with X_t netted out on both sides.

Proposition 4 (Nesting LR-BGM)**.** *Set $p_0 = 0$, so $X_t^c = 0$, $\Lambda_t = 1$, and $\delta_{e,t} = \delta_{t-1}$. For any exogenous sequences $\{L_t\}$ and $\{X_t\}$, the business formation block of the baseline model—equations (38), (39), and (36)—is isomorphic to LR-BGM evaluated at the same sequences.*

The isomorphism holds at the level of GDP (36) rather than gross output (40): vacancy costs X_t are intermediate inputs in both models and cancel from the resource constraint, leaving $Y_t = C_t + \nu_t^f N_t^e$ in each case. The sole departure of the baseline from LR-BGM is that L_t and X_t are endogenous, determined jointly by the DMP block (5)–(6). Restoring $p_0 > 0$ further adds endogenous exit to the LR-BGM core.

3.9. Shock transmission and the δ–s asymmetry

The equilibrium laws of motion imply qualitatively distinct labor-market dynamics under product destruction and under match separation. This section establishes the asymmetry analytically, in three steps. The first gives necessary and sufficient conditions for the Beveridge curve to slope in response to each shock in a special case of the AGS economy with infinitely elastic vacancy creation. In this case, the labor market is block recursive and we can obtain analytic policy functions for the log linearized system. The second is an accounting result at horizon $h = 1$ that holds for any entry elasticity. The third extends the separation-shock result to endogenous exit. Negative u–v comovement can come from δ shocks, which destroy positions together with matches, or from partial reposting, which turns part of each separation into position destruction. The model contains both, and the quantitative analysis assesses their relative contributions.

Free entry of vacancies. Consider the AGS economy of Proposition 2 with $\sigma = \kappa = 0$, which by Remark 1 is Gabrovski and Silva (2025) with aggregate shocks to δ_t and s_t. Its labor market block closes on its own. Vacancy creation is infinitely elastic, $Q_t = x_m$, and entry is interior, $e_t > 0$. Write $\delta_t = \bar{\delta} e^{\tilde{\delta}_t}$ and $s_t = \bar{s} e^{\tilde{s}_t}$, and consider one shock at a time, with persistence $\rho_x \in [0, 1)$ for $x \in \{\delta, s\}$. Both margins are present, $0 < \bar{\delta} < \bar{\tau}$. The results for the destruction shock also hold at $\bar{\delta} = \bar{\tau}$, where there is no separation margin.

Free entry pins the price of a vacancy, not its flow value. By (47),

$$K_t = x_m \frac{r + \delta_t}{1 + r}, \tag{52}$$

the user cost of an asset that destruction depreciates. In Gabrovski and Silva (2025) δ is constant, so free entry fixes K as well. Here K_t moves with the exogenous δ_t and with nothing else, so the job creation condition (46) contains only tightness and the shocks. This is what makes the free-entry economy tractable: tightness can be solved without reference to unemployment or the vacancy stock.

Lemma 1 (Tightness under free entry). *In the economy just described, suppose* $\phi(1-\bar{\delta})\bar{f} < \eta_L(2+r-\bar{\tau})$, *which holds whenever* $\phi \leq \eta_L$. *Then the log-linearized labor market block has a unique bounded solution, in which tightness depends only on the shocks,* $\hat{\theta}_t = c_\delta \tilde{\delta}_t + c_s \tilde{s}_t$. *The coefficients, given in closed form in* (H.1), *satisfy* $c_\delta < 0$ *and* $c_s \leq 0$, *with* $c_s = 0$ *if and only if* $\rho_s = 0$.

Both shocks lower tightness, for different reasons and by very different amounts. A rise in δ_t raises the cost of a hire directly. A vacancy is an asset that destruction depreciates, so its flow value (52) rises by $\bar{\delta}/(r+\bar{\delta})$ percent per percent of δ_t, the share of the user cost due to destruction. That share is 0.45 at the measured destruction rate. The value of a hire falls only through survival, by about $\bar{\delta}$ percent, which is small. For cost and value to stay equal, the expected time to fill a vacancy must fall, so tightness falls sharply. A rise in s_t does not depreciate the vacancy at all. It lowers tightness only by shortening the expected life of next period's matches, and only if it persists. At the measured destruction rate and $\rho_x = 0.592$, tightness falls by 0.79 percent per percent of a destruction shock and by 0.04 percent per percent of a separation shock. At the estimated $\rho_s = 0.874$ the separation response is 0.07.

Given tightness, unemployment follows its law of motion (49). In log deviations,

$$\hat{u}_t = \lambda_u \hat{u}_{t-1} + \ell_\delta \tilde{\delta}_{t-1} + \ell_s \tilde{s}_{t-1} - \omega_u \hat{\theta}_{t-1}, \qquad \ell_\delta \equiv \bar{f}\,\frac{\bar{\delta}}{\bar{\tau}}, \quad \ell_s \equiv (1-\bar{\delta})\bar{f}\Big(1-\frac{\bar{\delta}}{\bar{\tau}}\Big), \tag{53}$$

where $\lambda_u \equiv 1-(1-\bar{\delta})\bar{f}-\bar{\tau}$ is the persistence of unemployment and $\omega_u \equiv (1-\eta_L)(1-\bar{\delta})\bar{f}$ measures the hires lost when tightness falls. We assume $(1-\bar{\delta})\bar{f}+\bar{\tau} \leq 1$, so $\lambda_u \in [0,1)$. The direct effects ℓ_δ and ℓ_s split the separation flow into unemployment between the two shocks. Destruction is a small share of separations, $\bar{\delta}/\bar{\tau} = 0.087$, so $\ell_\delta = 0.036$. Match separations are the rest, so $\ell_s = 0.37$. Vacancies are then $\hat{v}_t = \hat{\theta}_t + \hat{u}_t$, and entry is the residual.

The two shocks are therefore mirror images. Per percent of the shock, destruction moves tightness by 0.79 and unemployment by 0.036. Separation moves tightness by 0.04 and unemployment by 0.37. Since $v = \theta u$, vacancies fall only if tightness falls by more than unemployment rises, so the first shock is built to produce negative u–v comovement and the second to produce positive comovement. The proposition states this precisely. Timing adds one element. Unemployment is predetermined and rises with a lag, so what matters is how much of the tightness fall survives into the periods in which unemployment has risen. That is why persistence enters.

Proposition 5 (Destruction and separation shocks under free entry). *Under the assumptions of Lemma 1, let shock $x \in \{\delta, s\}$ follow $\tilde{x}_t = \rho_x^t \tilde{x}_0$ with $\tilde{x}_0 > 0$, the other shock at its mean. To first order, $\hat{\theta}_t = c_x \rho_x^t \tilde{x}_0$ and $\hat{u}_t = (\ell_x + \omega_u |c_x|)\, \tilde{x}_0 \sum_{j=0}^{t-1} \lambda_u^{t-1-j} \rho_x^j$, so $\hat{u}_0 = 0$ and $\hat{u}_t > 0$ for all $t \geq 1$, and $\hat{v}_t = \hat{\theta}_t + \hat{u}_t$.*

1. *$\hat{v}_1 < 0$ if and only if $|c_x|(\rho_x - \omega_u) > \ell_x$. If $\hat{v}_t \geq 0$ for some $t \geq 1$, then $\hat{v}_{t'} > 0$ for all $t' > t$.*
2. *The slope of the impulse response in the (u, v) plane, $\sum_{t\geq 0}(v_t - \bar{v})(u_t - \bar{u}) / \sum_{t \geq 0}(u_t - \bar{u})^2$, equals*
$$\bar{\theta}\left[1 - \frac{\rho_x(1-\lambda_u^2)}{1+\rho_x\lambda_u}\,\frac{|c_x|}{\ell_x + \omega_u|c_x|}\right], \tag{54}$$
and is negative if and only if $|c_x|\big[\rho_x(1-\lambda_u^2)/(1+\rho_x\lambda_u) - \omega_u\big] > \ell_x$.
3. *If $(1-\bar{\delta})\bar{f}\big[\eta_L + \phi(1-\bar{\delta})\bar{f}/\bar{\tau}\big] \geq 1$, then after a separation shock $\hat{v}_t > 0$ for every $t \geq 1$ and the slope (54) is positive, for every $\rho_s \in (0,1)$.*

If $\rho_x = 0$, both conditions fail for the destruction shock, and the separation shock leaves tightness unchanged at every date. In both cases $v_t = \bar{\theta} u_t$ after impact and the slope equals $\bar{\theta} > 0$.

See Appendix H for the proofs.

Table 3: Impulse responses under free entry at the measured destruction rate, in percent per percent of the shock.

	Destruction, $\rho_\delta = 0.592$			Separation, $\rho_s = 0.874$		
t	$\hat{\theta}_t$	$\hat{u}_t$	$\hat{v}_t$	$\hat{\theta}_t$	$\hat{u}_t$	$\hat{v}_t$
0	−0.79	0.00	−0.79	−0.07	0.00	−0.07
1	−0.47	0.17	−0.30	−0.06	0.39	0.32
2	−0.28	0.19	−0.09	−0.05	0.55	0.50
3	−0.16	0.16	0.00	−0.05	0.60	0.56
4	−0.10	0.13	0.03	−0.04	0.60	0.55

Table 3 shows the contrast. After a destruction shock, tightness falls by much more than unemployment rises, and vacancies stay below steady state while unemployment builds up. They turn positive only once tightness has recovered, and then only slightly. After a separation shock tightness barely moves, so vacancies follow unemployment up almost one for one from the

first month. The Beveridge slopes summarize the two paths: -0.24 for the destruction shock and $+0.47$ for the separation shock.

The conditions show how robust each result is. For the destruction shock, vacancies fall at $h = 1$ whenever $\rho_\delta > 0.21$, and the slope over the whole path is negative whenever $\rho_\delta > 0.37$, at the monthly frequency. Moderate persistence suffices. For the separation shock, the sufficient condition in Part 3 equals 4.05 at the calibration, so no persistence delivers negative comovement. Persistence is necessary for the destruction shock, however. After an iid shock, tightness is back at its steady state in the month after the shock, and from then on vacancies move with unemployment, with slope $\bar{\theta} = 0.51$.

The measured destruction rate matters through the direct effect ℓ_δ. In the parameterization of Coles and Moghaddasi Kelishomi (2018), all separations destroy positions, $\bar{\delta} = \bar{\tau}$. The direct effect on unemployment then rises from 0.036 to 0.42, while the tightness response rises only from 0.79 to 1.46, because the user-cost share $\bar{\delta}/(r + \bar{\delta})$ saturates once $\bar{\delta}$ is large relative to r. Vacancies still dip at $h = 1$, but the slope over the path is $+0.17$, and no persistence makes it negative. This is the free-entry result of Coles and Moghaddasi Kelishomi (2018). Their need for a finitely elastic creation technology repairs a consequence of having a single separation margin. The durable vacancy is essential as well. With a flow-cost vacancy, holding K fixed, the user-cost term disappears from tightness, and the slope after a destruction shock is $+0.49$ at $\rho_\delta = 0.592$.

Any entry elasticity. With finitely elastic creation, entry $e = G(Q)$ links the predetermined vacancy stock to Q, tightness depends on the states, and the closed form is lost. The comovement at $h = 1$ can still be characterized by accounting.

Lemma 2 (Comovement at $h = 1$ for any entry technology)**.** *Suppose* $p_0 = 0$.

1. (s shock.) *A positive* s_t *shock leaves* u_t *unchanged and moves* v_t *only through entry* e_t. *At horizon* $h = 1$ *it raises unemployment and vacancies, so* u *and* v *comove positively, if and only if entry does not offset the reposting inflow* $d \equiv (1 - \delta_t)(1 - u_t)$*:*

   (i) $(1 - \delta_t)(1 - \eta_L)q(\theta_t)\,\partial e_t/\partial s_t < d$, *and*

   (ii) $-(1 - \delta_t)\big[1 - (1 - \eta_L)q(\theta_t)\big]\partial e_t/\partial s_t - \partial e_{t+1}/\partial s_t < d$.

2. (δ shock.) *A positive* δ_t *shock raises unemployment and reduces vacancies at* $h = 1$, *so* u *and* v *comove negatively, if entry over* t *and* $t + 1$ *does not offset the destruction of*

pre-committed vacancies,

$$(1-\delta_t)\big[1-(1-\eta_L)q(\theta_t)\big]\frac{\partial e_t}{\partial \delta_t}+\frac{\partial e_{t+1}}{\partial \delta_t}<(1-q(\theta_t))v_t+s_t(1-u_t),$$

and the matching term from entry at t does not offset the destruction of matches, $(1-\delta_t)(1-\eta_L)q(\theta_t)\,\partial e_t/\partial\delta_t<(1-s_t)(1-u_t)+f(\theta_t)u_t$.

See Appendix I for the proof.

Part 1 rests on accounting. Every separation at a surviving product line adds one worker to unemployment and returns one position to the vacancy pool, so unemployment and pre-committed vacancies rise by the same inflow d, apart from what the period-t entry response creates. Conditions (i) and (ii) say that this response does not overturn the inflow, for any ρ_s and any entry technology. Entry responds because a persistent s shock shortens expected match duration, the standard DMP channel, and because the state deviations the shock leaves, with feedback through consumption and variety, move the value of future matches.

Conditions (i) and (ii) are very easily satisfied when $p_0 = 0$. Two cases sign them analytically. Both require $\rho_s = 0$ and the conditions of Proposition 1, under which the labor market block closes on its own, and in both tightness stays at $\bar{\theta}$ from period t on, so entry does not respond at t. Under free entry this is the $\rho_s = 0$ case of Proposition 5, and vacancies rise by $\bar{\theta}d$. With $\xi < \infty$, the case $\bar{\theta} = 1$ is also exact: equal increases in u and pre-committed vacancies leave entry unchanged and propagate as equal deviations. Outside these cases the conditions are quantitative, and at the calibrated $p_0 = 0$ benchmark they are far from binding. With $\sigma = 1$ and variety effects active, entry at t falls in every specification we consider, so (i) holds automatically, and in (ii) entry over t and $t+1$ offsets at most 2.6 percent of the reposting inflow (0.7 percent at the measured $\bar{\delta}_e$). This holds for $\rho_s \in [0, 0.95]$, including the estimated ρ_s, and across entry elasticities $1/\xi \in \{0.5, 1, 2\}$. Part 1 is therefore a sharp benchmark: without endogenous exit or costly reposting, a separation shock moves unemployment and vacancies together.

In Part 2, a positive δ_t shock destroys a fraction of the stock of vacancies carried into $t+1$; see Lemma 3 in Appendix I. Whether total vacancies fall depends on the entry response. Because δ_t is known at the start of t, entry responds within the period, and vacancies created at t and $t+1$ partially backfill the destroyed ones. The size of that cushion is governed by the job creation condition (E.16). Destruction lowers the dividend on a vacancy through survival and through

the surplus, and raises it only by slackening the market, since a fall in θ makes a new position fill faster.

Remark 2 (The entry cushion at finite ξ)**.** *In the economy of Proposition 5 with a finite entry elasticity, the condition in Part 2 of Lemma 2 holds, for $\bar{\delta}$ between 1 percent a year and $\bar{\tau}$, if and only if $\bar{\delta}$ is below a threshold $\bar{\delta}^*(\xi, \rho_\delta)$. At $\rho_\delta = 0.592$ the threshold is 3.8 percent a year at $\xi = 1$, 2.7 percent at $\xi = 2$, and 1.6 percent at $\xi = 10$. At the measured 3.2 percent, entry offsets 85 percent of the first-month destruction at $\xi = 1$ and more than all of it at $\xi = 2$. At $\bar{\delta} = \bar{\tau}$ it offsets the destruction several times over at every $\xi \leq 2$. The relationship is not monotone in ξ. Once ξ is large, a persistent shock collapses entry through the user-cost channel of Proposition 5, and vacancies fall at every $\bar{\delta}$ between 1 percent a year and $\bar{\tau}$. After an iid shock, entry refills the stock as $\xi \to \infty$ and vacancies rise, whatever $\bar{\delta}_e/\bar{\tau}$.*

The two results bracket the entry elasticity. At a moderate finite ξ, a low $\bar{\delta}_e/\bar{\tau}$ delivers the negative first-month response, because it leaves the standing vacancy stock overwhelmingly inherited and only inherited vacancies are exposed to the shock. As $\xi \to \infty$, only persistence does. The first-month sign is fragile at the model's monthly frequency, because entry responds in the same month. At the $p_0 = 0$ benchmark with the measured $\bar{\delta}_e/\bar{\tau} \approx 0.087$ and the estimated ρ_δ, entry offsets 81 percent of the first-month destruction, and at the superseded value of 0.21 it more than offsets it. The cushion is short-lived, however. At the measured $\bar{\delta}_e$, vacancies fall from the second month on and in every quarterly average, for $\rho_\delta \in \{0, 0.592, 0.9\}$ and entry elasticities $1/\xi \in \{0.5, 1, 2\}$. Negative u–v comovement at the quarterly frequency of Sections 4 and 5 is therefore robust. The two parts of Lemma 2 mirror each other: on impact s adds positions to the vacancy pool and δ destroys them, and the sign holds unless entry overturns it.

Endogenous exit. With $p_0 > 0$, the cutoff, product variety and retail employment at $t+1$ are determined jointly. Define the continuation profit threshold $g(N, L^c) \equiv \rho(N) z L^c (\mu(N) - 1)/(\mu(N)N) + f_e \rho(N)/\mu(N)$, the right side of the cutoff condition (41) with $Y^c = \rho(N) z L^c$. A firm survives whenever its continuation-cost draw satisfies $\chi \leq g$.

Proposition 6 (Separation shocks with endogenous exit)**.** *Suppose $p_0 > 0$ and $\alpha \to 0$, and let the exit margin satisfy*

$$(1 - \bar{\delta}) F'(\bar{\chi}^c) \bar{B}\, g_N + g_L F'(\bar{\chi}^c) \bar{E}^b < 1, \tag{55}$$

where $g_L > 0$ is the derivative of g in retail employment, $\bar{B} = \bar{N} + \bar{N}^e$ is the stock of product lines entering a period before exit, and $\bar{E}^b = (1-\bar{u})/\bar{\Lambda}$ is employment before exit. To first order in $F'(\chi^c)$, the result of Part 1 of Lemma 2 extends, with d replaced by $(1-\delta_t)\Lambda_{t+1}(1-u_t)$, provided this reposting inflow weakly dominates the endogenous exit drain $(1-\delta_t)F'(\chi^c_{t+1})|\partial\chi^c_{t+1}/\partial s_t| \cdot [(1-q(\theta_t))v_t + s_t(1-u_t)]$, together with the corresponding period-t exit term. If dominance is strict at $\rho_s = 0$, it holds for all $\rho_s \in [0, \bar{\rho}_s)$ for some $\bar{\rho}_s > 0$.

See Appendix I for the proof.

The drain is proportional to $F'(\chi^c)$, the density of continuation costs at the cutoff, while the reposting inflow is not. Dominance therefore holds whenever the exit margin is inelastic enough. Under DS-CES, g_N balances two opposing forces. Gross flow profit per line is proportional to $N^{(2-\varepsilon)/(\varepsilon-1)}$. It combines a market-share dilution effect $(1/N)$ and a love-of-variety demand expansion $(\rho(N) \propto N^{1/(\varepsilon-1)})$, with net exponent $(2-\varepsilon)/(\varepsilon-1)$ negative for $\varepsilon > 2$, so dilution dominates and profit falls in N. The option value of entry $f_e\rho(N)/\mu \propto N^{1/(\varepsilon-1)}$ always rises in N. At standard calibrations the option value dominates, so $g_N > 0$. Condition (55) does not require a sign for g_N. It requires the exit margin to be inelastic enough that the joint determination of exit, product variety, and employment is well behaved: both feedback terms are proportional to $F'(\chi^c)$. At the calibrated baseline the left-hand side of (55) is between 0.002 and 0.005.

Partial reposting. Lemma 2 and Proposition 6 hold with costless reposting. Under $p_0 = 0$ this is automatic: F collapses to a point mass at zero, so every reactivation draw is free, $\Lambda_{r,t} = 1$, and $Q^{rep}_t = Q_t$. With $p_0 > 0$ and $\alpha > 0$, the reposting inflow in (6) shrinks to $(1-\delta_{e,t})\Lambda_{r,t}s_{t-1}(1-u_{t-1})$, and the complement $1-\Lambda_{r,t}$ of separations destroys positions at surviving product lines. An s shock then has a destructive component like the one that drives the destruction-shock results, so the sign of its u–v comovement is no longer fixed. These results therefore do not rule out a role for s shocks in the Beveridge curve. Under costless reposting, they confine that role to cases where the endogenous-exit drain of Proposition 6 dominates. Partial reposting is the direct route: it lets a separation destroy the position as well as the match.

3.10. Steady state

A steady state is a fixed point of the equilibrium system in which all variables are constant and shocks are at their long-run means $(\bar{z}, \bar{\delta}, \bar{s})$. We characterize it in two blocks. The *exit block* in (χ^c, δ_e) space determines the exit threshold and total destruction rate given aggregate profitability. The *labor market and entry block* in (θ, N) space determines market tightness and product variety given δ_e. The two blocks interact through δ_e and $\rho(N)$, and must be solved jointly. We focus on the (θ, N) block for the diagrammatic analysis, treating δ_e as a parameter; the full steady-state system and key derivations are in Appendix D.

Key steady-state ratios. Setting $\delta_t = \bar{\delta}$ and $s_t = \bar{s}$, the Euler equations yield $N^e/N = \delta_e/(1-\delta_e)$. Retail profits as a fraction of consumption output are

$$\pi_s \equiv \frac{N d^f}{Y^c} = \frac{1}{\varepsilon(N)} - \frac{X^c}{Y^c}$$

The profit share π_s captures the net return to operating a product line after continuation costs: it falls with the elasticity of substitution (competition erodes markups) and with the continuation cost burden X^c/Y^c. The equilibrium relationship between total employment and product lines is:

$$N = \frac{\pi_s z L (1-\delta_e)}{f_e\left(\frac{r+\delta_e}{\mu(N)} + \delta_e \pi_s\right)} \tag{56}$$

At a given profit share, a higher destruction rate δ_e reduces product lines relative to employment by raising the effective discount rate on future profits and increasing the entry required to maintain the product-line stock. The full derivation of steady-state output shares and labor allocation is in Appendix D.

Four steady-state curves. The *exit threshold curve* gives the locus (χ^c, δ_e) consistent with the cutoff condition (41) at steady state:

$$\chi^c = \frac{\rho(N) f_e}{\mu(N)} \frac{1+r}{(1-\delta_e)\big(1-\psi_c\,\omega(\delta_e)\big)}, \qquad \omega(\delta_e) \equiv 1 - \frac{(1-p_0)(1-\bar{\delta})}{1-\delta_e}, \tag{57}$$

where ω is the share of surviving lines that drew a positive continuation cost, $p_0(\chi^c/\chi_m)^\psi/\Lambda$ in (33), written with $\Lambda = (1-\delta_e)/(1-\bar{\delta})$. A higher destruction rate requires higher per-period profits to compensate, raising χ^c: the denominator falls in δ_e, at rate $1-\psi_c$. Higher

variety N shifts the curve upward through $\rho(N)$. The *product destruction curve* follows from $\Lambda(\chi^c)(1-\delta) = (1-\delta_e)$:

$$\chi^c = \chi_m \left(\frac{(1-\delta_e)/(1-\delta) - 1 + p_0}{p_0} \right)^{1/\psi} \tag{58}$$

It is negatively sloped: a higher cutoff χ^c raises the endogenous survival probability and therefore lowers δ_e.

For the labor market block, express e and K as functions of tightness θ using the steady-state unemployment and vacancy laws of motion:

$$e(\theta) = \frac{\delta_e\big(\tau\theta + (1-\delta_e)f(\theta)\big) + (1-\delta_e)^2(1-\Lambda_r)sf(\theta)}{\tau + (1-\delta_e)f(\theta)}, \qquad K(\theta) = \frac{r+\delta_e}{1+r}\, e(\theta)^{1/\xi}\, x_m \tag{59}$$

The second term in $e(\theta)$ is entry that replaces positions retired at surviving product lines; it vanishes with costless reposting ($\Lambda_r = 1$). With $\alpha > 0$, $\Lambda_r = F(Q/\alpha)$ depends on $Q = e^{1/\xi}x_m$, so $e(\theta)$ is defined implicitly. The *job creation curve* is the steady-state JCC (42):

$$\left(\kappa + \frac{K(\theta)}{q(\theta)}\right)(r + \tau + (1-\delta_e)\phi\theta q(\theta)) = (1-\delta_e)(1-\phi)\left(\frac{\rho(N)z}{\mu(N)} - K(\theta) - b - s\mathcal{R}(\theta)\right) \tag{60}$$

where $\mathcal{R}(\theta) = (1-\delta_e)\big(Q(\theta) - Q^{rep}(\theta)\big)/(1+r)$ is the steady-state reposting shortfall (20), which depends on tightness through Q. Its right-hand side is increasing in N through $\rho(N)$: more product lines raise recruiter compensation, stimulating vacancy creation — the employment–variety feedback emphasized by Schaal and Taschereau-Dumouchel (2019) and derived here in a more flexible setting. Under general CES with $\zeta = 0$, $\rho = 1$ and (60) determines θ independently of N, yielding a vertical line in (θ, N) space. The *resource constraint curve* combines the aggregate resource constraint, free entry (31), and the N-L relationship (56):

$$N = \frac{(\mu(N)-1)zL(\theta)(1-\delta_e)}{f_e(\delta_e\mu(N) + r)} \tag{61}$$

Employment $L(\theta)$ is increasing in θ, so (61) is increasing and bounded above by $\bar{N} = (\mu - 1)z(1-\delta_e)/(f_e(\delta_e\mu + r))$, the product variety consistent with full employment.

Remark 3 (Variety effects and elasticity of substitution)**.** *At $p_0 = 0$, the elasticity of substitution enters the exit threshold curve (57), the job creation curve (60), and the resource constraint curve (61) through the markup $\mu(N)$. The benefit of variety $\zeta(N)$ enters only the first two, through $\rho(N)$, which is absent from (61). The product destruction curve (58) depends on neither. Its parameters are p_0, ψ, χ_m, and $\bar{\delta}$.*

Under DS-CES, $\zeta = 1/(\varepsilon - 1)$, so the two cannot be moved separately. The distinction matters under general CES, where ζ and ε are separate parameters.

The number of steady states depends on the matching technology. We impose the following regularity conditions.[16]

Assumption 1 (Matching technology)**.** *The matching function has constant returns to scale, so the job-finding rate $f(\theta) \equiv m(1,\theta)$ and the vacancy-filling rate $q(\theta) \equiv f(\theta)/\theta$ depend on tightness alone, and f is continuously differentiable.*

(M1) Bounded probabilities: $f(\theta) \le \min\{1,\theta\}$ *for all* $\theta > 0$.

(M2) Diminishing elasticity: $\varepsilon_f(\theta) \equiv \theta f'(\theta)/f(\theta)$ *lies in* $(0,1)$ *and is nonincreasing in* θ.

(M1) says that f and q are probabilities. By (M2), f rises and q falls with tightness, so $q(0^+) \in (0,1]$ and $q(\theta) \le 1/\theta \to 0$. (M2) also makes $\ln L$ concave in $\ln\theta$, because the elasticity of L is $\varepsilon_f(\theta)$ times the unemployment rate $\tau/(\tau + (1-\delta_e)f(\theta))$, and both fall in θ.

Under DS-CES, and with $p_0 \to 0$ so that $\mathcal{R} = 0$, the job creation curve (60) can be solved for the recruiter compensation it requires:

$$\frac{z}{\mu} N^{1/(\varepsilon-1)} = \Phi(\theta) \equiv b + K(\theta) + \frac{\big(\kappa + K(\theta)/q(\theta)\big)\big(r + \tau + (1-\delta_e)\phi f(\theta)\big)}{(1-\delta_e)(1-\phi)}. \tag{62}$$

$\Phi(\theta)$ is the break-even recruiter compensation at tightness θ: the compensation at which posting a vacancy just pays. It adds the worker's outside option b, the flow value of a vacancy $K(\theta)$, and the expected hiring cost $\kappa + K/q$, annuitized at the rate $r + \tau$ at which matches end and grossed up for the worker's bargaining share. The term $(1-\delta_e)\phi f(\theta)$ is the part of the saved hiring cost that the worker captures through the outside option, which improves as the market tightens. The last condition concerns Φ:

(C) $\ln\Phi$ is convex in $\ln\theta$.

Condition (C) says that the elasticity of break-even compensation with respect to tightness, $\theta\Phi'(\theta)/\Phi(\theta)$, does not fall as the market tightens. Each further percent of tightness requires a

[16] The soft-min Cobb–Douglas function of Silva (2026), $m(u,v) = s\,a\,uv/[(av)^p + u^p]^{1/p}$, satisfies them whenever $s \le 1$ and $sa \le 1$, with elasticity $\varepsilon_f(\theta) = 1/[(a\theta)^p + 1]$ strictly decreasing, and it nests the den Haan, Ramey, and Watson (2000) function at $s = a = 1$. The Cobb–Douglas function of our calibration violates (M1) for $\theta < A^{1/\eta_L}$ but respects it near the calibrated steady state, so the high equilibrium below is unaffected. Under every bounded specification we check, that equilibrium is feasible, with $q \approx 1$ and unemployment near one third.

weakly larger percentage rise in compensation. Since $\ln N_{JC} = (\varepsilon - 1)\ln\Phi + \text{const}$, (C) makes the job creation curve convex in log–log space, while (M2) makes the resource constraint curve concave, so the two cross at most twice.

Assumption 1 delivers most of (C). Under (M2), $1/q$ is log-convex in $\ln\theta$, and sums and products of log-convex functions are log-convex. Two pieces of Φ fall outside it. The term in $\phi f(\theta)$ is log-concave under (M2), and $K(\theta)$ depends on the steady-state entry schedule $e(\theta)$ in (59), which is an equilibrium object. We therefore state (C) separately. It holds at our calibration for every bounded-probability specification we have checked, at each value of the entry elasticity $\xi \in \{0.5, 1, 2\}$.

Proposition 7 (Steady-state equilibria under DS-CES)**.** *Restrict to DS-CES preferences ($\rho(N) = N^{1/(\varepsilon-1)}$, $\mu = \varepsilon/(\varepsilon-1)$ constant) and let $p_0 \to 0$ so that δ_e is a parameter. Suppose Assumption 1 and condition (C) hold, and $b > 0$ or $\kappa > 0$. Then the job creation curve (60) satisfies*

$$N_{JC}^0 \equiv \lim_{\theta\to 0} N_{JC}(\theta) = \left(\frac{\mu}{z}\left(\frac{\kappa(r+\tau)}{(1-\delta_e)(1-\phi)} + b\right)\right)^{\varepsilon-1} > 0 \tag{63}$$

and $N_{JC}(\theta) \to \infty$ as $\theta \to \infty$, while the resource constraint curve (61) satisfies $N_{RC}(0) = 0$ and $N_{RC}(\theta) \to \bar{N} < \infty$. The two curves cross either never or, generically, exactly twice. Two equilibria exist if and only if $N_{JC}(\theta) < N_{RC}(\theta)$ for some θ, and one has strictly higher tightness and product variety than the other. In the double limit of Proposition 3, the job creation curve is vertical and the equilibrium is unique if and only if $z > \Phi(0)$.

Figure 4 illustrates the two equilibria. A positive productivity shock z shifts both curves outward: higher z raises recruiter compensation $w^{int} = \rho(N)z/\mu$, increasing the surplus from posting vacancies and shifting (60) rightward; simultaneously higher aggregate income supports more product lines for a given employment level, shifting (61) upward. The high equilibrium moves from E_2 to E_2' with both higher tightness and more product lines, as illustrated in Figure 5.

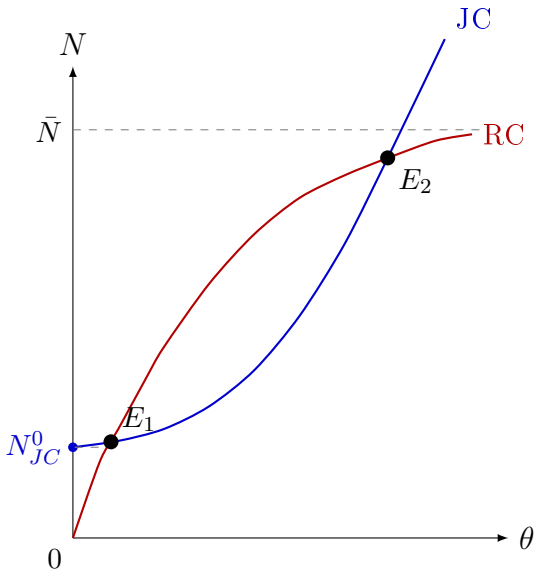


Figure 4: Steady-state equilibria in (θ, N) space. The job creation curve (JC) is increasing and intersects the vertical axis at $N^0_{JC} > 0$ whenever $b > 0$ or $\kappa > 0$; see (63). The resource constraint curve (RC) is increasing, starts at the origin, and is bounded above by $\bar{N} = (\mu - 1)\bar{z}(1 - \bar{\delta}_e)/(f_e(\bar{\delta}_e\mu + r))$. E_1 is the low and E_2 the high tightness–variety equilibrium. Schematic.

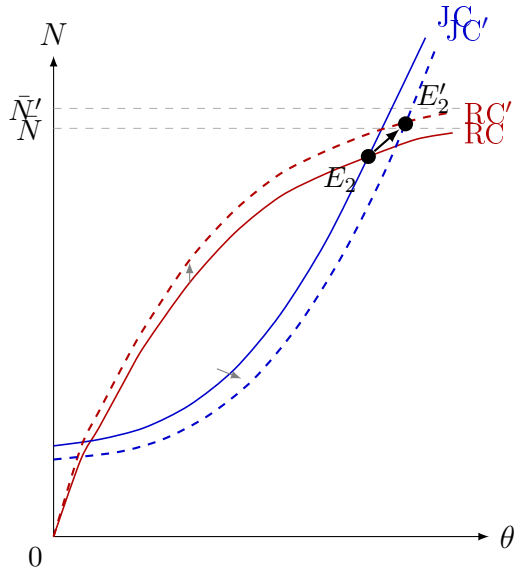


Figure 5: Effect of a positive productivity shock $z \uparrow$ on the steady-state equilibrium. Higher z raises recruiter compensation $w^{int} = \rho(N)z/\mu$, shifting the job creation curve rightward (JC to JC′); simultaneously higher income supports more product lines for given employment, shifting the resource constraint curve upward (RC to RC′, with $\bar{N}' > \bar{N}$). The high equilibrium moves from E_2 to E_2', with strictly higher market tightness and product variety.

4. Empirical Evidence and Bartik Local Projections

4.1. Overview and identification strategy

The cross-state evidence in Section 1 establishes that states with structurally higher establishment exit rates accumulate larger unemployment gaps during recessions, but the correlation conflates the causal effect of exit shocks with persistent differences in industrial composition, institutional environment, and average firm age across states. A state panel regression of labor market outcomes on local exit rates would confound the structural δ shock with the local demand conditions that drive both exits and unemployment simultaneously — the classic omitted variable in any cross-sectional or within-state identification strategy. The standard panel remedy of adding state and time fixed effects absorbs permanent cross-state heterogeneity and aggregate time variation but leaves untouched the within-state, within-time comovement between local exit activity and local business conditions.

We address this using a Bartik shift-share design (Bartik, 1991; Goldsmith-Pinkham, Sorkin, and Swift, 2020). The instrument for state s at quarter t is a weighted average of *national* industry-level establishment exit rates, where the weights are the state's pre-determined industry employment shares. To see why this breaks the confound, consider Michigan in 2008: its instrument is large because manufacturing nationally — not just in Michigan — was experiencing a wave of permanent closings, and Michigan had a high pre-recession exposure to manufacturing. The national exit rate is computed leaving out Michigan's own establishments (leave-one-out), so the instrument cannot be mechanically driven by Michigan-specific conditions. The instrument thus isolates variation in exit exposure driven by a state's inherited industry structure interacting with national industry cycles, rather than by local demand or policy.

The identifying assumption is that, conditional on state and time fixed effects and lagged controls, the Bartik instrument is uncorrelated with state-specific shocks to unemployment and vacancy creation beyond the exit channel. Section 4.2 describes instrument construction in detail; Section 4.3 states the local projection specification; Section 4.4 reports results. Section 4.4 provides two validation exercises: a severity placebo that tests whether the vacancy IRF is driven by recession depth rather than exit exposure, and a joint local projection that checks robustness to controlling simultaneously for the layoff-and-discharge instrument.

4.2. Instrument construction

BED data and the permanence problem. The underlying data source is the BLS Business Employment Dynamics (BED) survey, which tracks establishment-level employment via state unemployment insurance payroll records at quarterly frequency from 1992Q3. BED dates each establishment by third-month employment. An establishment is classified as a "closing" in a given quarter if it reported positive employment in the third month of the previous quarter and reports zero employment in the third month of the current quarter. Coverage is all private-sector establishments across 12 supersectors and all 50 states.

Raw BED closings conflate two economically distinct events: *permanent exits* — establishments that dissolve irreversibly, destroying the associated product line and vacancy slot — and *temporary shutdowns*, establishments that suspend operations for one to three quarters before reopening. The model's δ shock corresponds exclusively to permanent exit: only an irreversible dissolution removes a product line from the variety space and eliminates the recruiter's vacancy. Using raw closings as the shock proxy therefore overstates the true destruction margin and introduces attenuation bias when the temporary-shutdown share varies over time. The contamination is most severe during the 2020Q2 lockdown, where roughly 400,000 of the recorded closings reflected government-mandated suspensions that subsequently reopened, but it is present at lower intensity throughout the sample: seasonal businesses, firms temporarily suspending operations during downturns, and establishments undergoing ownership transitions all appear as closings in BED without constituting permanent product-line destruction.

BED Deaths as the measure of permanent exits. We address this using a second BED data series that resolves the permanence problem directly. In addition to closings, BED separately reports *deaths*, which by construction exclude temporary shutdowns. Let q_0 denote the last quarter in which an establishment reports positive third-month employment. The establishment is a closing in $q_0 + 1$. It is a *death* only if it reports zero employment in the third month of each of the four consecutive quarters $q_0 + 1$ through $q_0 + 4$. A death is therefore a closing that is followed by three further quarters of confirmed zero employment.[17] Since a temporarily closed establishment re-enters the payroll base within one to three quarters, it never accumulates four

[17] BED data classes: 06 (Closings) and 08 (Deaths). Deaths $\subseteq$ Closings in every industry×quarter cell, so the ratio is bounded above by unity by construction.

consecutive zero quarters and therefore never appears in BED Deaths. Our δ instrument uses BED Deaths, not raw closings, as the numerator of the national exit rate.

Two features of this definition matter for the empirical design. First, the four-quarter rule is a *confirmation* requirement, not a dating rule. BLS attributes the death to the quarter in which the establishment closed, so the series is correctly timed even though it can only be finalized four quarters later. The rule induces a publication lag and vintage revisions. It does not shift the destruction series forward in time, and so cannot by itself generate delayed dynamics in the local projections of Section 4.4. Second, the criterion is stricter than the annual Census Business Dynamics Statistics (BDS) measure of establishment deaths, which compares employment in the first quarter of consecutive years and therefore still counts an establishment that is idle at the reference date but later reopens. BED Deaths applies the spirit of the BDS permanence screen at quarterly frequency, with a tighter permanence criterion.

Figure 6 shows employment-weighted mean BED Closings and Deaths rates across 12 supersectors from 1992Q3 to 2019Q4, together with 10th–90th industry percentile bands. The Deaths/Closings ratio (lower panel) averages 0.63 across the full sample and is remarkably stable over the business cycle, dipping modestly during the GFC but remaining well below unity throughout. This stability implies that the fraction of closings that are temporary does not vary sharply over time, and that using Deaths directly gives a cleaner series than either raw closings or a time-varying permanence correction.

The Deaths/Closings ratio exhibits substantial cross-sector heterogeneity. Mining has the lowest mean ratio (0.37), and Construction is similarly low (0.52), consistent with the prevalence of project-based and seasonal establishments that suspend operations temporarily without constituting permanent product-line destruction. Leisure and hospitality (0.58) and Other services (0.59) are also below the sample average. At the other end, Manufacturing (0.71), Retail trade (0.70), and Information (0.70) have the highest ratios, reflecting structurally lower rates of temporary closure in those sectors. No supersector reaches the theoretical upper bound of unity in any quarter of the sample. The employment-weighted mean across all 12 supersectors is 0.63 (2001–2019); supersector time series are shown in Appendix C.1.

Most papers using BED data for shock identification (e.g., Elsby and Michaels, 2013) implicitly treat all closings as permanent. We are not aware of a prior paper using the BED Deaths series to align the instrument with the structural concept of irreversible product-line

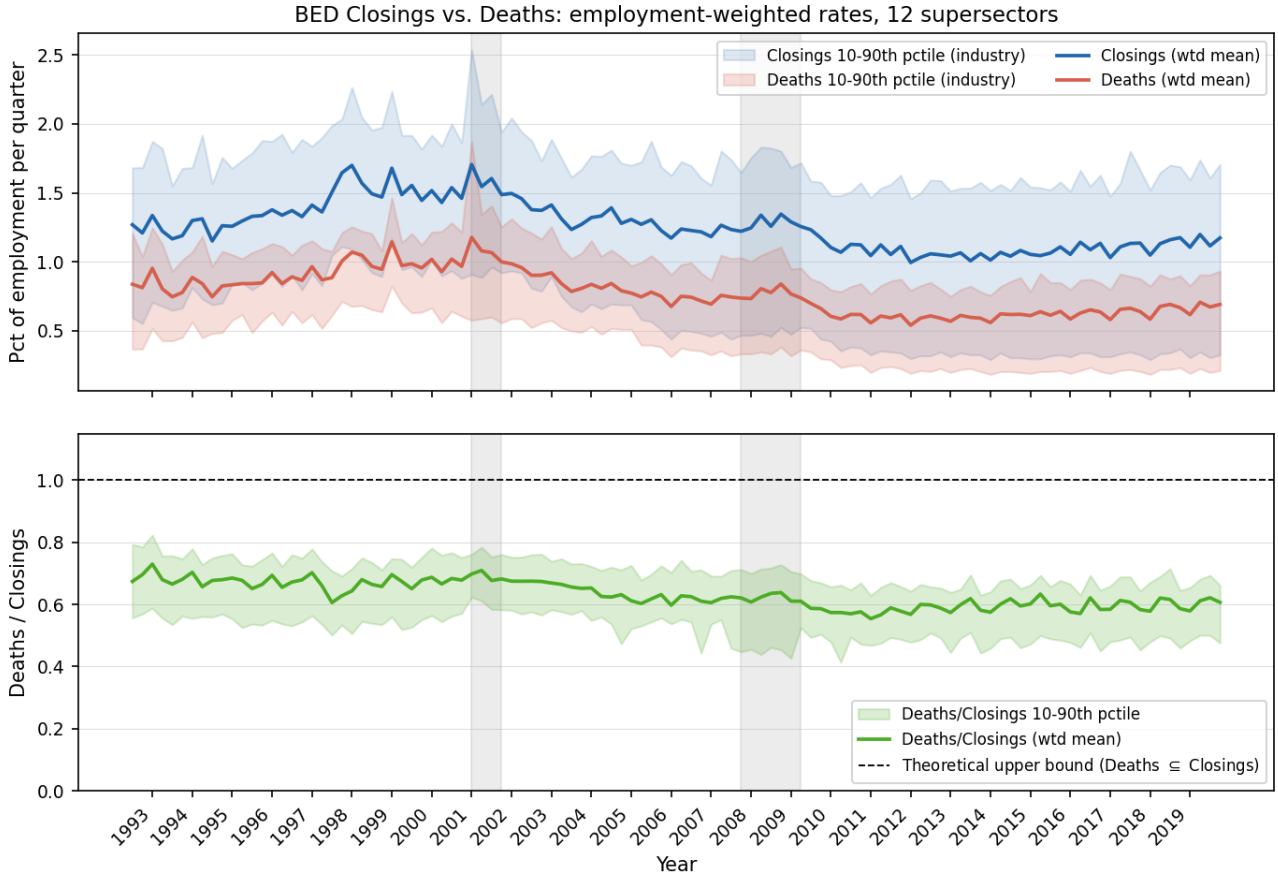


Figure 6: BED Closings and Deaths: employment-weighted rates across 12 supersectors, 1992Q3–2019Q4. *Top panel:* mean closing rate (blue) and death rate (red) as percent of employment per quarter, with 10th–90th industry percentile bands. Grey shading denotes NBER recessions. *Bottom panel:* Deaths/Closings ratio (dashed line at 1.0 marks the theoretical upper bound; ratio is bounded above by unity since Deaths $\subseteq$ Closings in every industry×quarter cell).

destruction.

Leave-one-out construction and base-year shares. The leave-one-out (LOO) national permanent exit rate for state s, supersector j, quarter t is

$$g^{\delta}_{-s,j,t} \;=\; \frac{\text{deaths}_{j,t}}{E^{\text{nat}}_{-s,j,t_0}},$$

where $E^{\text{nat}}_{-s,j,t_0} = E^{\text{nat}}_{j,t_0} - E_{s,j,t_0}$ is base-year national employment in supersector j minus state s's own base-year employment. The LOO correction in the denominator ensures that state s's own exit activity cannot mechanically inflate its instrument value. Numerator LOO — subtracting state s's own deaths from the national count — is not implemented because BED state-level deaths by supersector are not publicly available. Employment shares $\omega_{s,j}$ are drawn from the

2006 QCEW (agglvl=54, private sector), a stable pre-GFC year that predates the bulk of the identifying variation. The Bartik instrument is:

$$B^{\delta}_{s,t} \;=\; \sum_{j} \omega_{s,j} \cdot g^{\delta}_{-s,j,t}. \tag{64}$$

Residualization. Raw BED exit rates co-move with industry-specific demand conditions: when a sector contracts nationally, establishments exit at higher rates *and* surviving firms simultaneously reduce vacancy posting and raise layoffs. An instrument built on raw $g^{\delta}_{j,t}$ would therefore partly identify the effect of industry demand contractions rather than the structural δ shock. We address this by projecting each log shock rate on predetermined demand controls before Bartik aggregation, retaining the residual as the instrument input.

The baseline residualization (v2) runs a within-industry OLS regression:

$$\log g^{\delta}_{j,t} \;=\; \alpha_j + \gamma\, \Delta\log p_t + \lambda\, \Delta\log VA_{j,t-1} + \nu^{\delta}_{j,t}, \tag{65}$$

where $\Delta\log p_t$ is aggregate nonfarm business productivity growth (FRED: OPHNFB) and $\Delta\log VA_{j,t-1}$ is real value-added growth in supersector j lagged one quarter (BEA via FRED series RVAM, RVAC, etc.). Both regressors are predetermined relative to the current-period shock draw: aggregate productivity is contemporaneous but common to all industries and thus absorbed by the time fixed effects in the second stage; industry VA is lagged one quarter to ensure strict predetermination. The residual $\nu^{\delta}_{j,t}$ is then Bartik-aggregated in place of the raw rate, yielding $\tilde{B}^{\delta}_{s,t}$.

The FRED BEA real value-added series begin in 2005Q1, which would restrict the δ residualization sample to 2005Q2. To recover the full 1997Q1+ sample, we extend the quarterly VA series back to 1992Q1 using a constrained Chow-Lin (1971) temporal disaggregation. An indicator regression estimated on the 2005Q1+ post-sample relates quarterly sectoral VA growth to real GDP growth: $\Delta\log VA_{j,t} = \alpha_j + \beta_j\, \Delta\log \mathrm{GDP}_t + \varepsilon_{j,t}$. The pre-2005 quarterly path is recovered by applying the implied growth rates $\hat{\alpha}_j + \hat{\beta}_j\, \Delta\log \mathrm{GDP}_t$ recursively from the 2005Q1 anchor level. The backcast is then constrained to match published BEA annual benchmarks for 1997–2004 via a proportional Denton adjustment: for each year A with annual benchmark $Y_{j,A}$, quarterly values are scaled by $Y_{j,A}/\overline{VA}^{\,\text{unconstrained}}_{j,A}$ so that the within-year average equals the published figure exactly. Industries with low GDP–VA correlation (R^2 below threshold) receive $\hat{\beta}_j = 0$, using only the sector-specific mean growth rate to avoid injecting noisy cyclical variation into the pre-period.

After v2 residualization the cross-state correlation between the δ and layoffs-and-discharges (LD) Bartik instruments falls from approximately 0.54 on raw rates to 0.33.[18] The persistence of this residual correlation after controlling for industry demand cycles and aggregate productivity is informative: the shared variation is not a demand artifact but most likely reflects industry-specific financial conditions that move both exit rates and layoff rates simultaneously. This motivates the joint local projection robustness check in Section 4.4.

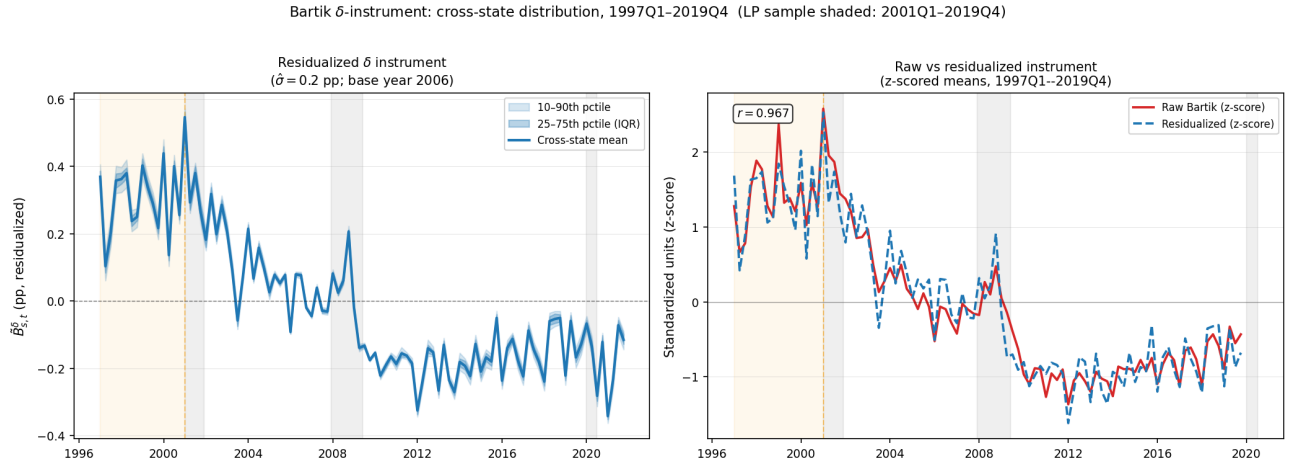


Figure 7: Cross-state δ Bartik instrument distribution, 1997Q1–2019Q4. *Left:* residualized instrument $\tilde{B}^{\delta}_{s,t}$ (pp of base-year employment); shading shows 25th–75th (dark) and 10th–90th (light) cross-state percentile bands. *Right:* z-scored cross-state means of raw (red) and residualized (blue dashed) instruments; $r = 0.905$. Orange shading (1997Q1–2000Q4): pre-LP period available for $\delta \to u$ but not vacancy or LD/QU. Grey shading: NBER recessions.

Figure 7 documents the cross-state distribution of the instrument and the effect of residualization. Both panels extend to 1997Q1, the start of the Chow–Lin-extended value-added series; the orange-shaded region (1997Q1–2000Q4) marks the pre-JOLTS period that is available for the $\delta \to u$ specification but not for the vacancy LP. The left panel shows substantial cross-sectional dispersion throughout, with the interquartile range widening sharply during the GFC. Negative values in some states and quarters reflect lower-than-predicted closing rates given prevailing aggregate conditions and industry composition; since the local projection absorbs the

[18] As a robustness check we further augment equation (65) with a monetary policy sensitivity term $\psi(\phi_j \times \Delta MP_t)$, where ϕ_j is log average establishment size in supersector j (Census CBP, 2000) and ΔMP_t is the quarterly change in the Wu–Xia (2016) shadow federal funds rate (v3 specification). The v3 results are reported in **??**; v2 is the baseline throughout.

cross-state mean via time fixed effects, identification comes from cross-sectional variation in residualized exposure, which remains well-defined regardless of sign.

The right panel compares the z-scored cross-state means of the raw and residualized instruments ($r = 0.905$). The two series track each other closely throughout the sample, indicating that residualization does not materially alter identifying variation in normal times. Two episodes diverge. During the GFC, the residualized mean rises somewhat more than the raw instrument at 2008Q4 ($z = 1.51$ vs. 0.95), consistent with residualization removing attenuating aggregate and industry-level variation. The more striking divergence would appear in 2020Q2 (not shown, excluded from LP sample): the raw Bartik spikes to $z = 2.33$ while the residualized instrument falls to $z = -0.45$, because 2020Q2 BED closings were dominated by government-mandated temporary shutdowns that correlate strongly with industry composition and aggregate conditions; residualization correctly removes this variation, leaving below-average permanent destruction. This episode illustrates precisely why the LP sample is capped at 2019Q4.

The cross-state mean of the instrument exhibits a pronounced secular decline over the sample, falling roughly one-third from the late 1990s to the late 2010s. This mirrors the well-documented decline in aggregate labor market fluidity: the Shimer (2005) quarterly separation rate fell proportionally over the same period, and the ratio of the two series is nearly constant throughout (coefficient of variation 14%), indicating that permanent establishment destruction has declined in tandem with broader separation margins rather than becoming relatively more or less important over time.

Two features of the left panel deserve note. First, the cross-sectional percentile bands are narrow relative to the time-series movement of the mean: the within-quarter p90–p10 spread averages 0.045 pp, a coefficient of variation of about 15%. The pooled standard deviation across all state-quarters ($\hat{\sigma} = 17.4$ pp) is therefore predominantly a time-series number, reflecting the common component of the instrument that moves all states together. The time fixed effects in the local projection absorb this common component; identification comes from the within-period cross-sectional deviations, which have a typical magnitude of roughly 0.02 pp. Second, some state-quarter cells take negative values, reflecting lower-than-predicted closing rates given industry composition and aggregate conditions; these are well-defined as deviations from the cross-state mean and cause no identification problem.

4.3. Local projection specification

For each horizon $h = 0, 1, \ldots, 20$ quarters, the panel local projection estimating equation is

$$y_{s,t+h} - y_{s,t-1} \;=\; \alpha_s + \alpha_t + \beta_h \, \tilde{B}^{\delta}_{s,t} + \gamma_1 \, y_{s,t-1} + \gamma_2 \, \log L_{s,t-1} + \varepsilon_{s,t+h},$$

where $y_{s,t}$ is the labor market outcome of interest for state s in quarter t, $\tilde{B}^{\delta}_{s,t}$ is the residualized δ Bartik instrument constructed in Section 4.2, α_s and α_t are state and time fixed effects, $y_{s,t-1}$ is the lagged outcome, and $L_{s,t-1}$ is the lagged labor force. The left-hand side measures the cumulative h-period change relative to the quarter before the shock, so the sequence $\{\hat{\beta}_h\}$ traces the impulse response directly. Standard errors are clustered at the state level ($n = 50$).

The two outcomes are the state unemployment rate $u_{s,t}$ and vacancy rate $v_{s,t}$, both in percentage points (unemployed persons or vacancies as a share of the labor force). Concretely, $v_{s,t} = 100 \times V_{s,t}/L_{s,t}$, where $V_{s,t}$ is the quarterly average of monthly JOLTS job-openings stocks (seasonally adjusted) and $L_{s,t}$ is the LAUS labor force. The two specifications differ only in the autoregressive control $y_{s,t-1}$.

The estimation sample runs from 2001Q1, when JOLTS state vacancy data become available, through 2019Q4; observations with $t + h >$ 2019Q4 are dropped to exclude COVID-19.[19] The instrument $\tilde{B}^{\delta}_{s,t}$ has cross-sectional standard deviation $\hat{\sigma}_{\tilde{B}^{\delta}} = 17.4$ pp; all reported $\hat{\beta}_h$ are scaled by $\hat{\sigma}^{-1}_{\tilde{B}^{\delta}}$ so that the impulse responses correspond to a one-standard-deviation shock.

We also construct Bartik instruments for layoffs-and-discharges $\tilde{B}^{\mathrm{LD}}$ (JOLTS LD) and quits $\tilde{B}^{\mathrm{QU}}$ (JOLTS quits), using the same 2006 QCEW base-year shares and v2 residualization. Neither is treated as a well-identified separate estimation target. $\tilde{B}^{\mathrm{LD}}$ conflates genuine match dissolution at surviving establishments with vacancy withdrawals at partially closing firms, mixing s- and δ-type events in proportions that vary across industries and cycles; moreover, $\tilde{B}^{\delta}$ Granger-causes future LD through $h = 12$ while the reverse predictability fades by $h = 8$, indicating that LD is partly endogenous to destruction. $\tilde{B}^{\mathrm{QU}}$ is procyclical and driven largely by job-to-job transitions, which represent a worker gain for one employer offset by a hire at another and have no clear mapping to the model's match separation shock s; we include it as

[19] The δ unemployment equation can be estimated back to 1997Q1 using the Barnichon (2010) Composite Help-Wanted Index as a pre-JOLTS vacancy proxy; extending the sample attenuates the IRF modestly (peak +1.07 pp vs. +1.50 pp) because the 1997–2000 expansion years exhibit faster reabsorption of displaced workers. We use the 2001Q1 baseline throughout to maintain a common sample across both outcomes.

a diagnostic instrument only.[20] Both instruments and their local projections are reported in Appendix C.

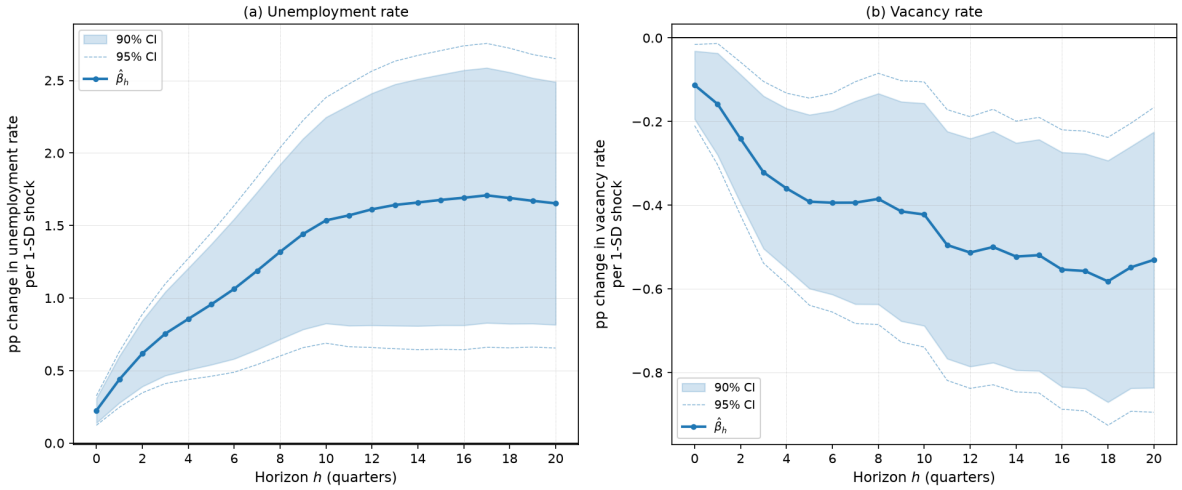


Figure 8: Impulse responses of the state unemployment rate (left) and vacancy rate (right) to a one-standard-deviation increase in the residualised δ Bartik instrument ($\hat{\sigma}_{\tilde{B}^\delta} = 17.4$ pp, 2001Q1–2019Q4). Both outcomes are estimated from panel local projections with state and time fixed effects; standard errors are clustered at the state level ($n = 50$). Shaded band: 90% confidence interval; dashed lines: 95% confidence interval. The instrument is BED establishment deaths (equation (64)), residualized on aggregate productivity growth and lagged industry value-added growth (equation (65))

4.4. Results

Figure 8 reports the impulse responses of the state unemployment rate and vacancy rate to a one-standard-deviation increase in the residualized δ Bartik instrument ($\hat{\sigma}_{\tilde{B}^\delta} = 17.4$ pp, 2001Q1–2019Q4 sample). The unemployment response rises steadily from 0.22 pp at $h = 0$ to a plateau of +1.71 pp by $h = 17$–20, remaining highly significant throughout ($p < 0.001$ from $h = 0$). The vacancy response falls persistently from −0.11 pp at $h = 0$ to a trough of −0.58 pp at $h = 18$, significant at $p < 0.01$ from $h = 2$ onward and remaining so through $h = 20$. The simultaneous and persistent rise in unemployment alongside the fall in vacancies traces the outward leg of a Beveridge loop, consistent with the model's central prediction that permanent product-line destruction depletes the vacancy stock rather than merely suspending it.

[20] In particular, a high quit rate in an industry does not imply that vacancies go unfilled — the quitting worker is typically replaced by a job-to-job mover from elsewhere, so the net effect on aggregate vacancies is near zero. This is conceptually distinct from the s shock in the model, which destroys a match and triggers a vacancy repost.

A notable feature of the two IRFs is their asymmetric dynamics at medium horizons: the vacancy response stabilizes around $h = 12$–14 while the unemployment response continues to rise through $h = 20$. This pattern is consistent with a recovery path in which new vacancy creation resumes before the unemployment pool fully clears. Whether the calibrated model replicates this relative timing is testable and will be assessed in Section 5.

A natural concern is that the δ Bartik instrument may inadvertently capture recession severity: states with structurally high establishment turnover may also experience worse aggregate downturns, so the IRF could reflect demand-driven labor-market deterioration rather than product-line destruction per se. We address this with a severity placebo that augments the baseline local projection with two interactions: $\tilde{B}_t^{\delta} \times \bar{u}_t^{\text{nat}}$, a level proxy for underlying economic slack, and $\tilde{B}_t^{\delta} \times \Delta\bar{u}_t^{\text{nat}}$, a flow proxy for cyclical momentum. If the instrument were a severity proxy, these interactions should absorb the baseline IRF. Instead, both interaction coefficients are insignificant at every horizon $h = 0, \ldots, 20$ for the vacancy outcome, and the baseline $\delta \to v$ coefficients are unchanged in magnitude and significance. This is the cleanest external validity check in the paper: the vacancy response to establishment destruction is not explained by where the aggregate economy happens to stand in the cycle. The severity placebo also passes for the unemployment outcome, though the interpretation is less sharp given the natural correlation between any adverse state-level shock and local unemployment. Robustness results for the alternative instruments and the joint local projection are collected in Appendix C.

5. Quantitative analysis

5.1. Relationship to Coles and Kelishomi

Our framework nests Coles and Moghaddasi Kelishomi (2018) as a special case, as formalized in Proposition 4. Relative to their calibration, we make two departures. First, we set δ to the BED Deaths employment-weighted exit rate ($\bar{\delta} \approx 1.1\%$ per quarter) rather than equating it to the full aggregate separation rate; Gabrovski and Silva (2025) show this correction materially improves the model's fit on the unemployment–productivity correlation. Second, we expand the estimation moment set to include the permanent exit rate δ and the entry flow N_t^e (proxied by BFS high-propensity business applications), and supplement unconditional second moments with identified IRFs from the Bartik local projections of Section 4 to separately discipline the propagation dynamics of δ and s shocks.

5.2. Calibration and estimation

This section sets out our estimation and calibration strategy, the taxonomy of parameters, and the explicit mapping from empirical targets to model objects.

Estimation framework. We estimate the model by Bayesian simulated method of moments, combining two complementary sets of moment conditions that together over-identify the parameter vector.

The first set consists of unconditional second moments of HP-filtered aggregate time series, extending Coles and Moghaddasi Kelishomi (2018). We target three groups: (i) standard deviations (m_1), (ii) contemporaneous pairwise correlations (m_2), and (iii) first-order autocorrelations (m_3) for unemployment, vacancies, separations, the job-finding rate, the permanent exit rate, business applications, and labor productivity. The full empirical moment matrix is reported in Table 4. All series are log-transformed and HP-filtered with smoothing parameter $\lambda = 1{,}600$ before computing moments; the identical filter is applied to simulated model data when evaluating $m(\theta)$. This symmetric treatment renders HP-filtered second moments a valid indirect inference target: Canova and Ferroni (2011) establish that applying the same filter to data and simulated output controls for any filter-induced distortion, even when the filter introduces spurious dynamics in isolation. We prefer $\lambda = 1{,}600$ over the $\lambda = 100{,}000$ convention of Coles and Moghaddasi Kelishomi (2018) because the higher smoothing parameter strips out the business-cycle frequency variation in δ at which the destruction mechanism operates, reversing the sign of $\text{cor}(\delta, v)$ and driving $\text{cor}(\delta, u)$ to near zero — an artifact of over-smoothing rather than a feature of the data. Appendix C.6 documents this explicitly: across the Hamilton (2018) regression filter, first differences, and HP with $\lambda = 1{,}600$, the correlations $\text{cor}(\delta, u) > 0$ and $\text{cor}(\delta, v) < 0$ are robust; $\lambda = 100{,}000$ is the anomalous specification.

Table 4: Business Cycle Moments

	$\sigma(x)$	Contemporaneous correlations							$\rho_1(x)$
		u	v	τ	f	δ	N^e	z	
Unemployment	0.1496	1							0.792
Vacancies	0.1419	-0.804	1						0.902
Separations	0.1019	0.577	-0.422	1					0.250
Job-finding rate	0.0977	-0.791	0.620	0.020	1				0.808
Exit rate	0.0793	0.310	-0.246	0.681	0.213	1			0.114
Entry flow (N^e)	0.0813	-0.105	0.482	-0.345	-0.226	-0.496	1		0.545
Labor productivity	0.0128	-0.009	0.306	-0.228	-0.212	-0.250	0.381	1	0.766
Derived statistics									
$\sigma(\theta)/\sigma(z)$	11.70								

Note: HP filter with $\lambda = 1{,}600$ applied to log-levels of each series. Correlations use pairwise-complete observations; the shortest series (N^e, entry flow) begins 2004Q3. Sources — u: BLS LAUS; v: Barnichon (2010) composite HWI spliced with BLS JOLTS at 2001Q1; s, f: Shimer (2012) continuous-time flows from CPS; δ: BLS BED Deaths (dataclass 08), employment-weighted; N^e: BFS high-propensity business applications (BAWBATOTALSAUS) per capita, proxy for model entry flow; z: BLS PRS85006163 (output per hour, nonfarm business).

Table 4 highlights several salient features of the data. Unemployment and vacancies are highly volatile relative to labor productivity ($\sigma(u)/\sigma(z) \approx 11.7$, $\sigma(v)/\sigma(z) \approx 11.1$) and display a pronounced Beveridge curve ($\text{cor}(u, v) = -0.80$). The unemployment-labor productivity correlation is near zero ($\text{cor}(u, z) = -0.01$), consistent with weak technology-shock transmission in models with endogenous separation. The exit rate co-moves strongly with the match separation rate ($\text{cor}(\delta, \tau) = 0.68$), reflecting a common cyclical component of firm destruction and worker-job dissolution. Importantly, the exit rate is positively correlated with unemployment ($\text{cor}(\delta, u) = 0.31$) and negatively correlated with vacancies ($\text{cor}(\delta, v) = -0.25$), consistent with the simultaneous vacancy-destruction mechanism central to our model.[21] Labor market variables and entry display strong persistence ($\rho_1 \in [0.55, 0.90]$), while separations and the exit

[21] These signs reverse under $\lambda = 10^5$, an artifact of over-smoothing documented in Appendix C.6.

rate exhibit substantially less ($\rho_1 \approx 0.11$–0.25), consistent with their role as shock-propagation mechanisms rather than persistent states.

The second set consists of empirically identified impulse responses from the Bartik local projections of Section 4: the $h = 0, \ldots, H$ quarter paths of unemployment and vacancies following a one-standard-deviation δ shock. Following Christiano, Eichenbaum, and Trabandt (2016), we form a Gaussian quasi-likelihood based on the sampling distribution of the LP estimator:

$$\log \mathcal{L}_{\text{IRF}}(\theta) = -\tfrac{1}{2}(\hat{\beta} - \beta(\theta))' \Omega_{\beta}^{-1} (\hat{\beta} - \beta(\theta)),$$

where $\hat{\beta}$ stacks the two estimated IRF paths, $\beta(\theta)$ is the corresponding model-implied path computed from the linearized state-space representation, and Ω_{β} is the sampling covariance estimated by wild cluster bootstrap with $n = 50$ state clusters.

The quasi-likelihood is a limited-information Bayesian update: rather than conditioning on the raw panel data, it conditions on $\hat{\beta}$ as a sufficient statistic for the model-implied path $\beta(\theta)$. Writing the full likelihood $\mathcal{L}(\text{data}|\theta)$ for the panel would require the model to generate 50 state-level employment series, cross-state variation in industry composition, and the full Bartik instrument—objects entirely outside a representative-agent aggregate economy. Even setting this aside, any reduced-form likelihood over $\hat{\beta}$ would require distributional assumptions on the LP residuals, the instrument's conditional distribution, and the finite-sample behavior of a clustered two-stage estimator—auxiliary assumptions with no grounding in the structural model. The quasi-likelihood avoids both problems by relying only on the asymptotic normality of $\hat{\beta}$, which follows from the LP moment conditions alone (Chernozhukov and Hong, 2003).

What is the value of using the IRF block instead of relying on the structural likelihood on aggregate time series? The answer is identification. The full likelihood recovers structural parameters from the time-series covariance structure of observables, but in a three-shock model the unconditional second moments cannot separately discipline the propagation of δ and s shocks: both raise unemployment, and their vacancy responses—the qualitative distinction between them—are collinear in second moments because they depend jointly on shock persistence, sunk-cost dynamics, and the entry margin. The Bartik design provides the only clean exogenous variation in δ and therefore the only moment condition that separately identifies the parameters governing the model's central mechanism. Omitting the IRF block would leave the asymmetric vacancy-destruction channel weakly identified from time-series covariances alone.

A filtering asymmetry between the two blocks is present by design: unconditional moments

are statistics of HP-filtered data, while IRF coefficients are estimated from level regressions. This is resolved on the model side. Model moments $m(\theta)$ are computed from simulated-then-filtered data, mirroring the data construction. Model IRFs $\beta(\theta)$ are read directly from the state-space impulse response without filtering, mirroring the LP estimation. The two objects therefore target genuinely distinct features of the model—cyclical second-moment structure and shock-specific propagation dynamics—in a way that is internally consistent on both the data and model sides.

The two blocks provide complementary identification. Unconditional moments discipline the level and cyclical volatility of the destruction and separation processes, the amplification and comovement structure, and the persistence of key aggregates. The IRF block separately identifies the dynamic transmission of δ shocks—in particular, the simultaneous elimination of vacancies that distinguishes δ from s and generates persistent Beveridge-curve dislocations—identification unavailable from unconditional moments alone, which conflate the contributions of all three shocks.

To prevent the IRF block from mechanically dominating the joint objective by virtue of moment count, we normalize each block's quadratic form by its dimension. Letting K_β denote the number of IRF moment conditions and K_m the number of unconditional moments, the log-posterior kernel is

$$\log p(\theta \mid \text{data}) \propto \log \pi(\theta) - \frac{1}{2K_\beta}(\hat{\beta} - \beta(\theta))'\Omega_\beta^{-1}(\hat{\beta} - \beta(\theta)) - \frac{1}{2K_m}(\hat{m} - m(\theta))'\Omega_m^{-1}(\hat{m} - m(\theta)),$$

where $\hat{m}$ collects the empirical moments from Table 4, $m(\theta)$ the corresponding model-implied moments from simulated HP-filtered data, and Ω_m a HAC covariance estimated by block bootstrap with block length eight quarters. The HP filter induces serial correlation in cyclical residuals at all lags, so HAC correction is required regardless of the choice of λ; the block length of eight quarters is sufficient to cover the effective filter width at $\lambda = 1{,}600$. The degrees-of-freedom normalization ensures each block contributes equally in expectation when the model is correctly specified, since each quadratic form is approximately $\chi^2(K)$ with expectation K under the null (Dridi, Guay, and Renault, 2007). **??** reports robustness to rescaling the unconditional moment block by factors of one-half and two.

Parameter taxonomy. We partition the full parameter space Θ into three blocks.

Estimated block Θ_e: structural parameters and shock processes whose identification comes from business-cycle moments. This block contains the replacement value b; the sunk-cost share x_v of total recruiting expenditure (which maps to the fixed matching cost κ in Stage 4 below); the inverse elasticity of entry with respect to vacancy value ξ^{-1}; the endogenous exit share $\omega_\delta \equiv (1 - F(\chi^c))/\delta_e$; the mass weight p_0 on the continuous part of the continuation-cost distribution F; the inverse intertemporal elasticity of substitution σ; and the persistence ρ_x and conditional standard deviation σ_x of each shock $x \in \{z, \delta, s\}$.

External block: parameters set directly from microeconomic evidence or long-run data. We fix the annual discount rate $r = 4\%$ and the matching-function elasticity $\eta_L = 0.6$. We set the elasticity of substitution $\varepsilon = 4.3$, implying a gross markup $\mu = \varepsilon/(\varepsilon - 1) \approx 1.30$. We calibrate the product destruction rate to the same exit measure that identifies the δ shock in Section 4. Employment-weighted BED establishment deaths average 3.2% per year over 1993–2019, or $\delta_e = 0.271\%$ per month. Against an aggregate monthly separation rate of $\tau = 3.1\%$, this implies $\bar{\delta}_e/\bar{\tau} = 0.087$. Firm exit accounts for about one separation in twelve.

Reconciling $\bar{\delta}_e$ with product-level destruction rates. Our value lies well below the 10% annual rate in Bilbiie, Ghironi, and Melitz (2012). The gap is informative, and it reflects a difference in what the parameter is asked to measure. BGM calibrate a single destruction shock to the product-level evidence of Bernard, Redding and Schott (2010), which records the value lost through product destruction at both continuing and exiting firms. Over a five-year horizon that evidence attributes 44 percent of output to product destruction, of which 30.4 points occur at firms that continue to operate. Netting out the continuing-firm component leaves roughly 2.7 percent per year attributable to exiting firms, close to the 3.2 percent we measure. BGM's higher rate is therefore not a different reading of the same margin. Their single δ must carry product churn inside surviving firms as well as destruction at exiting ones.

Our model faces the same constraint and resolves it in the opposite direction. Under symmetry one δ_e serves two roles. It is the fraction of product lines withdrawn from the variety space, and it is the fraction of employment destroyed by exit. These differ in the data, because dying establishments are smaller than average and because product lines also die inside surviving firms. We match the employment margin. The δ shock in this model destroys filled jobs and pre-committed vacancies, and that destruction is what the asymmetry in Proposition 5 and Lemma 2 turns on. The Bartik instrument is also built from employment-weighted establish-

ment deaths, so calibrating to a different margin would leave the empirical and quantitative sections describing different objects.

The cost is that the variety channel is calibrated conservatively. Product destruction inside continuing firms withdraws varieties without appearing in establishment deaths, so δ_e understates the movement in N that a broader product-level measure would deliver. Our results should be read as a lower bound on the amplification contributed by the variety channel. We regard this as the right trade for a model whose contribution is on the labor market side, and we prefer it to introducing a second destruction margin disciplined only by the ratio of two exit measures.

Dependent block Θ_d: parameters implied jointly by the normalizations, the external block, and a draw from Θ_e via the four-stage calibration procedure described below. We normalize the steady-state mass of firms $N = 1$ and the wage $w = 1$. The remaining parameters—z, ϕ, f_e, A, s, ψ, χ_m, x_m, κ, δ—are recovered from the following targets:

Table 5: Calibration targets for the dependent block Θ_d. Job-finding and vacancy-filling rates are gross JOLTS rates, corrected by $(1 - \delta_e)$ in Stage 1. X/Y and X^c/Y are external targets; all other Θ_e parameters listed in the estimated block are drawn by the SMM sampler.

Target	Value	Source	Parameters identified
Ann. destruction rate δ_e^{ann}	7.54%	BED Deaths	Monthly δ_e; with $\omega_\delta \in \Theta_e$: exogenous δ
Job-finding rate f^{emp}	41%	JOLTS	Corrected rate f, tightness θ, level A
Vacancy-filling rate q^{emp}	80%	JOLTS	Corrected rate q, tightness θ
Separation rate τ	3.1%	Shimer (2005)	Match separation $s = (\tau - \delta_e)/(1 - \delta_e)$
Recruiting share X/Y	1.5%	CET 2016	Vacancy value Q; then κ, z, f_e, x_m, ϕ
Fixed cost share X^c/Y	10%	See text	Retail ratio X^c/Y^c, profit share π_s
Wage $w = 1$	—	—	Productivity z
Firm mass $N = 1$	—	—	Entry cost f_e

Calibration algorithm. Given a draw from Θ_e and the external block, we recover Θ_d in four sequential stages.

Stage 1: direct conversions (targets → fundamentals). The destruction rate target δ_e^{ann} and endogenous share ω_δ pin

$$\delta_e = 1 - (1 - \delta_e^{ann})^{1/12}, \qquad \delta = (1 - \omega_\delta)\,\delta_e.$$

The match separation rate follows from the aggregate target:

$$s = \frac{\tau - \delta_e}{1 - \delta_e}.$$

The empirical job-finding and vacancy-filling rates are corrected for destruction before use: $f = f^{emp}/(1 - \delta_e)$, $q = q^{emp}/(1 - \delta_e)$. Market tightness $\theta = f/q$, the unemployment rate $u = \tau/(\tau + (1 - \delta_e)f)$, and the matching-function level parameter $A = f/\theta^{1-\eta_L}$ are then computed analytically.

Stage 2: profit share (root-find X^c/Y^c). The fixed-cost share of retail output X^c/Y^c is not directly observed; we recover it by solving

$$\frac{X^c/Y}{(Y^c/Y^{Gross})(Y^{Gross}/Y)} = \frac{X^c}{Y^c},$$

where $Y^c/Y^{Gross} = (r + \delta_e)/(r + \delta_e + \delta_e\pi_s)$ and $Y^{Gross}/Y = 1 + X/Y + X^c/Y$. The profit share $\pi_s = 1/\varepsilon - X^c/Y^c$ is a residual: retail value added equals the Lerner markup $1/\varepsilon$ after paying stochastic fixed costs.

Stage 3: cost-distribution parameters (root-find ψ). Given π_s, we recover the mass fraction of continuing cost draws, *cons*, from

$$\pi_s = \frac{\mu - 1}{\mu}\,(1 - cons)\,\frac{r + \delta_e}{r + \delta_e + cons\,(1 - \delta_e)},$$

then map *cons* to the power-law shape parameter $\psi = (cons/p_0)\,/\,(1 - cons/p_0)$ and location parameter $\chi_m = \chi^c/\varsigma^{1/\psi}$, where $\varsigma = ([(1 - \delta_e)/(1 - \delta)] - (1 - p_0))/p_0$ is the quantile of the exit threshold within the continuous part of the cost distribution. The mass weight $p_0 \in \Theta_e$ is estimated: given π_s, only the product $cons = (\psi/(1 + \psi))p_0$ is pinned by Stage 3, so p_0 and ψ are not separately identified from steady-state targets alone and require business-cycle moment restrictions.

Stage 4: vacancy value (root-find Q). We solve for the steady-state vacancy value Q by requiring the recruiting-cost share to equal its target: $X/Y = 1.5\%$. Given a trial value Q, the contact-rate value $K = (r+\delta_e)Q/(1+r)$ and the fixed matching cost $\kappa = (1-x_v)/x_v \cdot K/q$ follow from the sunk-cost split x_v. Total recruiting expenditure $X = e/(1+\xi^{-1})\cdot Q + \kappa q v$, the interior wage w^{int}, and firm-level output $Y^c = \rho z L^c$ are computed sequentially. The value of Q is chosen so that output implied by the recruiting-cost target, $Y = X/(X/Y)$, coincides with expenditure-side output $Y = C + \nu^f N^e$. The remaining dependent parameters are then recovered analytically:

$$z = (\mu/\rho)\, w^{int}, \tag{66}$$

$$f_e = \pi_s\, z\, L^c\, (1-\delta_e)\, \mu \,/\, (N\,(r+\delta_e)), \tag{67}$$

$$\phi = \frac{w-b}{w^{int} - K + \theta(K+q\kappa) - s\mathcal{R} - b}, \tag{68}$$

$$x_m = Q \,/\, e^{\xi^{-1}}.$$

The normalizations provide clean intuition: $w = 1$ pins z via (66) because the interior wage $w^{int} = \rho z/\mu$ is proportional to productivity; $N = 1$ pins f_e via (67) because free entry ties the mass of product lines to the entry cost; and the recruiting-cost share target $X/Y = 1.5\%$, once Q is found, pins bargaining power ϕ via (68). There $\mathcal{R}$ is the steady-state reposting shortfall, which is zero under costless reposting.

Discussion of b and ϕ. In our labor market framework, the labor share and the search wedge w/w^{int} both depend on Nash bargaining power ϕ and the worker outside option b. Shimer (2005) interprets b as unemployment insurance and calibrates it to 0.4, finding that technology shocks generate an order-of-magnitude too little unemployment and vacancy volatility, attributing this to excessively wage-elastic labor markets. Hagedorn and Manovskii (2008) show that even adjusting ϕ to match the mild empirical wage cyclicality leaves tightness volatility unchanged; what matters is the proportional response of profits to productivity. We resolve this debate by estimating b via simulated method of moments and, given the draw of b, selecting ϕ internally to satisfy (68)—ensuring that the wage normalization and labor-share implications are mutually consistent. Our model links market power with search frictions, making the labor share a natural discipline on both ϕ and b. Matching the consumption share and labor share jointly also implies a realistic stock-market capitalization-to-output ratio without sacrificing fit on business-cycle moments.

5.3. Inspecting the mechanism

Here we quantify novel features of the model mechanism. Across specifications, we set replacement-income ratio $b/w^{int} = 0.9$, a sunk-cost share of recruiting expenditure $x_v = 0.5$, and an endogenous exit share $\omega_\delta = 0.5$, so that half of steady-state destruction operates through the continuation-cost margin. The exit-elasticity target is $\psi\varsigma/(1-\varsigma) = 5$—the percent increase in the endogenous exit rate for a one-percent rise in the continuation-cost threshold—which pins the shape parameter $\psi \approx 0.033$ analytically.[22] The resulting steady state has a monthly destruction rate $\delta_e \approx 0.653\%$, exogenous component $\bar{\delta} \approx 0.326\%$, total separation rate $\tau = 3.1\%$, and unemployment $u \approx 7.0\%$. Two implied durations organize everything that follows. A firm's expected life is $1/\delta_e \approx 154$ months, so the firm stock is a slow-moving capital stock: the half-life of a deviation of N_t from its steady state is $\ln(0.5)/\ln(1-\delta_e) \approx 106$ months, or roughly nine years. The shocks, by contrast, are fleeting — the destruction shock has a half-life of 1.3 months and the technology shock 6.7 months (Section 5.3 below). The firm stock is therefore some eighty times more persistent than the δ shock that disturbs it. Essentially all of the persistence in the model's unemployment response is inherited from N_t rather than from the driving process, which is what allows a shock that dies within a month to move unemployment for years.

Shock calibration. The stochastic processes for technology and firm destruction are disciplined by a bivariate structural VAR(1) estimated at quarterly frequency on the HP($\lambda = 1{,}600$)-filtered cyclical components of log labor productivity z_t (BLS nonfarm business output per hour, PRS85006163) and log aggregate firm destruction δ_t (BLS BED Deaths, national employment-weighted exit rate) over 1992Q3–2019Q4 ($T = 110$).[23] The reduced-form system is

$$\begin{pmatrix} \log z_t \\ \log \delta_t \end{pmatrix} = A_1 \begin{pmatrix} \log z_{t-1} \\ \log \delta_{t-1} \end{pmatrix} + \mathbf{u}_t, \qquad \mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \Sigma),$$

[22] With the survival quantile $\varsigma \approx 0.993$ at the calibrated exit rate, feasibility limits the destruction elasticity to roughly 10 before implied fixed-cost shares exceed 18 % of output.

[23] The δ series is the national employment-weighted BED Deaths exit rate—the same underlying series that enters the Bartik instrument's national rates—aggregated directly rather than Bartik-weighted. Using the Bartik aggregate would conflate the shock process with the instrument construction.

with estimated companion matrix and pre-identification innovation correlation:

$$\hat{A}_1 = \begin{pmatrix} 0.735 & -0.002 \\ -1.050 & 0.207 \end{pmatrix}, \qquad \widehat{\text{corr}}(u_t^z,\, u_t^\delta) = -0.284.$$

Structural shocks are identified by a Cholesky decomposition ordering technology first, consistent with the within-period timing in which aggregate productivity is realized before firm-level continuation decisions. With P the lower-triangular Cholesky factor of $\hat{\Sigma}$, the structural innovations are $\varepsilon_t^z = u_t^z/P_{11}$ and $\varepsilon_t^\delta = (u_t^\delta - P_{21}\varepsilon_t^z)/P_{22}$, so $\text{Cov}(\varepsilon^z, \varepsilon^\delta) = 0$ by construction.

The two off-diagonal entries of $\hat{A}_1$ are excluded from the model's exogenous shock processes, for distinct reasons. The entry $\hat{a}_{21} = -1.050$ (HC3 SE = 0.830) reflects the tendency of exit to rise following productivity declines; this channel is generated endogenously by the model through the JCC and the continuation-cost threshold χ_t^c, so including it in the exogenous process would double-count the mechanism under study. The entry $\hat{a}_{12} = -0.002$ is two orders of magnitude smaller in absolute value and is consistent with z being an autonomous technology process unaffected by market-structure shocks. Treating both off-diagonal terms as equilibrium outcomes rather than primitive forcing processes leaves the diagonal entries as the persistence parameters of the exogenous shocks.

The quarterly diagonal persistence parameters are converted to monthly via $\rho^m = (\rho^Q)^{1/3}$, reflecting the cube-root relationship between quarterly and monthly AR(1) coefficients. Innovation standard deviations are rescaled to match the unconditional variance of the quarterly process: $\sigma^m = \sigma^Q\sqrt{(1-(\rho^Q)^2)/(1-(\rho^m)^2)}$. This yields $\rho_z = 0.902$, $\sigma_z = 0.0092$ for technology and $\rho_\delta = 0.592$, $\sigma_\delta = 0.0669$ for destruction, entering equations (7) and (8) directly. The destruction shock has a half-life of about 1.3 months and the technology shock about 6.7 months, against a firm-stock half-life of roughly nine years. Prolonged unemployment responses to δ shocks therefore reflect propagation through the firm and vacancy stocks rather than the persistence of the driving process itself: by the time unemployment peaks, the shock that caused it is long gone.

Choice of shocks and comparisons. We report impulse responses to the z and δ shocks throughout and abstract from the s shock. Under costless reposting, Proposition 5 and Lemma 2 establish that a positive s shock generates positive u–v comovement, the opposite of the Beveridge curve, because every separation returns its position as a vacancy. In that case the s shock does not engage the vacancy-destruction and variety channels that are central to the model's

amplification. Partial reposting changes this, since separations that are not reactivated destroy positions. The comparisons in this section are computed under costless reposting, where the s shock is a theoretical benchmark rather than a quantitative focus. The z and δ shocks, by contrast, interact with the firm-stock state N_t through the variety channel in equation (27) and through the firm and vacancy laws of motion in equations (38) and (6), which is precisely where the model departs from the augmented Gabrovski-Silva benchmark.

We organize the comparisons around three counterfactuals. *Comparison A* examines the role of the level of δ_e at steady state. To facilitate comparison, it abstracts from endogenous entry entirely: $p_0 = 0, \delta_e = \delta$. The baseline sets ($\delta_e = \bar{\delta}$ and replaces the endogenous component by s: $s = (\tau - \delta_e)/(1 - \delta_e)$. The high-$\delta$ counterfactual sets $\delta_e = \tau$, and thus $s = 0$, which replicates the Coles and Moghaddasi Kelishomi (2018) setup in which all separations are from destruction events. fixed. This comparison extends Gabrovski and Silva (2025), who document the LOM-multiplier mechanism for technology shocks, to the δ-shock dimension. *Comparison B* isolates the endogenous exit margin: we set $p_0 = 0$ and equate the exogenous destruction parameter $\bar{\delta}$. We keep the aggregate separation rate τ the same, which requires s in the counterfactual to compensate for the loss of endogenous destruction. *Comparison C* sets $\zeta = 0$ so that $\rho_t \equiv 1$ regardless of N_t, removing the $N_t \to \rho_t \to w_t^{int} \to$ JCC channel and quantifying what the baseline offers beyond the augmented Gabrovski-Silva framework without product variety. The markup $\mu = \varepsilon/(\varepsilon - 1)$ is preserved in both specifications; only the love-of-variety externality is shut off.

Comparison A: the level of destruction. Figures 9 and 10 report impulse responses to one-standard-deviation shocks. To place both on a common scale in which unemployment rises, the z figure shows a *negative* technology shock and the δ figure shows a positive destruction shock. Each panel traces the transmission chain from labor market outcomes (rows 1–2) through the extensive margin (row 3) to the vacancy-creation chain (row 4). Appendix E.5 derives the log-linearized propagation equations that underlie the discussion below.

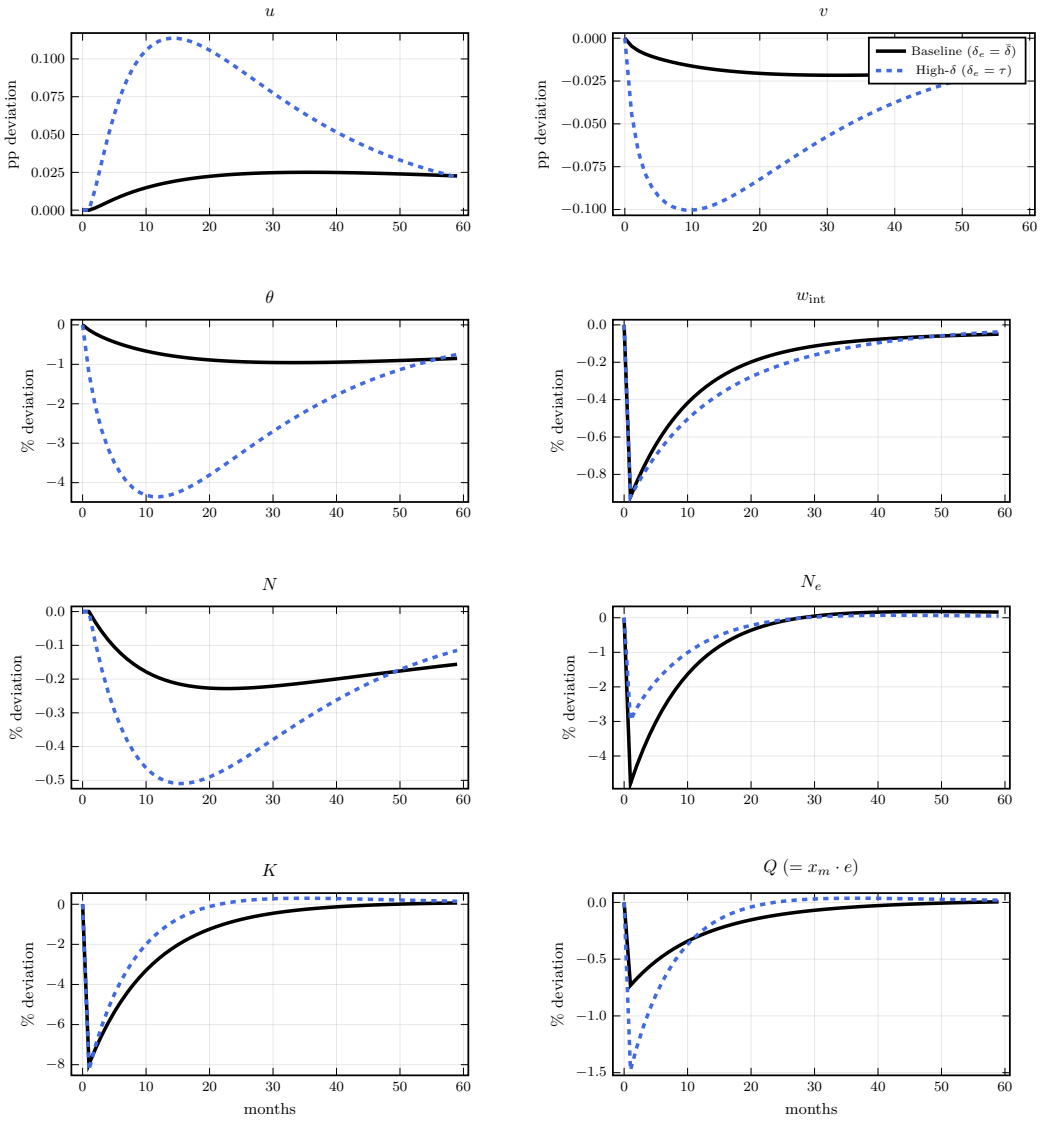


Figure 9: Comparison A — impulse responses to a one-standard-deviation *negative* technology shock. Solid black: baseline ($\delta_e = \bar{\delta} \approx 0.653\%$/month). Dashed blue: high-δ variant ($\delta_e = \tau = 3.1\%$/month; CK timing, $p_0 = 0$). Row 1 in percentage-point level deviation from steady state; rows 2–4 in $100 \times$ log deviation.

Under the *negative technology shock* (Figure 9), the high-δ variant generates a substantially larger response of N_t despite a nearly identical response of the flow vacancy value K_t. Log linearize the firm law of motion (38)

$$\hat{N}_{t+1} = (1 - \delta_e)\,\hat{N}_t \;+\; \delta_e\,\hat{N}^e_t, \tag{69}$$

where hats denote log-deviations from steady state and we use the steady-state relation $N^e/N = \delta_e/(1-\delta_e)$. The loading on the entry flow equals δ_e itself, so the nearly five-fold increase in δ_e from baseline to the high-δ variant (0.65% $\to$ 3.1% per month) translates a given entry response

into a commensurately larger accumulation of product lines. The point is worth stating plainly: δ_e is not only the rate at which product lines are destroyed, it is also the rate at which they are replaced, since $N^e/N = \delta_e/(1-\delta_e)$ in steady state. A low-δ_e economy is one whose firm stock both erodes slowly and repairs slowly. Indeed the entry flow $\hat{N}^e_t$ actually responds *more* in the baseline (peak 4.80% versus 2.96%); it is the loading δ_e, not the entry response itself, that delivers the larger movement in N_t. A complementary channel operates through the vacancy value. Log-linearizing (43) and using $\mathbb{E}_t\hat{Q}_{t+1} = \rho\hat{Q}_t$, which holds exactly under a one-shock approximation by undetermined coefficients,[24] yields

$$\hat{Q}_t = \underbrace{\frac{r+\delta_e}{1+r-\rho(1-\delta_e)}}_{\mathcal{D}(\delta_e,\rho)}\hat{K}_t,$$

where $\beta = 1/(1+r)$ and $\rho \in \{\rho_z, \rho_\delta\}$ is the persistence of the driving shock. The multiplier $\mathcal{D}$ is strictly *increasing* in δ_e, since $\partial\mathcal{D}/\partial\delta_e = (1+r)(1-\rho)/\text{den}^2 > 0$. The mechanism is that a higher survival discount $\beta(1-\delta_e)$ in the baseline causes the continuation term $\beta(1-\delta_e)\mathbb{E}_t\hat{Q}_{t+1}$ in (43) to nearly offset $\hat{Q}_t$, dampening the transmission from $\hat{K}_t$ to $\hat{Q}_t$. In the high-δ variant the continuation term is smaller, so $\hat{Q}_t$ — and through $\hat{e}_t = \xi\hat{Q}_t$ in (44), new postings — respond more strongly on impact, as visible in row 4 of Figure 9. The intuition is the familiar duration logic of asset pricing. A vacancy is a claim on a stream of future match surpluses that is extinguished when the product line dies, so its expected life is $1/\delta_e$: about thirteen years in the baseline, under three in the high-δ variant. A long-duration claim is priced mostly off the distant future, so a transitory disturbance to the current dividend barely moves it; a short-duration claim is close to a pure claim on near-term dividends and its price tracks current conditions almost one-for-one. This is why $\hat{Q}$ responds roughly twice as strongly in the high-δ variant (1.48% versus 0.73%) even though the dividend itself, $\hat{K}$, moves almost identically in the two (8.28% versus 7.98%). Both $\mathcal{D}$ and the LOM entry-loading δ_e in (69) therefore favor the high-δ variant on impact. However, the baseline generates more persistence via the survival term $(1-\delta_e) \approx 0.993$ in (69), which prolongs the $\rho_t \downarrow \rightarrow w_t^{int} \downarrow \rightarrow \theta_t \downarrow \rightarrow u_t \uparrow$ chain documented above.

Under the *destruction shock* (Figure 10), the same two forces operate as under the tech-

[24] Guess $\hat{Q}_t = \phi_Q\hat{\delta}_t$; then $\mathbb{E}_t\hat{Q}_{t+1} = \phi_Q\rho_\delta\hat{\delta}_t = \rho_\delta\hat{Q}_t$. For a z shock replace ρ_δ with ρ_z. The full perturbation solution additionally accounts for the N_t state but the qualitative result is unchanged.

nology shock, but now both scale with δ_e because the shock itself directly destroys product lines through (69). The high-δ variant therefore suffers a deeper N trough — five times deeper, -0.56% against -0.11%, since the destruction term is larger — but also recovers faster, since the replacement rate $N^e/N = \delta_e/(1-\delta_e)$ and the entry-loading $\mathcal{D}$ are both larger. The net effect on persistence is thus a race between a deeper initial hit and faster repair, and the figures show that recovery speed dominates: the high-δ variant converges markedly faster. This is consistent with the limiting case: as $\delta_e \to 0$ the δ shock becomes irrelevant regardless of persistence, so the baseline necessarily generates a smaller but longer-lived response. In the baseline, the monthly replacement rate is only 0.66% of the firm stock, so after a destruction impulse the variety depression is long-lived: ρ_t stays depressed, $w_t^{int} = \rho_t z_t/\mu$ remains low, and the job creation condition (42) stays tight, prolonging the recovery of θ_t and u_t. The central quantitative message of Comparison A is therefore the same across both shocks: a model calibrated to $\delta_e/\tau \approx 0.21$ — far below the Coles and Moghaddasi Kelishomi (2018) convention $\delta_e = \tau$ — necessarily operates with a low-churning firm stock, generating persistent responses through the slow N-recovery channel independently of endogenous exit or variety effects.

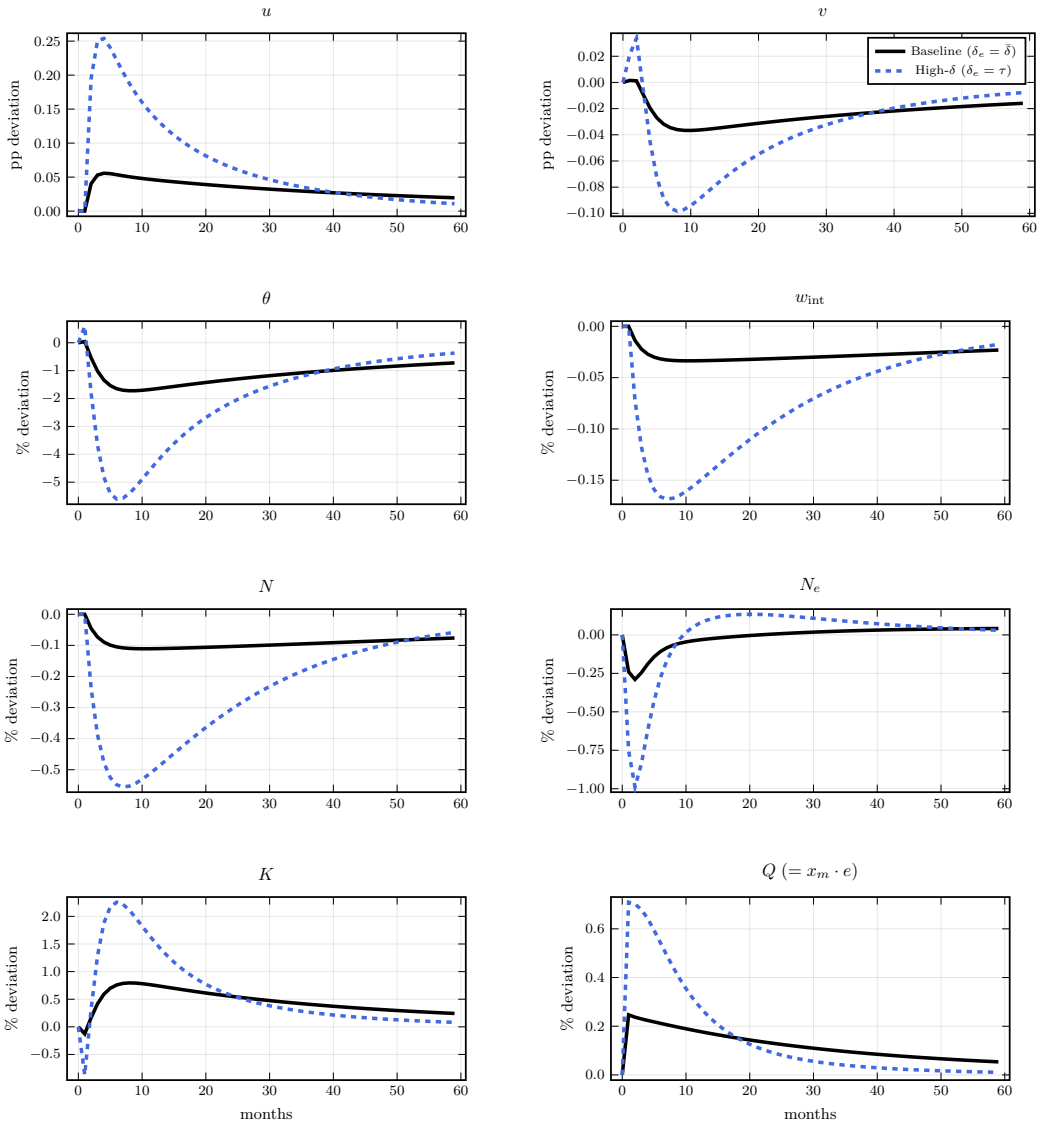


Figure 10: Comparison A — impulse responses to a one-standard-deviation destruction shock. Lines and units as in Figure 9.

Comparison B: endogenous versus exogenous exit. Figures 11 and 12 compare the full baseline with an exogenous-exit model in which $p_0 = 0$ and $\omega_\delta = 0$. The exogenous exit variant keeps the same rate of exogenous destruction $\bar{\delta}$ and aggregate separation τ, which requires raising s accordingly. By equations (38), (6), and (5), this keeps steady-state u, v, and θ equal so that percentage-point IRF differences are directly comparable.

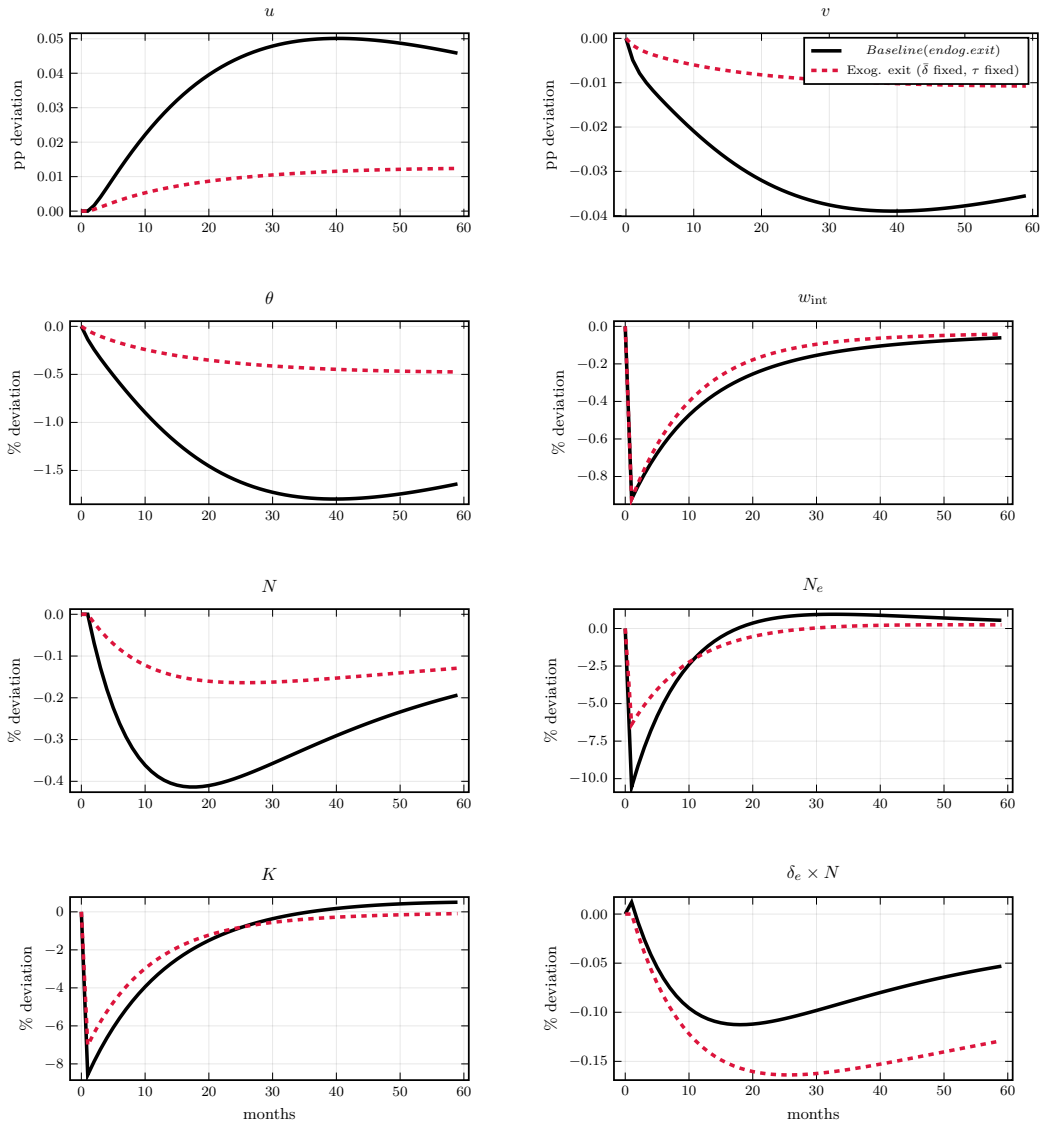


Figure 11: Comparison B — impulse responses to a one-standard-deviation *negative* technology shock. Solid black: full baseline (endogenous exit, $\omega_\delta = 0.5$, $\varepsilon_x = 5$). Dashed crimson: exogenous exit ($\bar{\delta}$ equated, τ fixed, s rises to compensate). Units as in Figure 9.

For the *negative technology shock* (Figure 11), the endogenous-exit model generates a larger response of N_t that builds over time. The continuation threshold satisfies

$$\chi_t^c = \underbrace{\frac{1}{\varepsilon}\frac{Y_t^c}{N_t}}_{\text{per-firm profit}} + \underbrace{\nu_{f,t}}_{\text{cont. value}}, \tag{70}$$

where $\nu_{f,t} = \rho_t f_e/\mu$ from free entry (31). A negative z shock lowers $Y_t^c = \rho_t z_t \bar{z} L_{c,t}$ immediately, compressing per-firm profit and lowering χ_t^c on impact, which induces exit. Two forces govern subsequent dynamics. First, note that a fall in N raises per-firm profit (market-share effect,

elasticity $(2-\varepsilon)/(\varepsilon-1)$) but simultaneously lowers the continuation value $\nu_f = \rho_t f_e/\mu$ through $\rho_t = N_t^{1/(\varepsilon-1)}$ (elasticity $1/(\varepsilon-1)$). At $\varepsilon \approx 4.3$, the profit-share elasticity (≈ -0.70) dominates the continuation-value elasticity (≈ 0.30). Hence, falling N tends to *raise* χ^c overall and constitutes a stabilizing feedback. Second, the endogenous model calibrates a substantially lower sunk entry cost f_e: with endogenous exit, steady-state profits are lower (some continuation costs are paid), so free entry requires a smaller f_e (13.7 versus 52.2, a factor of nearly four), amplifying entry responses at any given ν_f. Concretely, the endogenous-exit economy is one where setting up a firm is cheap and firms are correspondingly disposable, while the exogenous-exit economy is one of expensive, long-lived firms. Cheap entry means the entry margin does more of the adjustment. The exit-flow panel isolates the continuation margin directly. On impact the exogenous model is flat by construction — δ_e cannot move and N_t is predetermined — while the endogenous model shows exit rising immediately (+0.012%), which is the χ^c threshold responding to the profit compression. Thereafter the two paths are best read as a decomposition into rate and stock, since the exit flow is the product of the two. In the exogenous model the rate is fixed, so the flow simply tracks the shrinking firm stock and falls by 0.16% at its trough. In the endogenous model the stock falls considerably further (0.41%), yet the flow falls by *less* (0.11%), because the exit rate is climbing — by roughly 0.30% at the same horizon — as marginal firms are pushed over the threshold. A weaker economy sheds a larger fraction of a smaller stock. The elasticity $\varepsilon_x = \psi\varsigma/(1-\varsigma) = 5$ governs the local sensitivity of exit to the threshold throughout.

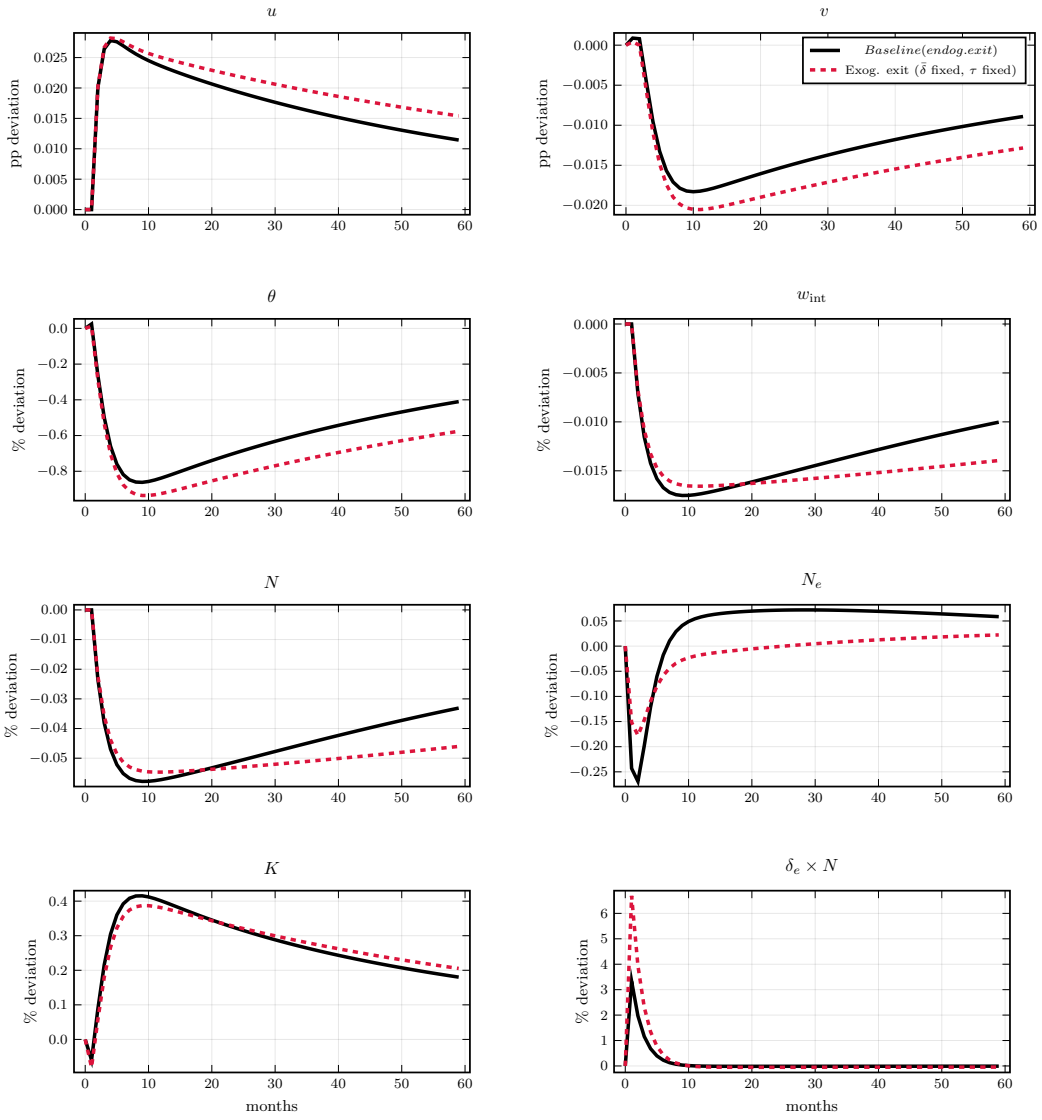


Figure 12: Comparison B — impulse responses to a one-standard-deviation destruction shock. Lines and units as in Figure 11.

For the *destruction shock* (Figure 12), the design equates $\bar{\delta}$ across both models so that a proportional δ shock delivers the same first-period impulse to $\delta_{e,t}$ in both *in levels*. Impact responses in u and N are therefore similar. The exit-flow panel needs one word of interpretation on this point. Because $\bar{\delta}$ is equated in levels while δ_e itself differs, the common level impulse corresponds to twice the *proportional* impulse in the exogenous model, whose δ_e is half as large. Plotted in log deviations, that panel accordingly shows the exogenous model jumping about twice as far on impact (6.7% against 3.3%); the two are nonetheless experiencing the same absolute wave of destruction, which is what makes the subsequent recovery paths comparable. Recovery, however, is markedly faster in the endogenous-exit model, for the same reason iden-

tified in Comparison A: the steady-state level of δ_e governs mean reversion. To equate $\bar{\delta}$ while eliminating endogenous exit, the exogenous model must set $\delta_{e,\text{exog}} = \bar{\delta}_{\text{endog}} \approx 0.326\%/\text{month}$, roughly half of $\delta_{e,\text{endog}} \approx 0.653\%$. From the N law of motion (69), the entry coefficient equals δ_e and the survival coefficient equals $(1 - \delta_e)$: higher δ_e in the endogenous model means both faster replenishment of the firm stock and stronger mean reversion of any N deviation. The exogenous model's low-churning firm stock therefore keeps N_t depressed for longer, sustaining the $\rho_t \downarrow \rightarrow w_t^{int} \downarrow \rightarrow \theta_t \downarrow$ chain and prolonging unemployment. Comparison B thus shows, as a mirror image of Comparison A, that equating $\bar{\delta}$ while reducing δ_e produces greater persistence — the endogenous-exit specification recovers faster precisely because its higher steady-state destruction rate implies a higher replacement rate.

The χ^c margin contributes a secondary channel for the destruction shock as well, though the direction is ambiguous. A $\delta \uparrow$ impulse directly destroys firms, lowering N_t; by (70) this raises per-firm profit (market-share effect) but lowers ν_f through ρ_t. The net effect on χ^c — and hence on additional endogenous exit — depends on which elasticity dominates along the transition, a quantitative question the IRFs resolve. The dominant effect of endogenous exit in Comparison B remains the structural difference in the replacement rate: the τ-fixing constraint halves δ_e in the exogenous model, slowing N-recovery and prolonging unemployment.

Comparison C: variety effects. The third comparison sets $\zeta = 0$ so that $\rho_t \equiv 1$ regardless of N_t, removing the link between product variety and the interior wage in equation (27). The markup $\mu = \varepsilon/(\varepsilon - 1)$ is unchanged; only the love-of-variety externality is shut off. Figures 13 and 14 report the results.

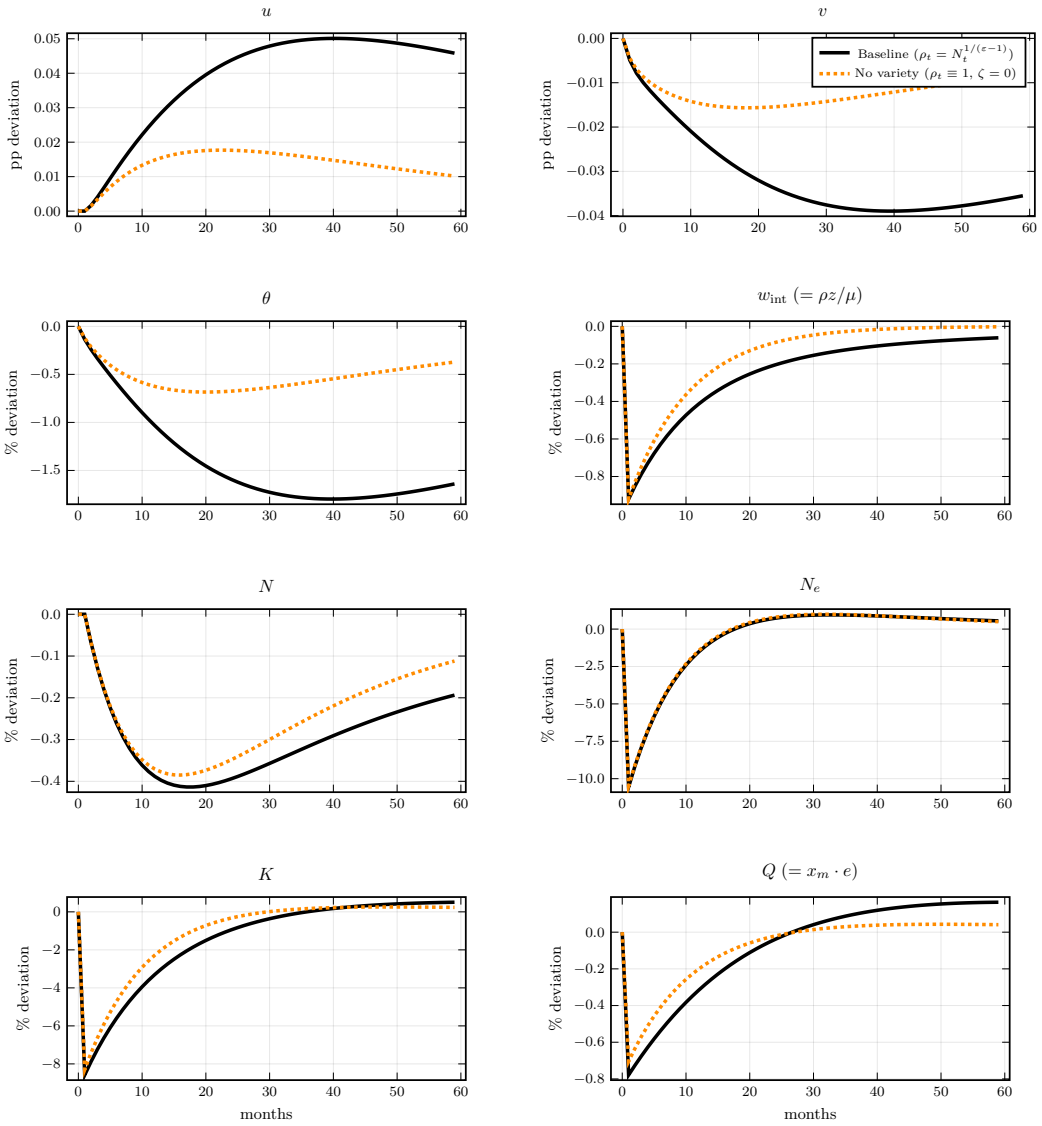


Figure 13: Comparison C — impulse responses to a one-standard-deviation *negative* technology shock. Solid black: baseline ($\rho_t = N_t^{1/(\varepsilon-1)}$, full variety). Dotted orange: no-variety counterfactual ($\rho_t \equiv 1$, $\zeta = 0$; markup μ unchanged). Row 1 in percentage-point level deviation from steady state; rows 2–4 in $100 \times$ log deviation.

Under the *negative technology shock* (Figure 13), the two models share the same direct $z \to w_t^{int} = \rho_t z_t/\mu$ channel, but diverge through the indirect variety channel. In the baseline, $z \downarrow$ reduces entry, N_t falls, $\rho_t = N_t^{1/(\varepsilon-1)}$ falls, and w_t^{int} declines by more than the direct z impulse alone. With $\rho_t \equiv 1$, the second-round compression of w_t^{int} is absent, and the gap between the two interior-wage paths is the pure $N \to \rho$ contribution. Because w_t^{int} pins the job-creation surplus (42), a smaller w_t^{int} response implies a smaller θ_t response, which attenuates both the unemployment rise and the vacancy fall. This comparison identifies the variety channel unusually cleanly. On impact the two specifications are *exactly* equal: N_t is predetermined, so

ρ_t has not yet moved, and $w_t^{int} = \rho_t z_t/\mu$ registers only the direct technology impulse in both models (both fall by 0.92%). The paths separate only from the second month, as the entry margin begins to erode N_t, and the gap then widens monotonically — reaching roughly a quarter of the baseline response within a year. Everything separating the two lines after impact is the variety externality, with no contamination from the direct effect of z.

The N and N_e panels show a further distinction. With $\rho_t \equiv 1$, the firm value $\nu_f = f_e/\mu$ is a constant: it does not rise when N recovers, so the entry Euler (30) provides no forward-looking amplification of the entry margin. The no-variety N response is consequently more compressed and recovers faster—entry is purely backward-looking in the counterfactual, governed only by current profits rather than the anticipated $\rho(N)$ appreciation.

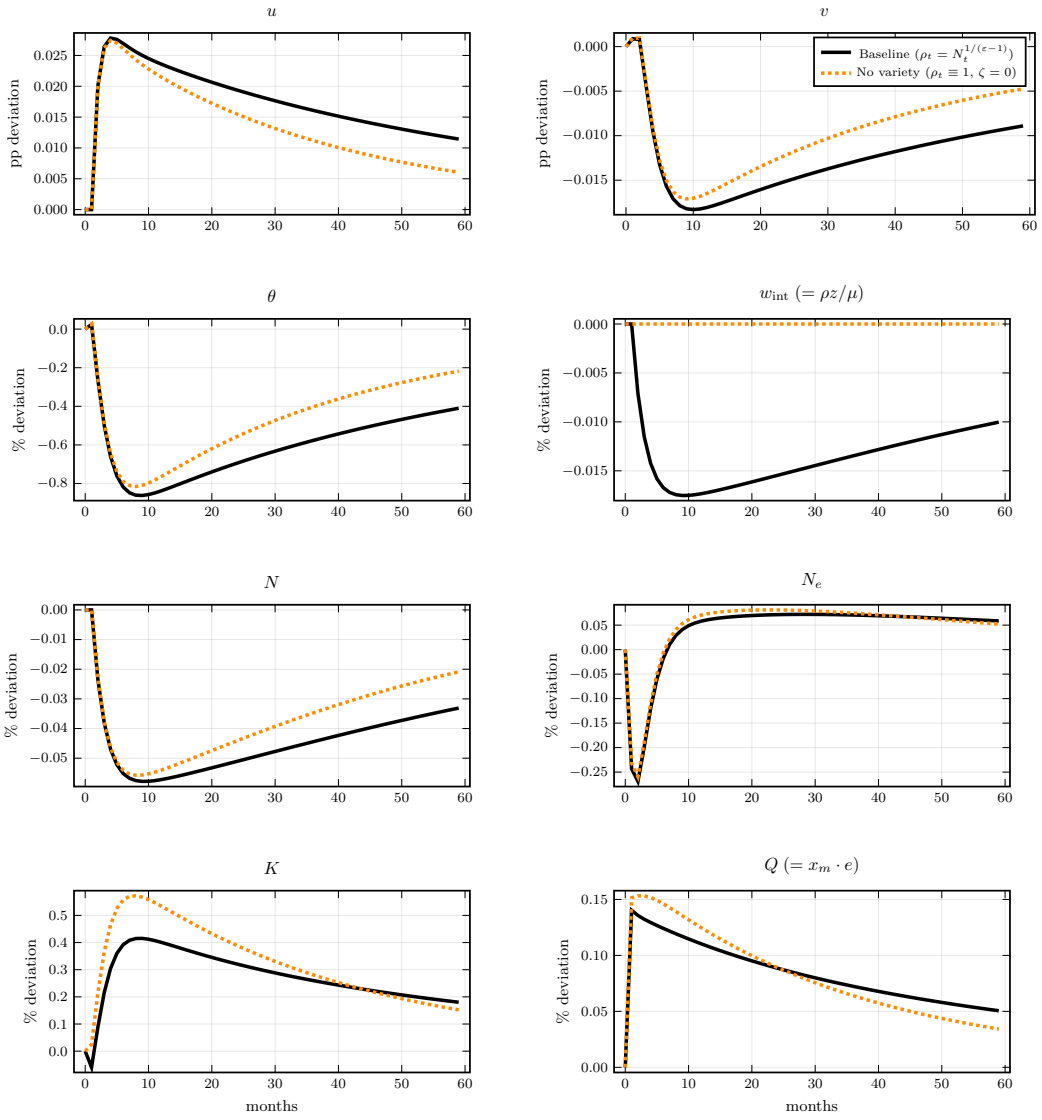


Figure 14: Comparison C — impulse responses to a one-standard-deviation destruction shock. Lines and units as in Figure 13.

Under the *destruction shock* (Figure 14), the variety channel works in the amplifying direction. A $\delta \uparrow$ impulse destroys firms, N_t falls, ρ_t falls, and w_t^{int} falls beyond what the exogenous destruction alone would imply. The lower w_t^{int} tightens the JCC—reducing the surplus firms extract from a filled job—and simultaneously depresses entry through the free-entry condition $\nu_f = \rho_t f_e/\mu$. Unemployment rises more and vacancies fall more in the baseline than in the no-variety counterfactual because the same N depression feeds back into wages and entry values in the full model. Absent taste for diversity, $\nu_f = f_e/\mu$ is fixed, so the entry Euler is unresponsive to the N depression: recovery is faster but the initial impact is also attenuated, since the $\rho \to \nu_f$ mechanism that amplifies destruction shocks into a prolonged Beveridge-curve response

is absent. In particular, w_t^{int} does not respond at all.

Taken together, Comparisons A–C decompose the model's amplification into three orthogonal channels: the level of δ_e governing asset duration and the LOM multiplier (A), the χ^c margin governing endogenous exit amplification (B), and the $N \to \rho$ variety externality governing the wage and entry-value response (C). Each channel is active in both the technology and destruction directions, but with different relative magnitudes that the IRF comparisons quantify.

5.4. Posterior estimates

6. Conclusion

Appendix A. Relation to literature

	Finitely elastic vacancies	Endogenous variety	Endogenous separation	Separate destruction margin
Bilbiie, Ghironi, and Melitz (2012)	NA	✓	NA	NA
Coles and Kelishomi (2018)	✓	×	×	×
Broer, Druedahl, Harmenberg, and Öberg (2025)	✓	×	✓	×
Gabrovski and Silva (2025)	✓	×	×	✓
Shao and Silos (2013)	✓	✓	×	×
Christiano, Eichenbaum, and Trabandt (2016)	×	×	×	×
Colciago, Fasani, and Rossi (2021)	×	✓	×	×
Cacciatore and Fiori (2016)	×	✓	✓	×
This paper	✓	✓	✓	✓

Table A.6: Comparison of model ingredients. *Finitely elastic vacancies*: sunk vacancy creation costs generating a positive vacancy value. *Endogenous variety*: equilibrium number of product lines responds to economic conditions. *Endogenous separation*: firm exit driven by idiosyncratic profitability draws. *Separate destruction margin*: exogenous product-line destruction distinguished from match separation.

Appendix B. Data sources

Table B.7: Data sources

Series	Source	Frequency & coverage	Unit
Panel A: Instrument construction			
BED establishment deaths (dc 08)	BLS BED	Quarterly, 1992Q3–2024Q2	National × supersector
QCEW employment shares	BLS QCEW	Annual, base year 2006	State × supersector
Layoffs & discharges (LD)	BLS JOLTS	Monthly, 2001Q1–2023Q1	National × supersector
Quits (QU)	BLS JOLTS	Monthly, 2001Q1–2023Q1	National × supersector
Panel B: Residualization controls			
Labor productivity (OPHNFB)	BLS, FRED	Quarterly, 1947Q1–present	National
Value added by industry	BEA, FRED	Quarterly, 2005Q1–present	National × supersector
Panel C: LP outcome variables			
Unemployment rate	BLS LAUS	Monthly, 1990M1–present	State
Civilian labor force	BLS LAUS	Monthly, 1990M1–present	State, thousands
Job openings	BLS JOLTS	Monthly, 2001M1–present	State, thousands
Panel D: Bayesian estimation moments			
Job-finding rate (f_t)	BLS CPS	Monthly, 1948M1–present	National
Separation rate (s_t)	BLS CPS	Monthly, 1967M1–present	National
Business formations (BF8)	Census BFS	Quarterly, 2004Q3–present	National

Panel A instruments are constructed as Bartik shift-share aggregates. BED deaths (dataclass 08) counts employment at establishments absent for four or more consecutive quarters, excluding temporary shutdowns by construction. QCEW shares are fixed at the 2006 base year and leave-one-out corrected. LD and QU are summed from monthly to quarterly; QU serves as a placebo instrument (Appendix C.5). Panel B controls enter the v2 residualization as first log-differences: aggregate productivity growth (OPHNFB) and lagged industry value-added

growth (12 BEA supersector series via FRED: VAPGDPIP, VAMFNIP, VACONIP, VATRDIP, VATRIIP, VAINIP, VAFNSIP, VAPBSIP, VAEDUHIP, VALAHIP, VAOSIP, VAGOSIP); the value-added series are extended to 2001Q1 via Chow–Lin temporal disaggregation using BLS employment as the indicator. Panel C outcomes are averaged from monthly to quarterly; the vacancy rate is defined as job openings divided by civilian labor force, expressed in percentage points. Panel D moments are national aggregates used as targets in Bayesian estimation. Job-finding and separation rates are constructed from CPS unemployment flows with the continuous-time correction for time-aggregation bias (Shimer, 2005). BF8 counts business applications with planned wages that appear in QCEW within eight quarters, and proxies for the model entry flow N_t^e.

Appendix C. Instrument diagnostics and robustness

Appendix C.1. Deaths/Closings ratio by supersector

Figure C.15 plots the Deaths/Closings ratio for each of the 12 BLS supersectors separately, divided into the six with the highest mean ratio (left panel) and the six with the lowest (right panel). Across all sectors and quarters the ratio remains strictly below unity, confirming that BED Deaths are always a subset of BED Closings. The cross-sector range is wide: Manufacturing, Retail, and Information cluster near 0.70, while Mining (0.37) and Construction (0.52) are substantially lower, consistent with the prevalence of project-based and cyclically suspended establishments in those sectors. The ratio is broadly stable within each supersector over time, with no systematic cyclical pattern, supporting the use of a fixed Deaths/Closings fraction as the implicit permanence assumption in our instrument.

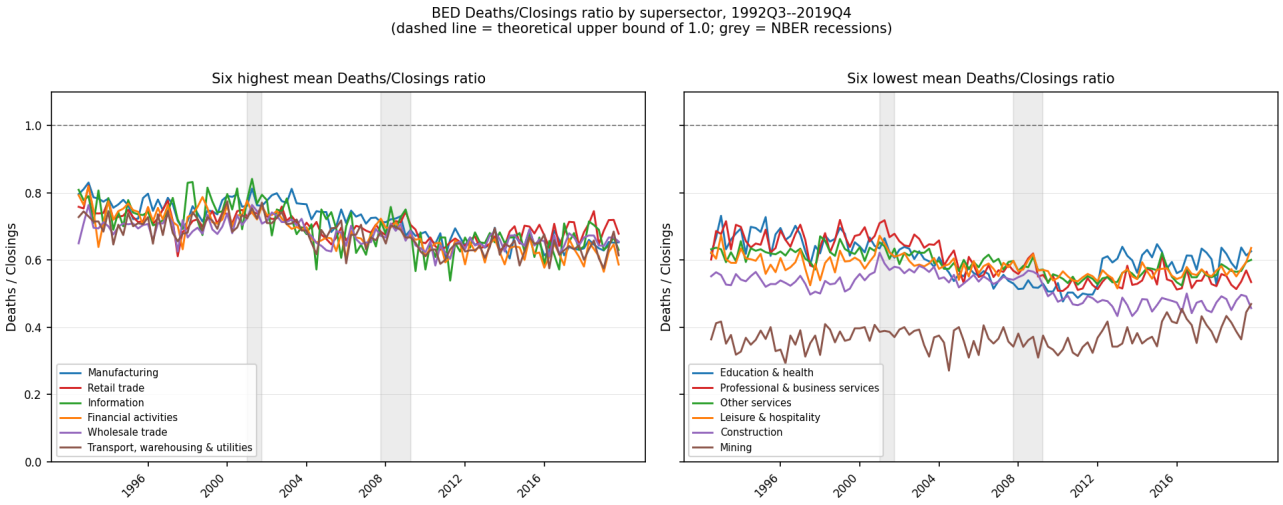


Figure C.15: BED Deaths/Closings ratio by supersector, 1992Q3–2019Q4. *Left panel:* six supersectors with highest mean ratio. *Right panel:* six supersectors with lowest mean ratio. Dashed horizontal line at 1.0 is the theoretical upper bound (Deaths $\subseteq$ Closings by construction). Grey shading denotes NBER recessions.

Appendix C.2. Residual δ–LD correlation

After v2 residualization the cross-state correlation between the δ and layoffs-and-discharges instruments is $r(\tilde{B}^{\delta}, \tilde{B}^{\mathrm{LD}}) = 0.334$, down from 0.54 on raw rates. The residualization controls for aggregate productivity growth, lagged industry value-added, and lagged labor-market tightness, ruling out the most obvious industry-demand and aggregate-cycle explanations for the shared variation. The leading interpretation is industry-specific financial conditions: when credit tightens in a sector, both product-line exit ($\uparrow \delta$) and layoffs at surviving establishments ($\uparrow$ LD) rise simultaneously, generating positive covariance without a structural link between the two shocks. Consistent with this, the Granger-causality evidence shows that δ predicts future LD at horizons $h = 4$–12 while LD predicts future δ only at $h = 0$–4; the long-horizon dependence runs from destruction to layoffs and not the reverse. A natural further step would be to residualize on a measure of sector-level credit supply, such as SLOOS net C&I tightening interacted with industry leverage exposure; we leave this for future work.

Appendix C.3. LD local projections

The separate LD local projections are subsumed by the joint LP figure (Figure C.16 below), which overlays the LD separate-specification IRF alongside the joint-LP coefficient for the same instrument. In the separate specification the LD unemployment response rises monotonically to a peak of +1.68 pp at $h = 20$, significant throughout, while the vacancy response is persistently

negative (trough -0.45 pp at $h = 20$). Both patterns survive the severity placebo. The monotonically rising unemployment IRF is anomalous given the estimated LD persistence $\hat{\rho}_{\text{LD}} \approx 0.3$: a transitory shock should produce a hump-shaped response, not a path that continues to rise five years out. The most likely explanation is GFC contamination — the 2008–09 recession produced a severe and prolonged layoff episode whose industry-composition variation is not fully absorbed by the v2 residualization — rather than a structural property of the s shock. We regard the LD IRFs as indicative rather than structural, and do not use them as estimation targets.

Appendix C.4. Joint local projection

Figure C.16 reports the joint versus separate local projections for the vacancy outcome. Each panel overlays the joint-LP coefficient (red, solid) against the corresponding separate-LP coefficient (blue, solid), with 90% confidence bands.

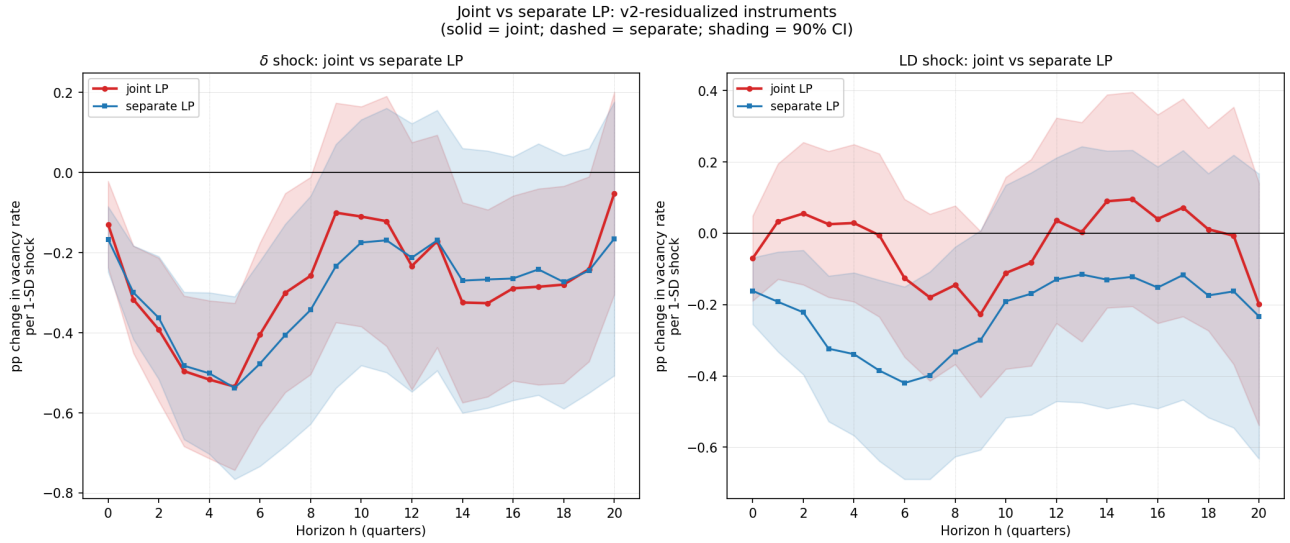


Figure C.16: Joint versus separate local projections: vacancy rate outcome. *Left panel:* δ instrument — joint LP (red) vs. separate LP (blue). *Right panel:* LD instrument — joint LP (red) vs. separate LP (blue). Shading denotes 90% confidence intervals; standard errors clustered at the state level ($n = 50$). Instruments are v2-residualized; sample 2001Q1–2019Q4.

Three findings emerge from the figure. First, the δ vacancy IRF is qualitatively stable across specifications: the joint and separate coefficients share the same sign, timing, and shape throughout, and the confidence bands overlap at most horizons. The trough in the joint specification (-0.535 pp at $h = 5$) is somewhat shallower and earlier than in the separate specification

(-0.582 pp at $h = 18$), reflecting partial absorption of the shared variation with LD, but the persistent negative response to product-line destruction is not an artifact of LD exposure.

Second, the δ unemployment IRF shrinks by roughly 0.6–0.8 pp in the joint specification relative to the separate LP, reflecting absorption of the variation that $\tilde{B}^{\delta}$ and $\tilde{B}^{\text{LD}}$ share. The joint peak of +1.15 pp remains positive and significant, so the unemployment result is robust, if attenuated.

Third, the LD vacancy coefficient collapses from its separate-LP value to approximately zero and is insignificant at all horizons once δ is controlled for (right panel). This collapse should *not* be read as evidence that δ subsumes a genuine LD effect on vacancies. Rather, it reflects that $r(\tilde{B}^{\delta}, \tilde{B}^{\text{LD}}) = 0.334$ is large enough that the two instruments do not span orthogonal directions in the shock space: the variation in $\tilde{B}^{\text{LD}}$ orthogonal to $\tilde{B}^{\delta}$ is insufficient to separately identify the LD vacancy response. The appropriate conclusion is that LD is not separately identified as a structural shock in this setting, not that separation shocks have zero causal effect on vacancies.

We therefore treat the separate δ LP as the primary specification. The joint LP establishes that the headline result — a persistent vacancy decline in response to product-line destruction — is robust to controlling for LD, and that the failure to separately identify LD is an instrument-space problem rather than a finding about vacancy dynamics.

Appendix C.5. QU placebo instrument

$\tilde{B}^{\text{QU}}$ is constructed from JOLTS quit rates using the same shares and residualization as the primary instruments. Quits represent voluntary job-to-job transitions: the quitting worker moves directly to another employer, leaving the origin firm with an unfilled position but supplying the destination firm with a worker. The net effect on aggregate unemployment and vacancies is therefore ambiguous and governed by matching frictions at the destination, not by the separation event at the origin. This is conceptually distinct from the model's s shock, which dissolves a match and triggers an unambiguous vacancy repost. Furthermore, the quit rate is procyclical — it rises when labor markets tighten — so $\tilde{B}^{\text{QU}}$ partly reflects the same industry-level demand cycle that v2 residualization attempts to remove.

Empirically, the QU instrument does not behave as a flat-zero placebo. Its unemployment and vacancy IRFs are nearly identical in magnitude to those of $\tilde{B}^{\delta}$, and the within-instrument correlation $\text{corr}(\hat{\beta}^{u}_{h}, \hat{\beta}^{v}_{h}) = -0.971$ across $h = 0, \ldots, 20$ — the same mechanical proportionality observed for $\tilde{B}^{\delta}$ (-0.948). This signature indicates that QU identifies industry-level demand

contractions (which move u and v symmetrically along the Beveridge curve) rather than a structurally distinct separation shock. The contamination is likely structural: high-quit industries tend to be procyclical, so the industry component of the quit rate is correlated with demand conditions even after residualization on aggregate productivity and lagged VA. The QU failure confirms that a well-identified s-type instrument remains an open challenge and motivates treating the LD results as indicative rather than structural.

Appendix C.6. Filter robustness for δ-related moments

Table 4 is computed with HP filter $\lambda = 1{,}600$, departing from the $\lambda = 10^5$ convention of (Shimer, 2005; Coles and Moghaddasi Kelishomi, 2018) and other papers. Here we show that smoothening at $\lambda = 10^5$ generates anomalous δ dynamic correlations with respect to other filters.

Table C.8 reports dynamic cross-correlations $\text{cor}(\delta_t, x_{t+h})$ for $x \in \{u, v\}$ and $h \in \{-4, \ldots, +4\}$ under four conventions: HP with $\lambda = 1{,}600$ and $\lambda = 10^5$; the Hamilton (2018) regression filter (projection of y_t on $y_{t-h}, \ldots, y_{t-h-3}$ with $h = 8$ quarters); and first differences. All series are log-transformed before filtering.

First, $\text{cor}(\delta_t, u_{t+h})$ is positive at $h = 0$ under every specification except $\lambda = 100^5$ (HP-100k: $+0.02$; HP-1600: $+0.39$; Hamilton: $+0.19$; FD: $+0.60$). Second, $\text{cor}(\delta_t, v_{t+h})$ is negative at $h = 0$ and $h = +1$ under HP-1600, Hamilton, and FD, while HP-100k delivers the opposite sign. HP-100k is flagged as an outlier at five of the nine horizons for the vacancy correlation.

Table C.8: Dynamic correlations $\text{cor}(\delta_t, x_{t+h})$ by detrending filter

	x = Unemployment (u)				x = Vacancies (v)			
h	HP-1600	HP-100k	Hamilton	FD	HP-1600	HP-100k	Hamilton	FD
−4	−0.265	−0.411	−0.332	−0.013	0.296	0.557	0.448	0.175
−3	−0.343	−0.435	−0.334	−0.115	0.211	0.490	0.357	0.068
−2	−0.271	−0.365	−0.257	−0.018	0.089	0.372	0.229	0.032
−1	−0.197	−0.309	−0.176	−0.414	−0.029	0.281	0.085	0.137
0	+0.386	+0.018	+0.185	+0.601	−0.225	+0.148	−0.057	−0.143
+1	+0.216	−0.048	+0.172	−0.129	−0.312	+0.065	−0.143	0.003
+2	+0.198	−0.031	+0.158	−0.026	−0.384	−0.005	−0.205	−0.197
+3	+0.201	−0.003	+0.133	−0.016	−0.292	+0.021	−0.144	−0.169
+4	+0.215	+0.031	+0.122	+0.127	−0.058	+0.133	−0.028	0.174

Note: All series log-transformed before filtering. Sample: 1992Q3–2021Q4 ($N = 118$); FD loses one observation. HP-1600: Hodrick-Prescott filter with $\lambda = 1{,}600$. HP-100k: HP filter with $\lambda = 10^5$. Hamilton: regression filter of Hamilton (2018), $h = 8$ quarters, $p = 4$ lags. FD: quarter-on-quarter first differences. Entries in **bold** indicate horizons where HP-100k has the opposite sign from all three other filters.

Appendix D. Steady state

We summarize the steady state of the system below:

$$
\begin{aligned}
\nu^f &= \rho(N) f_e / \mu(N) \\
w^{int} &= \rho(N) z / \mu(N) \\
\frac{d^f}{\nu^f} &= \frac{r + \delta_e}{1 - \delta_e} \\
\kappa + \frac{K}{q(\theta)} &= \frac{1 - \delta_e}{r + \tau}(w^{int} - w - K - s\mathcal{R}) \\
w &= \phi \left[w^{int} - K + \theta(K + q(\theta)\kappa) - s\mathcal{R} \right] + (1 - \phi) b \\
\mathcal{R} &= \frac{1 - \delta_e}{1 + r} \left(Q - Q^{rep} \right), \quad Q^{rep} = Q\Lambda_r - \psi_c Q(\Lambda_r - 1 + p_0), \quad \Lambda_r = F(Q/\alpha) \\
Q &= e^{1/\xi} x_m \\
1 - \delta_e &= (1 - \delta) F(\chi^c), \quad \tau = 1 - (1 - \delta_e)(1 - s) \\
K &= \frac{r + \delta_e}{1 + r} Q
\end{aligned}
$$

$$u = \frac{\tau}{\tau + (1-\delta_e) f(\theta)} \tag{D.1}$$

$$e = \delta_e(v + 1 - u) + (1-\delta_e)(1-\Lambda_r)s(1-u) \tag{D.2}$$

$$N^e = \delta_e N/(1-\delta_e)$$

$$L^c = \frac{r+\delta_e}{r+\delta_e+\delta_e\pi_s\mu}L, \quad L^e = \frac{\delta_e\pi_s\mu}{r+\delta_e+\delta_e\pi_s\mu}L \tag{D.3}$$

$$\chi^c = \frac{\rho f_e}{\mu}\left(\frac{r+\delta_e}{1-\delta_e}\frac{\mu-1}{\mu\pi_s}+1\right)$$

$$\pi_s = \frac{\mu-1}{\mu}\frac{(1-\psi_c\omega)(r+\delta_e)}{r+\delta_e+\psi_c\omega(1-\delta_e)}, \quad \omega = 1 - \frac{(1-p_0)(1-\delta)}{1-\delta_e}$$

$$Y^{Gross} = Y^C + \nu^f N^e = w^{int}L + Nd^f$$

$$Y = C + \nu^f N^e$$

$$X = e\frac{\xi}{\xi+1}Q + \kappa qv$$

$$X^c = N\psi_c\omega\chi^c$$

$$X^r = (1-\delta_e)s(1-u)\,\psi_c Q(\Lambda_r - 1 + p_0)$$

$$Y^c = \rho z L^c = C + X + X^c + X^r$$

Most of these equations are self-evident from the equilibrium system. For instance, evaluating (5) yields (D.1). The forms of Q^{rep} and X^r use the interior closed form of the reactivation-cost integral derived below (34). To find e, start with the steady-state vacancy law of motion:

$$[1-(1-\delta_e)(1-q)]v = (1-\delta_e)\Lambda_r s(1-u) + e$$

Expanding the left-hand side as $\delta_e v + (1-\delta_e)qv$ and rewriting $(1-\delta_e)qv = \tau(1-u)$, we have

$$\delta_e v + \tau(1-u) = (1-\delta_e)\Lambda_r s(1-u) + e$$

$$\delta_e v + [\tau - (1-\delta_e)\Lambda_r s](1-u) = e$$

and since $\tau - (1-\delta_e)\Lambda_r s = \delta_e + (1-\delta_e)(1-\Lambda_r)s$, we obtain (D.2). New entrants replace two kinds of lost positions: those destroyed with exiting product lines, $\delta_e(v+1-u)$, and those retired at surviving product lines because they were not reactivated. With costless reposting ($\Lambda_r = 1$) only the first remains, $e = \delta_e(v+1-u)$, and the contribution of entrants to the vacancy pool is governed by the total exit rate δ_e alone.

At $p_0 = 0$, where $X^c = 0$, labor in entry follows directly from the resource constraint curve:

$$\begin{aligned} L^e &= \frac{\delta_e}{1-\delta_e} N f_e / z \\ &= \frac{\delta_e}{1-\delta_e} \left[\frac{(\mu-1) z L (1-\delta_e)}{f_e(\delta_e \mu + r)} \right] f_e / z \\ &= \frac{\delta_e(\mu-1)}{\delta_e \mu + r} L \end{aligned}$$

The steady-state job creation and wage equations can be arranged as

$$\kappa + \frac{K}{q} = \frac{1-\delta_e}{r+\tau} \left[(1-\phi)(w^{int} - K - b) - \phi\theta q \left(\kappa + \frac{K}{q} \right) - (1-\phi) s \mathcal{R} \right] \tag{D.4}$$

Equation (D.4) equates the average recruiting cost to the present-discounted value of the match, adjusted for the wage bargain. The right-hand side incorporates a fraction $1-\phi$ of the flow surplus $(w^{int} - K - b)$, an adjustment for the outside option of the worker, and the recruiter's share $1-\phi$ of the expected reposting shortfall, which vanishes with costless reposting. Whereas in the standard DMP model, the analogous expression would characterize tightness, here K and w^{int} are additional endogenous variables. Moreover, the average recruiting cost has the compound form $\kappa + K/q$, and discounting is adjusted by the overall firm survival rate $1 - \delta_e$.

Appendix E. Derivations

Appendix E.1. Relationship between profit share and power law shape parameter ψ

Write $\omega \equiv 1 - (1-p_0)(1-\delta)/(1-\delta_e)$ for the steady-state share of surviving lines that drew a positive continuation cost, $p_0(\chi^c/\chi_m)^\psi/\Lambda$ with $\Lambda = (1-\delta_e)/(1-\delta)$, so that $X^c = N\psi_c \omega \chi^c$ by (33). The cutoff satisfies $\chi^c = Y^c(\mu-1)/(\mu N) + \rho f_e/\mu$, so $X^c = \psi_c\omega\big(Y^c(\mu-1)/\mu + N\rho f_e/\mu\big)$. Dividing by $Y^c = \rho z L^c$,

$$\frac{X^c}{Y^c} = \frac{\psi_c \omega}{\mu} \left(\mu - 1 + \frac{f_e N}{z L^c} \right).$$

The steady-state business formation Euler equation gives $d^f/\nu^f = (r+\delta_e)/(1-\delta_e)$, and with $N d^f = \pi_s Y^c$ this is $f_e N/(zL^c) = \mu\pi_s(1-\delta_e)/(r+\delta_e)$. Since $\pi_s = (\mu-1)/\mu - X^c/Y^c$,

$$\pi_s = \frac{\mu-1}{\mu}(1-\psi_c\omega) - \frac{\psi_c\omega(1-\delta_e)\pi_s}{r+\delta_e},$$

and solving for π_s,

$$\pi_s = \frac{\mu-1}{\mu} \frac{(1-\psi_c\omega)(r+\delta_e)}{r+\delta_e+\psi_c\omega(1-\delta_e)}. \tag{E.1}$$

Holding δ_e fixed, π_s falls with $\psi_c\omega$. For $\psi_c\omega = 0$, $\pi_s = (\mu-1)/\mu = 1/\varepsilon$, as in Bilbiie, Ghironi, and Melitz (2012). The cutoff in terms of the profit share is

$$\chi^c = \frac{\rho}{\mu}\left(\frac{zL^c(\mu-1)}{N} + f_e\right) = \frac{\rho f_e}{\mu}\left(\frac{r+\delta_e}{1-\delta_e}\frac{\mu-1}{\mu\pi_s} + 1\right) = \frac{\rho f_e}{\mu}\frac{1+r}{(1-\delta_e)(1-\psi_c\omega)},$$

which is (57).

Appendix E.2. Resource constraint curve

Write $c \equiv \psi_c\omega$. Rearranging (56) as $N = zL(1-\delta_e)/\big[f_e\big((r+\delta_e)/(\mu\pi_s)+\delta_e\big)\big]$ and substituting (E.1),

$$N = \frac{(\mu-1)(1-c)\,zL(1-\delta_e)}{f_e\big[\delta_e\mu + r + c(1-\mu\delta_e)\big]},$$

which reduces to (61) at $c = 0$. Labor in entry and in the consumption sector are then

$$L^e = \frac{\delta_e}{1-\delta_e}\frac{Nf_e}{z} = \frac{\delta_e(\mu-1)(1-c)}{\delta_e\mu + r + c(1-\mu\delta_e)}L, \qquad L^c = \frac{r+\delta_e+c(1-\delta_e)}{\delta_e\mu + r + c(1-\mu\delta_e)}L,$$

consistent with (D.3), and the continuation-cost share of gross consumption output is

$$\frac{X^c}{Y^c} = \frac{c\,(\mu-1)(1+r)}{\mu\big[r+\delta_e+c(1-\delta_e)\big]}.$$

Appendix E.3. Labor share of GDP

The aggregate resource constraint can be rearranged to pin down the income of recruiters:

$$w^{int}L = \frac{Y^c}{\mu} + \nu^f N^e$$

Thus, the income share of recruiters relative to gross output is

$$\begin{aligned}\frac{w^{int}L}{Y^{Gross}} &= \left(\frac{r+\delta_e}{r+\delta_e+\delta_e\pi_s}\right)\frac{1}{\mu} + \frac{\delta_e\pi_s}{r+\delta_e+\pi_s} \\ &= \frac{r+\delta_e+\delta_e\mu\pi_s}{\mu(r+\delta_e+\delta_e\pi_s)}\end{aligned}$$

The labor share of GDP is

$$\frac{wL}{Y} = (1 + \frac{X+X^c+X^r}{Y})\frac{w}{w^{int}}\frac{r+\delta_e+\delta_e\mu\pi_s}{\mu(r+\delta_e+\delta_e\pi_s)}$$

The steady-state labor share is composed of three factors. The first factor just reflects the wedge between gross output and GDP, which depends on the recruiting, fixed, and reactivation cost shares. The second and third comprise a search wedge w/w^{int} and a market power wedge

$w^{int}L/Y^{Gross}$. Through the latter, the labor share depends positively on δ_e and ρ, and negatively on π_s. Intuitively, a higher product destruction rate requires more labor resources, and thus income accruing to labor. A higher price markup raises profits and thus lowers the labor share. Indeed, in standard DSGE models with monopolistic competition, the labor share just the inverse of the price markup. Here that arises as a limiting case in which search frictions vanish ($w \to w^{int}$) and $\delta \to 0$. The latter entails eliminating the entry sector, which affects the labor share since only retailers charge markups. Note that the search wedge, in turn, depends on ϕ and b. Moreover, variety effects affect the search wedge through w^{int} but otherwise do not affect the steady-state shares.

The special case studied by Bilbiie, Ghironi, and Melitz (2012) arises if we shut down labor market frictions and fixed costs in production. In this limit $\delta_e = \delta$ (no endogenous exit), $w^{int} = w$, $X = X^c = 0$, and $\pi_s = 1/\varepsilon$, giving

$$\frac{wL}{Y} = \frac{(r+\delta)(\varepsilon-1)+\delta}{\varepsilon(r+\delta)+\delta} = \frac{C}{Y}\frac{1}{\mu} + \frac{\nu^f N^e}{C}$$

which equals the consumption share divided by the markup plus the entry share.

Appendix E.4. Wage determination

We derive the Nash wage (22). Start from the value of the large household:

$$W_t = \max_{c_t, a_{t+1}} \frac{c_t^{1-\sigma}}{1-\sigma} + \beta \mathbb{E}_t W_{t+1} \quad \text{s.t.}$$
$$c_t + \nu_t a_{t+1} = w_t L_t + b(1-L_t) + (\nu_t + d_t)a_t + D_t^{int} - T_t$$

which is (10) with $L_t = 1 - u_t$. Given the continuum of members in a household, the surplus of moving an unemployed member into employment is $\frac{\partial W_t}{\partial L_t}$. Nash bargaining solves

$$w_t = \arg\max \left[\frac{\partial W_t(w_t)}{\partial L_t}\right]^{\phi} [J_t(w_t) - Q_t]^{1-\phi} \tag{E.2}$$

For convenience, define the match surpluses $S_t^H \equiv \frac{\partial W_t}{\partial L_t}$ and $S_t^J \equiv J_t - Q_t$.

By the envelope theorem,

$$S_t^H = c_t^{-\sigma}\frac{\partial c_t}{\partial L_t} + \beta \mathbb{E}_t \left[S_{t+1}^H \frac{\partial L_{t+1}}{\partial L_t}\right]$$

From the budget constraint, $\frac{\partial c_t}{\partial L_t} = w_t - b$. Employment evolves as

$$L_{t+1} = (1-\delta_{e,t+1})\big[(1-s_t)L_t + f(\theta_t)(1-L_t)\big] \tag{E.3}$$

A match survives into $t+1$ only if its product line survives both exogenous destruction and the continuation-cost draw, with probability $1-\delta_{e,t+1} = (1-\delta_t)F(\chi^c_{t+1})$ from (3). Matches formed at t face product-line exit but not match separation before they first produce. This is the timing of the recruiter values (13) and (16) and of the vacancy law of motion (6). Partial reposting does not enter (E.3): it governs whether a vacated position returns as a vacancy, not whether the worker separates. Hence

$$\frac{\partial L_{t+1}}{\partial L_t} = (1-\delta_t)F(\chi^c_{t+1})\,(1-s_t-f(\theta_t))$$

and, since s_t, δ_t, and θ_t are known at t,

$$S^H_t = c_t^{-\sigma}(w_t-b) + (1-s_t-f(\theta_t))\,\beta(1-\delta_t)\mathbb{E}_t\left[F(\chi^c_{t+1})S^H_{t+1}\right] \tag{E.4}$$

The first-order condition of (E.2) is

$$\phi\frac{\frac{\partial S^H_t}{\partial w_t}}{S^H_t} + (1-\phi)\frac{\frac{\partial S^J_t}{\partial w_t}}{S^J_t} = 0$$

The next-period wage is re-bargained under next-period conditions, so the continuation term in (E.4) does not move with w_t and $\frac{\partial S^H_t}{\partial w_t} = c_t^{-\sigma}$. From (19), the recruiter surplus is

$$S^J_t = w^{int}_t - w_t - K_t + (1-s_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right) - s_t\mathcal{R}_t \tag{E.5}$$

where $\mathcal{R}_t$ is the expected discounted reposting shortfall (20). It depends only on period-$t+1$ objects, so $\frac{\partial S^J_t}{\partial w_t} = -1$, and the first-order condition gives the sharing rule

$$S^H_t = \frac{\phi}{1-\phi}c_t^{-\sigma}S^J_t \tag{E.6}$$

Forwarding (E.6) one period and using $m_{t+1} = \beta(c_{t+1}/c_t)^{-\sigma}$,

$$\beta(1-\delta_t)\mathbb{E}_t\left[F(\chi^c_{t+1})S^H_{t+1}\right] = \frac{\phi}{1-\phi}c_t^{-\sigma}\,\mathbb{E}_t m_{t+1}(1-\delta_t)F(\chi^c_{t+1})S^J_{t+1} = \frac{\phi}{1-\phi}c_t^{-\sigma}\left(\kappa+\frac{K_t}{q(\theta_t)}\right)$$

where the second equality is the job creation condition (21). Substituting into (E.4),

$$S^H_t = c_t^{-\sigma}\left[w_t - b + (1-s_t-f(\theta_t))\,\frac{\phi}{1-\phi}\left(\kappa+\frac{K_t}{q(\theta_t)}\right)\right]$$

Equating (E.5) with S^J_t from the sharing rule (E.6) gives

$$w^{int}_t - w_t - K_t + (1-s_t)\left(\kappa+\frac{K_t}{q(\theta_t)}\right) - s_t\mathcal{R}_t = \frac{1-\phi}{\phi}(w_t-b) + (1-s_t-f(\theta_t))\left(\kappa+\frac{K_t}{q(\theta_t)}\right)$$

Solving for the wage,

$$w_t = \phi\left(w_t^{int} - K_t + f(\theta_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right) - s_t\mathcal{R}_t\right) + (1-\phi)b$$

The household surplus carries no reposting term: a separated worker becomes unemployed whether or not the recruiter reactivates the vacated position. The shortfall therefore enters bargaining only through the recruiter's surplus, and the worker bears share ϕ of it through a lower wage. The term acts like a firing cost, except that separation is exogenous. Costless reposting ($\alpha \to 0$ or $p_0 = 0$) gives $\mathcal{R}_t = 0$ and recovers the wage without reposting.

Substituting the wage back into (E.5) gives the wage-substituted surplus

$$S_t^J = (1-\phi)\left(w_t^{int} - K_t - b\right) - \phi\theta_t\left(K_t + q(\theta_t)\kappa\right) + (1-s_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right) - (1-\phi)s_t\mathcal{R}_t$$

The recruiter bears only its bargaining share $1-\phi$ of the reposting shortfall.

Appendix E.5. Log-linearized propagation equations

This appendix collects log-linearized versions of the equations that govern impulse-response dynamics. Let a hat denote the log-deviation from the deterministic steady state: $\hat{x}_t \equiv \ln(x_t/\bar{x})$, so $x_t \approx \bar{x}(1+\hat{x}_t)$. For rates that are small at steady state ($\delta_e, s, \tau, u, v \ll 1$), we use the level deviation $\tilde{x}_t \equiv x_t - \bar{x}$ directly and note $\tilde{x}_t \approx \bar{x}\hat{x}_t$. We present equations in the order in which they feed into one another: the exit margin first, then the variety stock, the labor market, and finally the vacancy-value chain. The equations describe the model with costless reposting ($\alpha \to 0$), which is the model solved for the comparisons in Section 5.3. Partial reposting adds $\bar{\Lambda}_r$ to the reposting inflow in the vacancy law of motion and the shortfall $\mathcal{R}_t$ to the recruiter surplus.

1. Endogenous exit rate. The survival probability is $\Lambda_t \equiv F(\chi_t^c) = (1-p_0) + p_0(\chi_t^c/\chi_m)^\psi$, and the total exit rate is (3):

$$\delta_{e,t} = 1 - (1-\delta_{t-1})\,\Lambda_t,$$

which in steady state is $\bar{\delta}_e = 1-(1-\bar{\delta})\bar{\Lambda}$. Define $\varsigma \equiv \left(\frac{\bar{\chi}^c}{\chi_m}\right)^\psi$, so that $\bar{\Lambda} = (1-p_0) + p_0\varsigma$ and $1-\bar{\Lambda} = p_0(1-\varsigma)$.

Totally differentiating around steady state:

$$d\delta_{e,t} \;=\; \bar{\Lambda}\, d\delta_{t-1} \;-\; (1-\bar{\delta})\,\frac{d\Lambda_t}{d\chi_t^c}\, d\chi_t^c.$$

The derivative of Λ with respect to χ^c at steady state is

$$\left.\frac{d\Lambda}{d\chi^c}\right|_{\bar{\chi}^c} \;=\; \frac{p_0\,\psi\,\varsigma}{\bar{\chi}^c}.$$

Dividing through by $\bar{\delta}_e$ and writing $d\delta_{t-1}/\bar{\delta} = \hat{\delta}_{t-1}$, $d\chi^c/\bar{\chi}^c = \hat{\chi}_{c,t}$:

$$\hat{\delta}_{e,t} \;=\; \frac{\bar{\Lambda}}{\bar{\delta}_e}\,\bar{\delta}\,\hat{\delta}_{t-1} \;-\; \frac{(1-\bar{\delta})\,p_0\psi\varsigma}{\bar{\delta}_e}\,\hat{\chi}_{c,t}. \tag{E.7}$$

Define the *exit elasticity* ε_x as the elasticity of the endogenous exit probability $(1-\Lambda_t)$ with respect to χ^c:

$$\varepsilon_x \;\equiv\; \frac{d(1-\Lambda)}{d\chi^c}\,\frac{\bar{\chi}^c}{1-\bar{\Lambda}} \;=\; \frac{p_0\psi\varsigma}{p_0(1-\varsigma)} \;=\; \frac{\psi\varsigma}{1-\varsigma}.$$

Since $p_0(1-\varsigma) = 1-\bar{\Lambda}$, equation (E.7) becomes

$$\hat{\delta}_{e,t} \;=\; \frac{\bar{\Lambda}}{\bar{\delta}_e}\,\bar{\delta}\,\hat{\delta}_{t-1} \;-\; \frac{(1-\bar{\delta})(1-\bar{\Lambda})}{\bar{\delta}_e}\,\varepsilon_x\,\hat{\chi}_{c,t}.$$

Both coefficients are positive and $\hat{\chi}^c$ enters with a negative sign: a higher continuation threshold raises survival and lowers $\delta_{e,t}$. At the baseline calibration $\varepsilon_x = 5$, meaning a 1 % rise in χ^c raises the endogenous exit probability $(1-\Lambda_t)$ by 5 %.

2. Continuation threshold. From the cutoff condition (32) with μ constant and $\rho_t = N_t^{1/(\varepsilon-1)}$:

$$\hat{\chi}_{c,t} = \underbrace{\alpha_\pi\left(\hat{Y}_{c,t} - \hat{N}_t\right)}_{\text{profit component}} + \underbrace{(1-\alpha_\pi)\,\frac{1}{\varepsilon-1}\hat{N}_t}_{\text{continuation-value component}},$$

where $\alpha_\pi \equiv \bar{R}^f/\bar{\chi}^c \in (0,1)$ is the steady-state share of per-firm gross profit in the threshold. The profit component (elasticity $\alpha_\pi \cdot \frac{\varepsilon-2}{\varepsilon-1} < 0$ in N for $\varepsilon > 2$) rises when N falls, reflecting the market-share gain to surviving firms under DS-CES demand. The continuation-value component (elasticity $(1-\alpha_\pi)/(\varepsilon-1) > 0$ in N) falls when N falls, because the option value $\nu_f = \rho f_e/\mu$ shrinks with ρ. The two forces partially offset: whether $\hat{\chi}^c$ rises or falls in response to $\hat{N} < 0$ depends on calibrated α_π and ε.

3. Firm stock (variety LOM). Log-linearizing the firm law of motion $N_{t+1} = (1-\delta_{e,t})(N_t + N_t^e)$ around steady state (using $\bar{N}^e/\bar{N} = \bar{\delta}_e/(1-\bar{\delta}_e)$):

$$\hat{N}_{t+1} = (1-\bar{\delta}_e)\,\hat{N}_t \;+\; \bar{\delta}_e\,\hat{N}_t^e \;-\; \hat{\delta}_{e,t}, \tag{E.8}$$

where the last term uses $(1 + \bar{N}^e/\bar{N}) \approx 1/(1 - \bar{\delta}_e)$ so the exit rate enters with unit coefficient in absolute log-deviation terms. The three forces are: (i) *mean reversion* at rate $\bar{\delta}_e$ — the survival coefficient $(1 - \bar{\delta}_e)$ governs how long any N deviation persists; (ii) *entry amplification* — the loading $\bar{\delta}_e$ on $\hat{N}^e$ is larger when steady-state churning is higher, so the same entry impulse produces a larger N response; (iii) the *exit margin* $\hat{\delta}_{e,t}$ from (E.7), which feeds back through χ^c. Combining (i) and (ii): models with higher $\bar{\delta}_e$ show larger amplitude but faster convergence; models with lower $\bar{\delta}_e$ show smaller but more persistent deviations. This is the common mechanism underlying Comparisons A and B.

4. Interior wage (recruiter compensation). From $w_t^{int} = \rho_t z_t/\mu$ (27) with $\rho_t = N_t^{1/(\varepsilon-1)}$:

$$\hat{w}_t^{int} = \frac{1}{\varepsilon - 1}\,\hat{N}_t + \hat{z}_t. \tag{E.9}$$

This is the *variety channel*: a contraction of the firm stock N reduces the relative price ρ (fewer varieties lower the CES price index), compressing the marginal revenue product of labor w^{int} even holding productivity constant. The elasticity $1/(\varepsilon - 1) \approx 0.30$ at $\varepsilon = 4.3$ is moderate but cumulates over the slow N recovery.

5. Aggregate separation rate. From $\tau_t = \delta_{e,t} + s_{t-1}(1 - \delta_{e,t})$ (4), log-linearizing:

$$\hat{\tau}_t = (1 - \bar{s})\frac{\bar{\delta}_e}{\bar{\tau}}\,\hat{\delta}_{e,t} + \frac{\bar{s}(1 - \bar{\delta}_e)}{\bar{\tau}}\,\hat{s}_{t-1},$$

where the coefficients sum to $1 - \bar{s}\bar{\delta}_e/\bar{\tau}$, which is close to one, since $\bar{\tau} = \bar{\delta}_e + \bar{s}(1 - \bar{\delta}_e)$. With the calibrated ratio $\bar{\delta}_e/\bar{\tau} \approx 0.21$, firm exit contributes about one-fifth of aggregate separation variance; match-level separations account for the rest.

6. Unemployment LOM. From (5), log-linearizing around $(\bar{u}, \bar{\tau}, \bar{f})$:

$$\tilde{u}_t = \big(1 - (1 - \bar{\delta}_e)\bar{f} - \bar{\tau}\big)\,\tilde{u}_{t-1} + (1 - \bar{u})\,\tilde{\tau}_t + \bar{u}\bar{f}\,\tilde{\delta}_{e,t} - \bar{u}(1 - \bar{\delta}_e)\,\tilde{f}_{t-1}, \tag{E.10}$$

where $\tilde{x}_t = x_t - \bar{x}$ denotes the level deviation and $\tilde{f}_t = f(\theta_t) - \bar{f}$. The term in $\tilde{\delta}_{e,t}$ is the exit exposure of matches formed in $t - 1$, which have not yet produced. The autoregressive coefficient $(1 - (1 - \bar{\delta}_e)\bar{f} - \bar{\tau})$ governs unemployment persistence: with monthly $\bar{f} \approx 0.40$ and $\bar{\tau} \approx 0.031$, it equals approximately 0.57, giving an unemployment half-life of around two months in the absence of sustained τ or θ shocks. A δ shock is persistent precisely because it generates a sustained $\tilde{\tau}_t > 0$ through $\tilde{\delta}_{e,t} > 0$ and, via the variety channel (E.9), a sustained $\tilde{f}_{t-1} < 0$ through the depressed JCC.

7. Vacancy LOM. From (6), splitting into predetermined and new-entrant components:

$$\tilde{v}_t = (1-\bar{\delta}_e)\Big[(1-\bar{q})\,\tilde{v}_{t-1} + \bar{s}(1-\bar{u})\,\hat{s}_{t-1}\Big] - \bar{v}^{\text{pre}}\,\tilde{\delta}_{e,t} + \tilde{e}_t, \tag{E.11}$$

where $\bar{v}^{\text{pre}} = \bar{v} - \bar{e}$ is the steady-state predetermined vacancy stock. As in (E.10), δ_e enters in level deviations and s in log deviations, matching their coefficients: $\partial[(1-\delta_e)x]/\partial\delta_e = -x$, while the reposting term carries $\bar{s}$ already. The three terms capture: (i) *survival of existing vacancies*, which depends on $(1-\bar{\delta}_e)$ and amplifies the persistence asymmetry between high- and low-δ_e specifications; (ii) *destruction impulse* $\hat{\delta}_{e,t}$, which directly destroys the predetermined stock; and (iii) *new entrant vacancies* $\tilde{e}_t$, which respond contemporaneously to the vacancy-value signal Q_t. The sunk-cost predetermination means that $\tilde{v}_t^{\text{pre}}$ cannot adjust within the period, so the vacancy response to a δ shock is asymmetrically slow relative to unemployment, consistent with the empirical Beveridge curve dynamics in Section 4.4.

8. Vacancy value and new postings. From (43), log-linearizing under a one-shock approximation ($\mathbb{E}_t\hat{Q}_{t+1} = \rho_\omega \hat{Q}_t$, exact by undetermined coefficients for shock $\omega \in \{z, \delta\}$ with persistence ρ_ω):

$$\hat{Q}_t = \underbrace{\frac{r+\bar{\delta}_e}{1+r-\rho_\omega(1-\bar{\delta}_e)}}_{\mathcal{D}(\bar{\delta}_e,\,\rho_\omega)}\,\hat{K}_t, \tag{E.12}$$

where r is the steady-state discount rate. The duration multiplier $\mathcal{D}$ is strictly increasing in $\bar{\delta}_e$ (since $\partial\mathcal{D}/\partial\bar{\delta}_e = (1+r)(1-\rho_\omega)/\text{den}^2 > 0$ for $\rho_\omega < 1$) and decreasing in shock persistence ρ_ω. Intuitively, Q is the present value of the per-period dividend K: higher $\bar{\delta}_e$ shortens the effective duration of the vacancy asset, amplifying its proportional response to the same $\hat{K}$ innovation. New vacancy postings respond via (44):

$$\hat{e}_t = \xi\,\hat{Q}_t = \xi\,\mathcal{D}(\bar{\delta}_e, \rho_\omega)\,\hat{K}_t, \tag{E.13}$$

so the entry vacancy margin inherits both the duration amplification from $\mathcal{D}$ and the convexity of the posting cost through ξ.

9. Entry elasticity and the K decomposition. Equation (E.12) is derived holding $\delta_{e,t+1}$ fixed and so applies exactly to z shocks where $\hat{\delta}_{e,t+1} = 0$ on impact. For δ shocks, log-linearizing the full K definition (43), $K_t = Q_t - \beta(1-\delta_{e,t+1})Q_{t+1}$, with $\delta_{e,t+1}$ treated as a dynamic variable and imposing $\hat{Q}_{t+1} = \rho_\omega\hat{Q}_t$ gives

$$\hat{K}_t \;=\; \underbrace{\frac{\bar{\delta}_e}{r+\bar{\delta}_e}\,\hat{\delta}_{e,t+1}}_{(+)\text{ duration}} \;+\; \underbrace{\frac{1}{\mathcal{D}(\bar{\delta}_e,\rho_\omega)}\,\hat{Q}_t}_{\text{vacancy value}}, \tag{E.14}$$

or, rearranged for the vacancy value,

$$\hat{Q}_t \;=\; \mathcal{D}(\bar{\delta}_e,\rho_\omega)\,\hat{K}_t \;-\; \underbrace{\frac{\bar{\delta}_e}{1+r-\rho_\delta(1-\bar{\delta}_e)}}_{\phi_D(\bar{\delta}_e,\rho_\delta)>0}\hat{\delta}_{e,t+1}, \tag{E.15}$$

which reduces to (E.12) when $\hat{\delta}_{e,t+1}=0$. Equations (E.14) and (E.15) are the same relation: $\phi_D=\mathcal{D}\,\bar{\delta}_e/(r+\bar{\delta}_e)$ identically. It is an asset-pricing identity in two unknowns, so it signs neither $\hat{K}_t$ nor $\hat{Q}_t$ on its own. All it says is that $\hat{Q}_t>0$ requires $\hat{K}_t>[\bar{\delta}_e/(r+\bar{\delta}_e)]\hat{\delta}_{e,t+1}$, and reading the first display as raising $\hat{K}$ for given $\hat{Q}$ and then the second as raising $\hat{Q}$ for that $\hat{K}$ would use one equation twice. The second relation is the job-creation condition.

9a. The dividend from the job-creation condition. Let $C_t\equiv\kappa+K_t/q(\theta_t)$ be the expected cost of filling a position and S_{t+1} the bracket on the right of (42), so that $C_t=\mathbb{E}_t m_{t+1}(1-\delta_{e,t+1})S_{t+1}$ and $\bar{C}=\beta(1-\bar{\delta}_e)\bar{S}$. With $q(\theta)\propto\theta^{-\eta_L}$, so $\hat{q}_t=-\eta_L\hat{\theta}_t$, log-linearizing gives

$$\hat{K}_t \;=\; \underbrace{-\eta_L\hat{\theta}_t}_{\text{congestion relief}} \;-\; \underbrace{\frac{1}{\chi_K}\frac{\bar{\delta}_e}{1-\bar{\delta}_e}\,\hat{\delta}_{e,t+1}}_{\text{survival}} \;+\; \underbrace{\frac{1}{\chi_K}\Big(\hat{S}_{t+1}+\hat{m}_{t+1}\Big)}_{\text{surplus}}, \qquad \chi_K\equiv\frac{\bar{K}/\bar{q}}{\kappa+\bar{K}/\bar{q}} \tag{E.16}$$

A destruction shock moves all three terms. The *survival* term is negative and is the direct discounting effect: a position more likely to be withdrawn is worth less, which makes the vacancy asset cheaper, not dearer. The *surplus* term is also negative, since $w^{int}_{t+1}=\rho(N_{t+1})z_{t+1}/\mu(N_{t+1})$ falls with variety and the continuation terms fall with it. The one positive force is *congestion relief*: destruction raises unemployment and removes vacancies, so $\hat{\theta}_t<0$, a newly posted position fills faster, and the dividend a recruiter can support rises. This, and not duration, is what can make entry lean against the shock. Congestion relief enters with the matching elasticity η_L while the survival term carries $1/\chi_K$, so the cushion is weaker the smaller the posting cost is as a share of total hiring costs.

Combining (E.16) with (E.15) and the entry schedule (E.13) gives the entry response to a destruction shock,

$$\hat{e}^{\delta}_t \;=\; \xi\,\mathcal{D}\,\hat{K}^{\delta}_t \;-\; \xi\,\phi_D\,\hat{\delta}_{e,t+1}, \tag{E.17}$$

now with $\hat{K}^{\delta}$ supplied by (E.16) rather than assumed. Two of its three components are negative, so $\hat{e}^{\delta}>0$ requires congestion relief to dominate. The sign is quantitative and we read it from the solved model rather than from these coefficients; in the simulations of Appendix F.1 it does

dominate at the baseline calibration, giving $\hat{e}^{\delta} > 0$. For a technology shock the survival term is absent, a negative shock gives $\hat{K}^{z} < 0$ through the surplus channel, and $\hat{e}^{z} < 0$: entry contracts. The signs therefore differ across shocks — $\hat{e}^{\delta} > 0$, $\hat{e}^{z} < 0$ — and the role this plays in shaping amplification is discussed in Appendix F.

9b. Free entry. The exception is the free-entry limit $\xi \to \infty$, where the entry technology pins $Q = x_m$ so that $\hat{Q}_t = 0$. Then (E.14) alone signs the dividend, which is the user cost (52), and tightness falls. With $\sigma = \kappa = 0$ the labor market block is then block recursive, and Lemma 1 and Proposition 5 solve it in closed form, including the role of $\bar{\delta}_e/\bar{\tau}$ in the sign of the vacancy response.

Transmission chain summary. The equations above trace two distinct propagation chains from any aggregate shock. For a *destruction shock* ($\delta \uparrow$), the primary channel is direct vacancy and firm destruction, partially cushioned by an entry expansion driven by congestion relief:

$$\hat{\delta} \uparrow \xrightarrow{(E.7)} \hat{\delta}_e \uparrow \xrightarrow{(E.10),(E.11)} \tilde{u} \uparrow,\ \tilde{v} \downarrow \text{ (direct)} \xrightarrow{(E.16)} \hat{\theta} \downarrow \Rightarrow \hat{K} \uparrow \xrightarrow{(E.15)} \hat{Q} \uparrow \xrightarrow{(E.17)} \hat{e} \uparrow \xrightarrow{(E.11)} \tilde{v} \uparrow \text{ (partial off}$$

The secondary channel prolongs the unemployment response through variety:

$$\hat{\delta}_e \uparrow \xrightarrow{(E.8)} \hat{N} \downarrow \xrightarrow{(E.9)} \hat{w}^{int} \downarrow \xrightarrow{\text{JCC}} \hat{\theta} \downarrow \xrightarrow{(E.10)} \tilde{u} \uparrow \text{ (amplified).}$$

For a *technology shock* ($z \downarrow$):

$$\hat{z} \downarrow \xrightarrow{(E.9)} \hat{w}^{int} \downarrow \xrightarrow{\text{JCC}} \hat{K} \downarrow,\ \hat{\theta} \downarrow \xrightarrow{(E.12)} \hat{Q} \downarrow \xrightarrow{(E.13)} \hat{e} \downarrow \xrightarrow{(E.11)} \tilde{v} \downarrow \xrightarrow{(E.8)} \hat{N} \downarrow \xrightarrow{(E.9)} \hat{w}^{int} \downarrow \text{ (amplified).}$$

In both chains, the $N \to w^{int}$ link (E.9) generates a second-round amplification absent in models without product variety (Comparison C). The entry sign is opposite across shocks — $\hat{e}^{\delta} > 0$ cushions the direct vacancy destruction while $\hat{e}^{z} < 0$ compounds the JCC-driven vacancy contraction — a distinction that drives the asymmetric role of ξ documented in Appendix F.

Appendix F. Impulse responses: additional comparisons

Appendix F.1. Comparison D: role of entry elasticity (ξ)

Figures F.17 and F.18 compare impulse responses under the baseline ($\xi_{\text{inv}} = 1.0$, solid black) and a near-free-entry counterfactual ($\xi_{\text{inv}} = 0.1$, dashed blue). Both calibrations match the same empirical targets; ξ_{inv} is the only structural difference, so all parameter changes — lower

Q and K, higher Nash share ϕ — are endogenous to the recalibration. Sign conventions follow Comparisons A–C: the z figure shows a *negative* technology shock and the δ figure shows a positive destruction shock, placing both on a scale where unemployment rises. The log-linearized equations underlying the discussion are derived in Appendix E.5, equations (E.14)–(E.17).

The entry response $e_t = G(Q_t)$ is the mechanism that connects the model to Coles and Moghaddasi Kelishomi (2018). As shown in Remark 1, setting $s_t = 0$ and $\sigma = \kappa = 0$ recovers the Coles and Moghaddasi Kelishomi model exactly. In that limit, all vacancies arise from new postings, and ξ governs the elasticity of entry to the vacancy value Q. For destruction shocks the shock slackens the market, so a new position fills faster and K and Q rise, generating an entry surplus that partially cushions the direct vacancy destruction. The parameter ξ controls the strength of this cushion: higher ξ (lower ξ_{inv}) amplifies the entry response and weakens the net downward pressure on vacancies. Whether u and v comove positively or negatively after a destruction shock therefore depends on ξ — and, as discussed below, on the level of $\bar{\delta}_e$. Empirically, u and v comove *negatively* following destruction shocks (Beveridge curve recessions), which requires the direct vacancy destruction to dominate the entry cushion. This places a lower bound on ξ_{inv}: too close to free entry, and the model generates the wrong sign of u-v comovement for δ shocks.

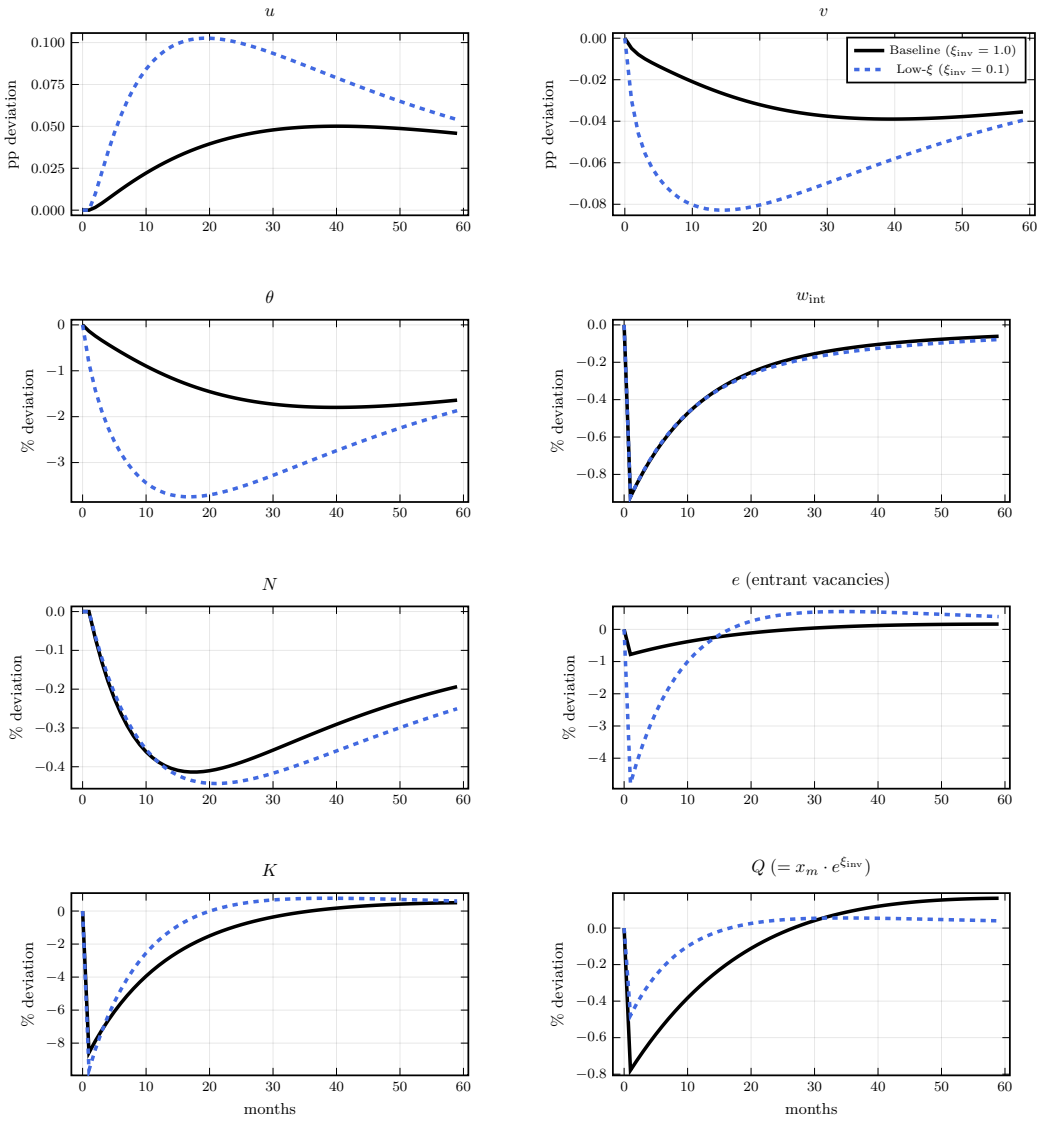


Figure F.17: Comparison D — impulse responses to a one-standard-deviation *negative* technology shock. Solid black: baseline ($\xi_{\text{inv}} = 1.0$). Dashed blue: near-free-entry ($\xi_{\text{inv}} = 0.1$, recalibrated). Row 1 in percentage-point level deviation from steady state; rows 2–4 in $100 \times$ log deviation.

Under the *negative technology shock* (Figure F.17), lowering ξ_{inv} substantially amplifies the labor-market response. The mechanism runs through entry. A negative z shock depresses w^{int}, compressing the JCC surplus and lowering K. Via (E.12), Q falls, and by (E.13) new vacancy postings contract: $\hat{e}^z < 0$. With the near-constant cost schedule ($\xi_{\text{inv}} = 0.1$, $\xi = 10$), any given $\hat{Q}$ movement generates a ten-fold larger $\hat{e}$ response. The realized amplification of entry is somewhat smaller than ten-fold — about six — because the flatter cost schedule also damps $\hat{Q}$ itself (0.48% versus 0.78%): with entry costs nearly constant, the vacancy value has less room to move. The net effect is still a far sharper collapse in postings. The vacancy pool contracts

further, θ falls more, and the $N \to \rho \to w^{int}$ feedback loop — N falling faster, $\rho = N^{1/(\varepsilon-1)}$ contracting, w^{int} falling further — amplifies unemployment substantially relative to the baseline.

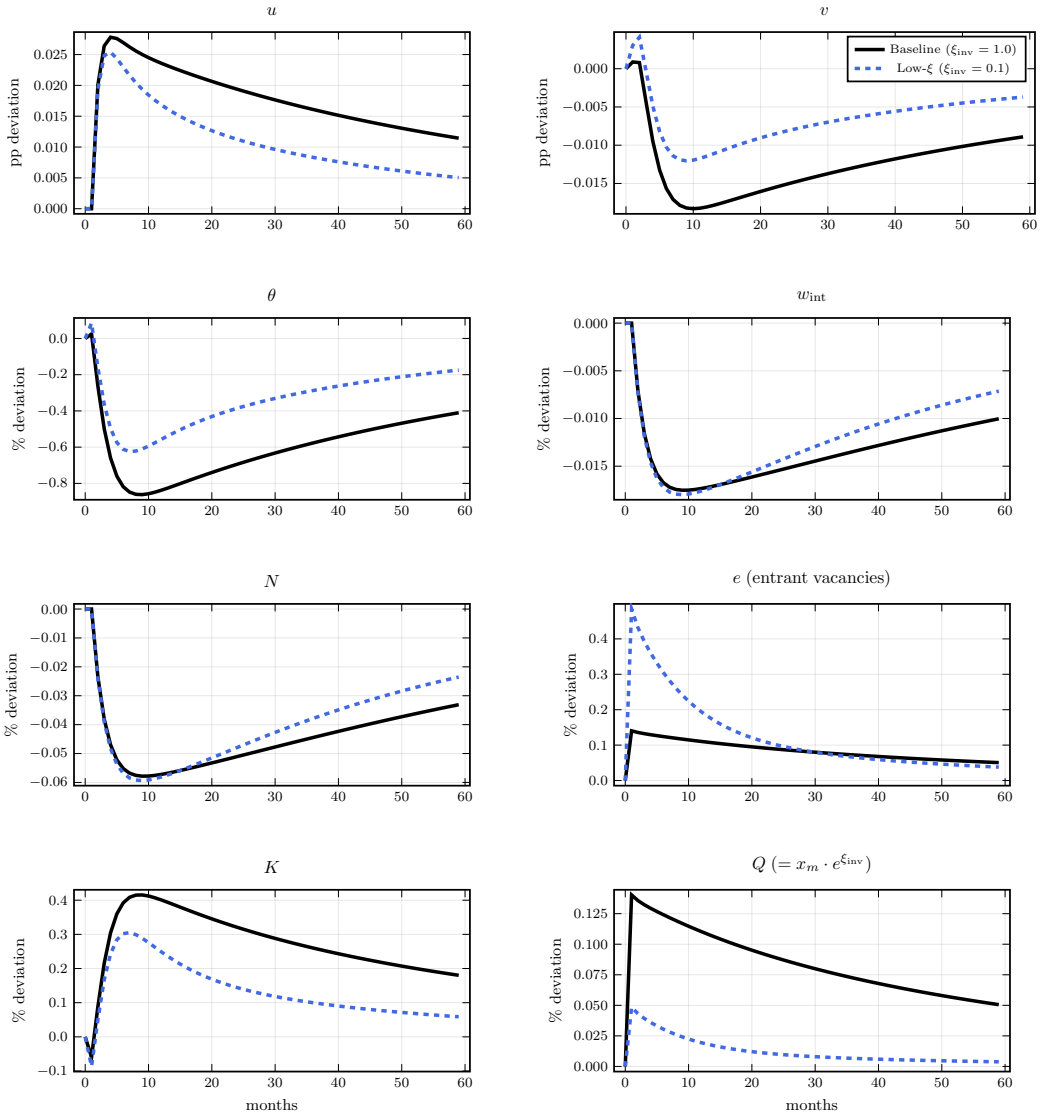


Figure F.18: Comparison D — impulse responses to a one-standard-deviation positive destruction shock. Lines and units as in Figure F.17.

Under the *destruction shock* (Figure F.18), the effect of ξ is reversed. A positive δ shock pulls the dividend on a vacancy in both directions (E.16). Lower survival and a lower match surplus both reduce it. Against these, destruction raises unemployment and removes vacancies, so tightness falls, a newly posted position fills faster, and the dividend a recruiter can support rises. At the baseline calibration this congestion relief dominates, so K and Q rise and entry *expands*: $\hat{e}^{\delta} > 0$. This entry expansion — the same $G(Q_t)$ channel as in Coles and Moghaddasi Kelishomi (2018) — partially backfills vacancies destroyed by the direct $\hat{\delta}_e \uparrow$ impulse, muting the decline in

θ and the rise in u. A larger ξ increases cushioning, so the near-free-entry calibration generates *less* unemployment amplification from destruction shocks.

The contrast between the two shocks is the cleanest way to read Comparison D: ξ is a single parameter playing opposite roles, and what flips its sign is the direction in which the shock moves the vacancy value. A technology decline lowers the surplus from a filled job, so Q falls, entry falls with it, and a high ξ makes matters worse — entry *amplifies*. A destruction shock slackens the labor market: unemployment rises and the vacancy stock falls, so a newly posted position fills faster and the return to creating one *rises*. Entry then leans against the shock, and a high ξ makes matters better — entry *cushions*. Firms respond to a wave of destruction by building, precisely because destruction has made building more attractive at the margin. This is the sense in which the entry margin is stabilizing for δ shocks and destabilizing for z shocks, and it is why no single statement about "the elasticity of entry" characterizes the model's amplification without naming the shock.

Whether the entry cushion is large enough to reverse the sign of u-v comovement is a joint function of ξ and $\bar{\delta}_e$, not of ξ alone. To see this, consider the free-entry limit $\xi_{\text{inv}} \to 0$. There $Q \approx x_m$ is pinned and $K = x_m(r+\delta_e)/(1+r)$ rises with δ_e (duration shortening); the JCC then determines θ, which *falls* — the LHS of the job-creation condition rises with K while the RHS, proportional to $(1-\delta_e)(w^{int} - K - b)$, falls on both margins. Total vacancies $v = \theta u$ therefore have an ambiguous sign: positive u-v comovement requires the u rise to exceed the θ fall, which depends on how large a share of total separation flows through the δ channel.

At Coles and Moghaddasi Kelishomi's calibration, $\delta_e = \tau$ (all separation is destruction), so a δ shock raises u sharply; the duration shortening simultaneously generates a large entry surplus. To prevent vacancies from comoving *positively* with unemployment, CK make vacancies partially inelastic.

The baseline calibration of the present model, disciplined by BED Deaths data, delivers $\bar{\delta}_e/\bar{\tau} \approx 0.21$: only one-fifth of total separation flows through firm destruction. At this low $\bar{\delta}_e$, the proportional u rise from a δ shock is small — most separations are not of the δ type — while the standing vacancy stock is overwhelmingly inherited rather than newly posted, and only inherited vacancies are exposed to the shock. The destruction impulse is therefore large relative to any entry response that could offset it, and vacancies fall following destruction shocks *even at free entry*. Figure F.18 confirms this: at $\xi_{\text{inv}} = 0.1$, vacancies rise for two months, turn

negative in the third, and trough at -0.012 pp, while the baseline at $\xi_{\text{inv}} = 1.0$ shows a stronger decline still (-0.018 pp).

This is a key virtue of disciplining δ with data rather than setting $\delta_e = \tau$ as in Coles and Moghaddasi Kelishomi (2018): the empirically calibrated destruction rate endogenously preserves negative u-v comovement across the entire range of ξ shown here, including free entry. Convex entry costs ($\xi_{\text{inv}} > 0$) are not needed to prevent the CK Beveridge curve shift; the low empirical $\bar{\delta}_e$ already rules it out. Paragraph 9c of Appendix E.5 gives the reason in closed form: the unemployment loading is proportional to $\bar{\delta}_e/\bar{\tau}$ and so vanishes linearly as the destruction share falls, while the tightness loading is governed by $\bar{\delta}_e/(r+\bar{\delta}_e)$, which saturates once $\bar{\delta}_e$ is large relative to r. Lowering $\bar{\delta}_e$ therefore weakens the unemployment response much faster than the tightness response. The role of ξ is then purely quantitative — governing the strength of the cushioning and hence the *degree* of negative comovement — rather than determining its sign.

In the limit $\xi \to \infty$, $\hat{e}^{\delta} = \xi \hat{Q} \to 0$: the entry cushion disappears and destruction shocks hit the vacancy stock with no replenishment. The central message of Comparison D is therefore the asymmetry across shock types together with this robustness result: the empirically calibrated $\bar{\delta}_e$ makes negative u-v comovement for δ shocks robust to the entry-elasticity specification, while ξ remains the key determinant of technology-shock amplification.

Appendix G. Limiting cases

Proof of Proposition 1

We establish the if-and-only-if by showing necessity and sufficiency of each condition separately.

Sufficiency. Set $\sigma = 0$, $\zeta = 0$, $p_0 = 0$. Under $\sigma = 0$, the stochastic discount factor reduces to $m_{t+1} = \beta$, eliminating dependence on consumption C_t from all intertemporal conditions. Under $\zeta = 0$, general CES preferences give $\rho(N_t) = 1$ for all N_t, so recruiter compensation $w_t^{int} = \rho(N_t) z_t / \mu(N_t) = z_t/\mu$ depends on N_t only through $\mu(N_t)$; but under general CES with $\zeta = 0$, μ is a constant independent of N_t, so $w_t^{int} = z_t/\mu$ is exogenous to the labor market block. Under $p_0 = 0$, $\Lambda_t = F(\chi_t^c) = 1 - p_0 = 1$ for all χ_t^c, so $\delta_{e,t} = 1 - (1-\delta_{t-1})\Lambda_t = \delta_{t-1}$ depends only on the exogenous shock.

Under $p_0 = 0$ the distribution F collapses to a point mass at zero, which removes the reposting margin as well as the exit margin. Every reactivation cost draw is $\alpha\chi = 0$, so

$\Lambda_{r,t} = F(Q_t/\alpha) = 1$, the mean cost $\alpha \int_0^{Q_t/\alpha} \chi \, dF = 0$, and hence $Q_t^{rep} = Q_t$ and $X_t^r = 0$. The reposting shortfall $\mathcal{R}_t$ (20) vanishes from the job creation condition (42), (6) reverts to full reposting, and X_t^r drops out of the resource constraint (40).

Under all three conditions, inspect the labor market block (42)–(45) and laws of motion (5)–(6). With $m_{t+1} = \beta$, $w_t^{int} = z_t/\mu$, $\delta_{e,t} = \delta_{t-1}$, and $\Lambda_{r,t} = 1$, these equations form a closed system in $\{u_t, v_t, \theta_t, e_t, Q_t, K_t\}_{t=0}^{\infty}$ given exogenous states $\{z_t, \delta_t, s_t\}$ and initial conditions $\{v_{-1}, u_{-1}\}$; J_t and w_t then follow from (23) and (22). No variable from the business formation block $\{N_t, C_t, Y_t^c, \chi_t^c\}$ appears. The labor market block is therefore self-contained.

Given the solution to the labor market block, the business formation block (38)–(41) determines $\{N_t, C_t, Y_t^c, \chi_t^c\}$ taking $\{u_t, X_t\}$ as given. This confirms the recursive structure.

Necessity. We show that relaxing any single condition breaks separability: a business formation variable then enters the labor market block, and for generic values of the remaining parameters it does not cancel.

$\sigma > 0$: With $\sigma > 0$, $m_{t+1} = \beta(C_{t+1}/C_t)^{-\sigma}$ depends on C_t, which is determined by the resource constraint (40) jointly with N_t, X_t^c, and Y_t^c. The job creation condition (42) and flow vacancy value (43) both contain m_{t+1}, so the labor market block depends on C_t and hence on the business formation block.

$\zeta > 0$: With $\zeta > 0$, $\rho(N_t) > 1$ for $N_t > 1$ and $\rho'(N_t) > 0$, so $w_t^{int} = \rho(N_t) z_t/\mu(N_t)$ depends on N_t. Since N_t is determined in the business formation block, the job creation condition (42) depends on N_{t+1} through w_{t+1}^{int}, breaking separability.

$p_0 > 0$: With $p_0 > 0$, $\Lambda_t = F(\chi_t^c) < 1$ for $\chi_t^c < \chi_m$, and χ_t^c satisfies the cutoff condition (41) which depends on Y_t^c, N_t, and $\mu(N_t)$ — all determined in the business formation block. Hence $\delta_{e,t} = 1 - (1 - \delta_{t-1})\Lambda_t$ depends on business formation variables. Since $\delta_{e,t}$ enters both laws of motion (5)–(6), the labor market block depends on the business formation block.

Partial reposting adds no further channel, so the three conditions remain necessary and sufficient. The reactivation rate $\Lambda_{r,t} = F(Q_t/\alpha)$ and the option value Q_t^{rep} depend only on Q_t, which is itself a labor market variable, and the aggregate reactivation cost X_t^r enters only the resource constraint (40) — a flow from the labor market block into the business formation block, in the same direction as X_t, which leaves the recursive structure intact.

Proof of Proposition 2

Under clone replacement, $N_t = 1$ and $\mu = 1$ are maintained by construction at all times. We verify that the business formation block drops out and the remaining system reduces to (46)–(49).

Business formation block. Substituting $N_t = 1$ and $\mu = 1$ into the baseline equilibrium definition:

The firm LOM (38): with $N_t = 1$ maintained by construction, this equation is satisfied trivially by the clone replacement assumption and imposes no restriction on other variables.

The business formation Euler (39): with $\rho(1) = 1$, $\mu(1) = 1$, $f_e\rho(N_t) = f_e$, and $(\mu(N_{t+1}) - 1) = 0$, the right-hand side reduces to $(1 - \delta_t)\mathbb{E}_t m_{t+1}\Lambda_{t+1}\{f_e\} = f_e \cdot (1 - \delta_t)\mathbb{E}_t m_{t+1}$. This is a condition on f_e alone and is satisfied by appropriate choice of f_e given β and $\bar{\delta}$; it imposes no restriction on labor market variables.

The cutoff condition (41): with $\mu = 1$, $(\mu - 1)/\mu = 0$, so $\chi_t^c = f_e\rho(N_t)/\mu(N_t) = f_e$. Since clone replacement maintains $N_t = 1$ regardless of profitability, the exit decision is moot; $\chi_t^c = f_e$ is a constant that imposes no restriction on aggregate allocations.

The resource constraint (40): clone replacement is taken to operate through $p_0 = 0$, so $X_t^c = 0$. This is worth stating explicitly, because endogenous exit is absent whenever $\Lambda_t = 1$ and there are two ways to obtain that. Either $p_0 = 0$, or the cutoff weakly exceeds the support of F, which under clone replacement means $f_e \geq \chi_m$ since $\chi_t^c = f_e$. The two routes are not equivalent once reactivation is present. Setting $p_0 = 0$ removes both margins, since the exit and reactivation thresholds are evaluated on the same distribution, whereas $f_e \geq \chi_m$ leaves $\Lambda_{r,t} = F(Q_t/\alpha) < 1$ whenever $Q_t < \alpha\chi_m$. Under the second route (48) would carry $\Lambda_{r,t}$, (50) would carry $-X_t^r$, and (46) would carry the reactivation shortfall. The proposition as stated assumes the first. With $\rho(1) = 1$ and $\mu(1) = 1$, $Y_t^c = z_t(1 - u_t)$ and $l_t^e = 0$, giving $C_t = z_t(1 - u_t) - X_t$, which is (50).

Labor market block. Substituting $\rho(N_t) = 1$, $\mu(N_t) = 1$, $\Lambda_t = 1$, and $\delta_{e,t} = \delta_{t-1}$ into (42)–(5) yields (46)–(49) directly. The stochastic discount factor $m_t = \beta(C_t/C_{t-1})^{-\sigma}$ is unchanged. The stochastic processes (7)–(9) are unchanged.

The resulting system (46)–(49) together with $e_t = G(Q_t)$, $m_t = \beta(C_t/C_{t-1})^{-\sigma}$, (50), and stochastic processes (7)–(9) closes the economy. Local determinacy is a separate question. With a finite entry elasticity the system has two predetermined states, u_t and the predeter-

mined vacancy stock, and two forward-looking variables, Q_t and K_t. We do not establish the Blanchard–Kahn condition analytically. At $\sigma = \kappa = 0$ it is verified numerically at every calibration and entry elasticity we consider: exactly two roots are stable. Under free entry ($\xi \to \infty$) with $\sigma = \kappa = 0$, the system is block recursive and the condition holds analytically: Lemma 1 gives one unstable root for tightness and one stable root for unemployment. For $\sigma > 0$, local uniqueness is maintained as a regularity condition and verified numerically as part of the baseline model solution.

Proof of Proposition 3

Consider the DS-CES equilibrium with $p_0 = 0$ and parameters (ε, f_e) satisfying $f_e = \bar{f}_e/\varepsilon$ for fixed $\bar{f}_e > 0$. Since $p_0 = 0$, $\Lambda_t = 1$ and $\delta_{e,t} = \delta_{t-1}$ throughout. We establish claims (iii), (iv), (i), and (ii) in that order, since (iii) is used in (iv) and (i) validates an assumption used in (iii).

Claim (iii): $w_t^{int} \to z_t$.. Under DS-CES, $\mu = \varepsilon/(\varepsilon - 1) \to 1$ and $\rho(N_t) = N_t^{1/(\varepsilon-1)}$. For any N_t satisfying $N_t = O(e^{o(\varepsilon)})$ as $\varepsilon \to \infty$ — a condition weaker than boundedness, verified below — $\ln \rho(N_t) = \ln N_t/(\varepsilon - 1) \to 0$, so $\rho(N_t) \to 1$. Hence $w_t^{int} = \rho(N_t) z_t/\mu(N_t) \to z_t$ along any such equilibrium path. □

Claim (iv): N_t drops out of aggregate allocations. We show that $\{\theta_t, u_t, v_t, C_t, Y_t^c\}$ are determined independently of N_t in the limit.

Labor market block. From claim (iii), $w_t^{int} \to z_t$. Since $p_0 = 0$ gives $\Lambda_t = 1$, the job creation condition (42), flow vacancy value (43), free entry (44), and laws of motion (5)–(6) converge to (46)–(49), which contain no reference to N_t.

Resource constraint. Since $p_0 = 0$, $X_t^c = 0$. Labor devoted to entry satisfies $l_t^e = f_e N_t^e/z_t = O(1/\varepsilon) \to 0$, established in claim (ii). Hence $L_t^c = L_t - l_t^e \to L_t = 1 - u_t$. Output per firm $y_t = z_t L_t^c/N_t \to z_t(1 - u_t)/N_t$, and $Y_t^c = \rho(N_t) N_t y_t \to 1 \cdot N_t \cdot z_t(1 - u_t)/N_t = z_t(1 - u_t)$. The resource constraint becomes $C_t = z_t(1 - u_t) - X_t$, independent of N_t.

Since all equations determining $\{\theta_t, u_t, v_t, C_t, Y_t^c\}$ are independent of N_t and of $\bar{f}_e$ in the limit, $\bar{f}_e$ can be chosen freely without changing aggregate allocations. By claim (i), $\bar{N}$ is pinned by $\bar{f}_e$, and $\bar{N} = 1$ corresponds to $\bar{f}_e = \bar{z}\bar{L}(1 - \bar{\delta})/(r + \bar{\delta})$. The normalization $\bar{N} = 1$ is therefore without loss of generality for aggregate allocations. □

Claim (i): the mass of product lines. Aggregate allocations do not depend on N_t by claim (iv), so the path of N_t does not affect the main result. We characterize it to confirm that N_t stays bounded away from zero and infinity, which claims (ii) and (iii) use.

Under DS-CES μ is constant, and with $p_0 = 0$, $\Lambda_t = 1$ and $X_t^c = 0$. Gross consumption output is $Y_t^c = \rho(N_t) N_t y_t = \rho(N_t) z_t L_t^c$. Dividing the business formation Euler equation (39) by $f_e \rho(N_t)$ gives

$$1 = (1 - \delta_t)\, \mathbb{E}_t\, m_{t+1} \frac{\rho(N_{t+1})}{\rho(N_t)} \left[1 + \frac{\mu - 1}{f_e} \frac{z_{t+1} L_{t+1}^c}{N_{t+1}} \right].$$

Along a path on which N_t stays in a compact subset of $(0, \infty)$, three limits hold as $\varepsilon \to \infty$. First, $(\mu - 1)/f_e = \varepsilon/[(\varepsilon - 1)\bar{f}_e] \to 1/\bar{f}_e$. Second, $\rho(N_{t+1})/\rho(N_t) = (N_{t+1}/N_t)^{1/(\varepsilon-1)} \to 1$. Third, $L_{t+1}^c = 1 - u_{t+1} - l_{t+1}^e \to 1 - u_{t+1}$, since bounded N_t and N_{t+1} give bounded entry N_t^e and hence $l_t^e = f_e N_t^e / z_t = O(1/\varepsilon)$. Hence

$$1 = (1 - \delta_t)\, \mathbb{E}_t\, m_{t+1} \left[1 + \frac{z_{t+1}(1 - u_{t+1})}{\bar{f}_e N_{t+1}} \right].$$

Both $N_{t+1} = (1 - \delta_t)(N_t + N_t^e)$ and u_{t+1} are determined at t, the latter by (5). They can therefore be taken outside the expectation, which gives (51). Entry absorbs the gap between the target and the surviving stock, $N_t^e = N_{t+1}/(1 - \delta_t) - N_t$, and the limit applies while this is positive. Under $\sigma > 0$ the discount factor depends on C_{t+1}, which is independent of N by claim (iv), so the rule contains only labor market objects and shocks. Since z_t, $1 - u_t$ and $\mathbb{E}_t m_{t+1}$ are bounded and $(1 - \delta_t)\mathbb{E}_t m_{t+1} < 1$, N_{t+1} stays in a compact subset of $(0, \infty)$. This confirms the premise of the argument and the condition $N_t = O(e^{o(\varepsilon)})$ of claim (iii).

At the steady state $m = \beta = 1/(1 + r)$, and (51) gives

$$\bar{N} = \frac{\beta(1 - \bar{\delta})\bar{z}\bar{L}}{\bar{f}_e[1 - \beta(1 - \bar{\delta})]} = \frac{\bar{z}\bar{L}(1 - \bar{\delta})}{\bar{f}_e(r + \bar{\delta})},$$

with $\bar{L} = 1 - \bar{u}$. This is the limit of the exact steady state at finite ε, $N = (\mu - 1) z L (1 - \delta)/[f_e(r + \delta\mu)]$, which is the resource constraint curve (61). Because the rule pins N_{t+1} period by period, there is no autonomous recursion for N_t and no stability condition is needed. □

Claim (ii): rates of convergence. Firm value: $\nu_t^f = \rho(N_t) f_e / \mu(N_t) \approx 1 \cdot (\bar{f}_e/\varepsilon) \cdot 1 = O(1/\varepsilon)$.

Investment: By claim (i), $N_t^e = N_{t+1}/(1 - \delta_t) - N_t = O(1)$ along the path, with steady-state value $\bar{\delta}\bar{N}/(1 - \bar{\delta})$. Hence $\nu_t^f N_t^e = O(1/\varepsilon) \cdot O(1) = O(1/\varepsilon)$.

Labor to entry: $l_t^e = f_e N_t^e / z_t = (\bar{f}_e/\varepsilon) \cdot O(1)/z_t = O(1/\varepsilon)$. □

Proof of Proposition 4

Set $p_0 = 0$. Then F is a point mass at zero, $F(\chi) = 1 - p_0 = 1$ for every $\chi \geq 0$, so no firm pays a continuation cost: $X_t^c = 0$, $\Lambda_t = 1$, and $\delta_{e,t} = \delta_{t-1}$ for all t. Every reactivation is free as well, so $X_t^r = 0$.

Firm LOM (38). With $\Lambda_t = 1$ and $\delta_{e,t} = \delta_{t-1}$, the baseline LOM becomes

$$N_t = (1 - \delta_{t-1}) \left(N_{t-1} + \frac{z_{t-1}(1 - u_{t-1})}{f_e} - \frac{Y_{t-1}^c}{f_e \rho(N_{t-1})} \right),$$

where $\{1 - u_{t-1}\}$ is treated as an exogenous sequence by assumption. Gross consumption output is $Y_t^c = \rho(N_t) N_t y_t = \rho(N_t) z_t L_t^c$, so $Y_{t-1}^c / [f_e \rho(N_{t-1})] = z_{t-1} L_{t-1}^c / f_e$. Labor not used in retail production is used in entry, $l_{t-1}^e = (1 - u_{t-1}) - L_{t-1}^c$, and $N_{t-1}^e = z_{t-1} l_{t-1}^e / f_e$. The parenthesized term is therefore $N_{t-1} + N_{t-1}^e$, and (38) is the BGM firm LOM $N_t = (1 - \delta_{t-1})(N_{t-1} + N_{t-1}^e)$.

Business formation Euler (39). With $\Lambda_{t+1} = 1$ and $X_{t+1}^c = 0$, the baseline Euler simplifies to

$$f_e \rho(N_t) = (1 - \delta_t) \, \mathbb{E}_t m_{t+1} \frac{\mu(N_t)}{\mu(N_{t+1})} \left\{ f_e \rho(N_{t+1}) + \frac{Y_{t+1}^c}{N_{t+1}} (\mu(N_{t+1}) - 1) \right\}.$$

Divide by $\mu(N_t)$ and use free entry (31), $\nu_t^f = \rho(N_t) f_e / \mu(N_t)$. This gives $\nu_t^f = (1 - \delta_t) \mathbb{E}_t m_{t+1} (\nu_{t+1}^f + d_{t+1})$, where $d_{t+1} = [(\mu(N_{t+1}) - 1)/\mu(N_{t+1})] \, Y_{t+1}^c / N_{t+1}$ is profit per product line, (28). This is the BGM asset-pricing equation for firm equity.

Resource constraint (36). With $X_t^c = X_t^r = 0$, (40) gives $Y_t^c = C_t + X_t$. Netting out X_t on both sides gives the GDP identity $Y_t = C_t + \nu_t^f N_t^e$, which is the BGM resource constraint. The household budget constraint $c_t = w_t(1 - u_t) + \pi_t - X_t$ closes the system identically in both models.

The two systems thus share the same three equations and the same exogenous sequences $\{z_t, \delta_t, 1 - u_t, X_t\}$. Since $1 - u_t$ and X_t are endogenous in the baseline but exogenous in LR-BGM, the proposition asserts isomorphism for any realization of those sequences, not that the equilibrium distributions coincide.

Appendix H. Proof of Lemma 1 and Proposition 5

Throughout, $\sigma = \kappa = 0$, $p_0 = 0$, $\mu = 1$, z is constant, and $Q_t = x_m$. Hats are log deviations from the steady state, and $\tilde{\delta}_t$, $\tilde{s}_t$ are the log deviations of the shocks. Under $p_0 = 0$, $\delta_{e,t+1} = \delta_t$ and $\tau_t = 1 - (1 - \delta_{t-1})(1 - s_{t-1})$, so $(1 - \bar{\delta})(1 - \bar{s}) = 1 - \bar{\tau}$ and $(1 - \bar{\delta})\bar{s} = \bar{\tau} - \bar{\delta}$.

Tightness coefficients. Define

$$D_\theta(\rho) \equiv \eta_L\big[1+r-\rho(1-\bar\tau)\big]+\rho\,\phi(1-\bar\delta)\bar f,$$

$$c_\delta \equiv -\frac{(1+r)\dfrac{\bar\delta}{1-\bar\delta}+\dfrac{\bar\delta}{r+\bar\delta}\Big[1+r-\rho_\delta(1-\bar\tau)+\rho_\delta\,\phi(1-\bar\delta)\bar f+\rho_\delta(1-\phi)(1-\bar\delta)\bar q\Big]}{D_\theta(\rho_\delta)}, \qquad c_s \equiv -\frac{\rho_s(1-\bar\delta)\bar s}{D_\theta(\rho_s)}. \tag{H.1}$$

The numerator of c_δ has a survival part, $(1+r)\bar\delta/(1-\bar\delta)$, and a user-cost part proportional to $\bar\delta/(r+\bar\delta)$. The numerator of c_s has neither. It contains only the effect of persistent separations on the continuation of next period's match.

The job creation condition. With $m_{t+1}=1/(1+r)$, multiply (46) by $1+r$ and substitute (52) at t and $t+1$:

$$\frac{x_m(r+\delta_t)}{q(\theta_t)} = (1-\delta_t)\,\mathbb{E}_t\Big\{(1-\phi)(z-b)+\frac{x_m(r+\delta_{t+1})}{1+r}\Big[\frac{1-s_{t+1}}{q(\theta_{t+1})}-\phi\,\theta_{t+1}-(1-\phi)\Big]\Big\}.$$

Only tightness and the shocks appear. The left side deviates by $[\bar\delta/(r+\bar\delta)]\tilde\delta_t+\eta_L\hat\theta_t$, using $d\ln(1/q)=\eta_L\hat\theta$. On the right, $\ln(1-\delta_t)$ deviates by $-[\bar\delta/(1-\bar\delta)]\tilde\delta_t$. Let Σ_{t+1} be the term in braces, with steady state $\bar\Sigma = x_m(r+\bar\delta)/[(1-\bar\delta)\bar q]$. Differentiating its second term and dividing by $\bar\Sigma$, with $\bar\theta\bar q=\bar f$,

$$\hat\Sigma_{t+1}=\frac{1}{1+r}\Big\{\big[1-\bar\tau-\phi(1-\bar\delta)\bar f-(1-\phi)(1-\bar\delta)\bar q\big]\frac{\bar\delta}{r+\bar\delta}\,\tilde\delta_{t+1}+\big[(1-\bar\tau)\eta_L-\phi(1-\bar\delta)\bar f\big]\hat\theta_{t+1}-(1-\bar\delta)\bar s\,\tilde s_{t+1}\Big\}.$$

Equating the two sides,

$$\eta_L\hat\theta_t=-\Big[\frac{\bar\delta}{r+\bar\delta}+\frac{\bar\delta}{1-\bar\delta}\Big]\tilde\delta_t+\mathbb{E}_t\hat\Sigma_{t+1}. \tag{H.2}$$

Proof of Lemma 1. Equation (H.2) contains neither u nor the predetermined vacancy stock, because free entry pins Q and K_t depends only on δ_t. Divide by η_L:

$$\hat\theta_t=\Phi_\theta\,\mathbb{E}_t\hat\theta_{t+1}+(\text{current and expected shocks}), \qquad \Phi_\theta\equiv\frac{1}{1+r}\Big[1-\bar\tau-\frac{\phi(1-\bar\delta)\bar f}{\eta_L}\Big].$$

Since $\phi\ge 0$, $\Phi_\theta<(1-\bar\tau)/(1+r)<1$. And $\Phi_\theta>-1$ if and only if $\phi(1-\bar\delta)\bar f<\eta_L(2+r-\bar\tau)$, which is the assumption. It holds whenever $\phi\le\eta_L$, because $(1-\bar\delta)\bar f<1<2+r-\bar\tau$. With $|\Phi_\theta|<1$, iterating forward and requiring a bounded solution leaves the forward sum of current and expected shocks as the unique solution. It does not depend on the endogenous states. With each shock an AR(1), every term of the sum is proportional to the current shock, so $\hat\theta_t=c_\delta\tilde\delta_t+c_s\tilde s_t$.

For c_δ, set $\tilde{s} \equiv 0$ and substitute $\hat{\theta}_t = c_\delta \tilde{\delta}_t$, $\mathbb{E}_t \hat{\theta}_{t+1} = c_\delta \rho_\delta \tilde{\delta}_t$ and $\mathbb{E}_t \tilde{\delta}_{t+1} = \rho_\delta \tilde{\delta}_t$ into (H.2). Multiplying by $1+r$ and collecting the terms in c_δ gives $c_\delta D_\theta(\rho_\delta)$ on the left and the negative of the numerator in (H.1) on the right. For c_s, set $\tilde{\delta} \equiv 0$ in the same way, which gives $c_s D_\theta(\rho_s) = -\rho_s(1-\bar{\delta})\bar{s}$. Since $\rho < 1$, $1 + r - \rho(1-\bar{\tau}) > 0$, so $D_\theta > 0$ and every term in the numerator of c_δ is positive. Hence $c_\delta < 0$, and $c_s \leq 0$ with equality if and only if $\rho_s = 0$.

The remaining equations are solved backward. The law of motion (49) gives (53), using $(1-\bar{u})\bar{\tau}/\bar{u} = (1-\bar{\delta})\bar{f}$ and combining the two channels through which δ_{t-1} raises u_t: lost matches, $\bar{f}\bar{\delta}$, and separations, $(1-\bar{\delta})\bar{f}(1-\bar{s})\bar{\delta}/\bar{\tau}$, which sum to $\bar{f}\bar{\delta}/\bar{\tau}$. Then $\hat{v}_t = \hat{\theta}_t + \hat{u}_t$, and entry is the residual of (48). The predetermined vacancy stock feeds into nothing except entry. The system therefore has one unstable root, $1/\Phi_\theta$, for one jump variable, and one stable root, λ_u, for one state, and the Blanchard–Kahn condition holds. □

Proof of Proposition 5. Since $0 < \bar{\delta} < \bar{\tau}$, $\ell_\delta > 0$ and $\ell_s > 0$. If $c_x = 0$, which happens only for the separation shock with $\rho_s = 0$, then $\hat{\theta}_t = 0$ at every date, $\hat{v}_t = \hat{u}_t > 0$ for $t \geq 1$, the condition of part 1 reads $0 > \ell_s$ and fails, and the slope is $\bar{\theta}$, as stated. Otherwise write $\Gamma_x \equiv \ell_x/|c_x| > 0$. The condition of part 1 is then $\rho_x > \omega_u + \Gamma_x$, that of part 2 is $\rho_x(1-\lambda_u^2)/(1+\rho_x\lambda_u) > \omega_u + \Gamma_x$, and (54) is $\bar{\theta}[1-\rho_x(1-\lambda_u^2)/((1+\rho_x\lambda_u)(\omega_u+\Gamma_x))]$. The innovation sets $\tilde{x}_t = \rho_x^t \tilde{x}_0$, so by Lemma 1 $\hat{\theta}_t = c_x \rho_x^t \tilde{x}_0$.

Step 1: unemployment. u_0 is predetermined, so $\hat{u}_0 = 0$. By (53), $\hat{u}_t = \lambda_u \hat{u}_{t-1} + (\ell_x + \omega_u|c_x|)\rho_x^{t-1}\tilde{x}_0$. Iterating gives $\hat{u}_t = (\ell_x + \omega_u|c_x|)\tilde{x}_0\, g_t$ with $g_t \equiv \sum_{j=0}^{t-1} \lambda_u^{t-1-j}\rho_x^j$. Since $\lambda_u, \rho_x \geq 0$ and $g_1 = 1$, $\hat{u}_t > 0$ for $t \geq 1$. When $c_x \neq 0$, $\ell_x + \omega_u|c_x| = |c_x|(\omega_u + \Gamma_x)$.

Step 2: part 1. With $g_1 = 1$, $\hat{v}_1 = |c_x|(\omega_u + \Gamma_x - \rho_x)\tilde{x}_0$, which is negative if and only if $\rho_x > \omega_u + \Gamma_x$. For $t \geq 1$ and $\rho_x > 0$, dividing $\hat{v}_t = |c_x|[-\rho_x^t + (\omega_u + \Gamma_x)g_t]\tilde{x}_0$ by $|c_x|\rho_x^t\tilde{x}_0$,

$$\hat{v}_t < 0 \iff \rho_x > (\omega_u + \Gamma_x)\sum_{i=0}^{t-1}\Big(\frac{\lambda_u}{\rho_x}\Big)^i.$$

The right side is strictly increasing in t, so once the inequality fails it fails at all later dates. If $\rho_x \leq \lambda_u$ the right side grows without bound. If $\rho_x > \lambda_u$ it converges from below to $(\omega_u + \Gamma_x)\rho_x/(\rho_x - \lambda_u)$, so vacancies stay below steady state at every horizon if and only if $\rho_x - \lambda_u \geq \omega_u + \Gamma_x$.

Step 3: part 2. In levels, $u_t - \bar{u} = \bar{u}|c_x|(\omega_u + \Gamma_x)\tilde{x}_0 g_t$ and $v_t - \bar{v} = \bar{v}|c_x|[-\rho_x^t + (\omega_u + \Gamma_x)g_t]\tilde{x}_0$, with $g_0 = 0$. For $\rho_x \neq \lambda_u$, $g_t = (\rho_x^t - \lambda_u^t)/(\rho_x - \lambda_u)$, and the case $\rho_x = \lambda_u$ follows by continuity.

Summing geometric series,

$$\sum_{t\geq 0}\rho_x^t g_t = \frac{\rho_x}{(1-\rho_x^2)(1-\rho_x\lambda_u)}, \qquad \sum_{t\geq 0} g_t^2 = \frac{1+\rho_x\lambda_u}{(1-\rho_x^2)(1-\lambda_u^2)(1-\rho_x\lambda_u)}.$$

The second uses $(1-\rho_x\lambda_u)(1-\lambda_u^2)-2(1-\rho_x^2)(1-\lambda_u^2)+(1-\rho_x^2)(1-\rho_x\lambda_u) = (\rho_x-\lambda_u)^2(1+\rho_x\lambda_u)$. Therefore

$$\frac{\sum_t (v_t-\bar v)(u_t-\bar u)}{\sum_t (u_t-\bar u)^2} = \frac{\bar v}{\bar u}\Big[1-\frac{1}{\omega_u+\Gamma_x}\frac{\sum_t \rho_x^t g_t}{\sum_t g_t^2}\Big] = \bar\theta\Big[1-\frac{\rho_x(1-\lambda_u^2)}{(1+\rho_x\lambda_u)(\omega_u+\Gamma_x)}\Big],$$

which is (54).

Step 4: part 3. By Lemma 1, $|c_s| = \rho_s(1-\bar\delta)\bar s/D_\theta(\rho_s)$, and $(1-\bar\delta)\bar s = \bar\tau(1-\bar\delta/\bar\tau)$, so $\Gamma_s = (1-\bar\delta)\bar f D_\theta(\rho_s)/(\rho_s\bar\tau)$. Since $(1+r)/\rho_s > 1+r$, $D_\theta(\rho_s)/\rho_s > \eta_L(r+\bar\tau)+\phi(1-\bar\delta)\bar f$, so

$$\Gamma_s > (1-\bar\delta)\bar f\Big[\eta_L\frac{r+\bar\tau}{\bar\tau}+\frac{\phi(1-\bar\delta)\bar f}{\bar\tau}\Big] > (1-\bar\delta)\bar f\Big[\eta_L+\frac{\phi(1-\bar\delta)\bar f}{\bar\tau}\Big] \geq 1.$$

Then for $t\geq 1$, $(\omega_u+\Gamma_s)g_t > g_t \geq \rho_s^{t-1} \geq \rho_s^t$, because g_t contains the term ρ_s^{t-1}, so $\hat v_t > 0$. And $\rho_s(1-\lambda_u^2)/(1+\rho_s\lambda_u) < \rho_s < 1 < \omega_u+\Gamma_s$, so the slope is positive.

Step 5: $\rho_x = 0$. For the destruction shock the left sides of both conditions are zero while $\omega_u+\Gamma_\delta > 0$, so both fail. Since $\hat\theta_t = 0$ for $t\geq 1$, $\hat v_t = \hat u_t$ after impact, and the slope is $\bar\theta$. For the separation shock $c_s = 0$, so $\hat\theta_t = 0$ at every date and the slope is again $\bar\theta$. □

Every closed form in this appendix is checked against a simulated impulse response of the same linear system in the replication programs.

Appendix I. Proofs of Lemmas 2 and 3 and Proposition 6

Throughout, the aggregate state, including s_t and δ_t, is known at the start of t, so period-t jump variables respond to period-t shocks, while the rates s_t and δ_t act on stocks only at the end of t. Statements are to first order around the steady state. Write $M_t \equiv f(\theta_t)u_t = q(\theta_t)v_t$ for matches at t, and let E_t^b and V_t^b denote employment and vacancies entering t before the period-t exit draw, so that $1-u_t = \Lambda_t E_t^b$ and $v_{pre,t} = \Lambda_t V_t^b$. We first establish the lemma on pre-committed vacancy destruction, which holds for $p_0 \geq 0$, then prove Lemma 2 and Proposition 6.

Lemma 3 (Pre-committed vacancy destruction)**.** *To first order in $F'(\chi^c)$, a δ_t shock changes the pre-committed vacancy stock at $t+1$ by*

$$\frac{\partial\, v_{pre,t+1}}{\partial\delta_t} = -\Lambda_{t+1}\big[(1-q(\theta_t))v_t + s_t(1-u_t)\big] + (1-\delta_t)\Lambda_{t+1}\big[1-(1-\eta_L)q(\theta_t)\big]\frac{\partial e_t}{\partial\delta_t}.$$

The first term, the destruction of the stock carried into $t+1$, is negative for any parameter values. The second is the part of period-t entry that is not matched at t.

Proof. The pre-committed vacancy stock is the component of v_{t+1} determined before entry e_{t+1} is chosen:

$$v_{pre,t+1} \equiv (1-\delta_t)\Lambda_{t+1}\big[(1-q(\theta_t))v_t + s_t(1-u_t)\big],$$

using $1-\delta_{e,t+1} = (1-\delta_t)\Lambda_{t+1}$ from (3). Differentiating the survival factor gives the first term; its derivative through Λ_{t+1} is proportional to $F'(\chi^c_{t+1})$ and is dropped at first order. In the bracket, u_t and $v_{pre,t}$ are predetermined up to the period-t exit response, which is also proportional to $F'(\chi^c_t)$. Hence $\Delta v_t = \Delta e_t$ and, since $q(\theta_t)v_t = M_t$, $\Delta[(1-q(\theta_t))v_t] = \Delta e_t - \Delta M_t$. With u_t fixed, $\Delta\theta_t = \Delta e_t/u_t$ and $\Delta M_t = u_t f'(\theta_t)\Delta\theta_t = (1-\eta_L)q(\theta_t)\Delta e_t$, which gives the second term. □

Proof of Lemma 2. Part 1: s shock. Under $p_0 = 0$, F is a point mass at zero. Hence $\Lambda_t = 1$ and $\delta_{e,t+1} = \delta_t$, and every reactivation draw is free, so $\Lambda_{r,t} = 1$ and $Q^{rep}_t = Q_t$: the reposting term in the surplus (23) vanishes. Start from the steady state and raise s_t. Because s_t is known at the start of t, the period-t jump variables, including θ_t and e_t, respond, while u_t and $v_{pre,t}$ are predetermined. So $\Delta u_t = 0$ and $\Delta v_t = \Delta e_t$ at $h = 0$, and, as in the proof of the Lemma, $\Delta M_t = (1-\eta_L)q(\theta_t)\Delta e_t$.

Step 1: unemployment. From (4), $\tau_{t+1} = \delta_t + s_t(1-\delta_t)$, so $\partial\tau_{t+1}/\partial s_t = 1-\delta_t$. From the unemployment law of motion (5),

$$u_{t+1} = u_t - (1-\delta_t)M_t + \tau_{t+1}(1-u_t),$$

so

$$\frac{\partial u_{t+1}}{\partial s_t} = d - (1-\delta_t)(1-\eta_L)q(\theta_t)\frac{\partial e_t}{\partial s_t}, \qquad d \equiv (1-\delta_t)(1-u_t),$$

which is positive if and only if condition (i) holds.

Step 2: pre-committed vacancies. From the vacancy law of motion (6) with $\Lambda_{r,t+1} = 1$,

$$v_{t+1} = \underbrace{(1-\delta_t)\big[(1-q(\theta_t))v_t + s_t(1-u_t)\big]}_{v_{pre,t+1}} + e_{t+1}. \tag{I.1}$$

Survival $1-\delta_t$ does not depend on s_t, and, as in the Lemma, $\Delta[(1-q(\theta_t))v_t] = \big[1-(1-\eta_L)q(\theta_t)\big]\Delta e_t$. Hence

$$\frac{\partial\, v_{pre,t+1}}{\partial s_t} = d + (1-\delta_t)\big[1-(1-\eta_L)q(\theta_t)\big]\frac{\partial e_t}{\partial s_t}.$$

The inflow d is common to Steps 1 and 2: each separation at a surviving product line adds one worker to unemployment and returns one position to the vacancy pool. The remaining terms are the matches and vacancies created by the period-t entry response. Steps 1 and 2 use only the laws of motion and the timing of s_t, so they hold for any ρ_s and any entry technology.

Step 3: vacancies. By Step 2 and (I.1),

$$\frac{\partial v_{t+1}}{\partial s_t} = d + (1-\delta_t)\big[1-(1-\eta_L)q(\theta_t)\big]\frac{\partial e_t}{\partial s_t} + \frac{\partial e_{t+1}}{\partial s_t},$$

which is positive if and only if condition (ii) holds. With Step 1, unemployment and vacancies comove positively at $h = 1$ under (i) and (ii). The entry responses are equilibrium objects: they depend on ρ_s, on the state deviations the shock leaves behind, and on feedback through the discount factor and product variety.

Step 4: two cases in which the conditions hold exactly. Let $\rho_s = 0$ and impose the conditions of Proposition 1 ($\sigma = 0$, $\zeta = 0$, $p_0 = 0$), so that $m_{t+1} = \beta$, $w_t^{int} = z_t/\mu$, and the labor market block closes on its own, with predetermined states $(u_t, v_{pre,t})$. The job creation condition (42) at any date involves s_{t+1} and later separation rates, which a period-t shock leaves at their mean in expectation. Under determinacy, a path that satisfies the block is the solution.

(a) Free entry ($\xi \to \infty$). Then $Q_{t+k} = x_m$ at every date, so $K_{t+k} = \bar{K}$ by (14), and $\theta_{t+k} = \bar{\theta}$ for all $k \geq 0$ satisfies (42). Since u_t and $v_{pre,t}$ are predetermined, $\Delta e_t = 0$, so (i) holds trivially. At $t+1$, entry adjusts to hold tightness fixed:

$$\frac{\partial v_{t+1}}{\partial s_t} = \bar{\theta} d > 0, \qquad \frac{\partial e_{t+1}}{\partial s_t} = (\bar{\theta} - 1)d > -d.$$

(b) Convex entry costs ($\xi < \infty$) with $\bar{\theta} = 1$. Guess that $\theta_{t+k} = \bar{\theta}$ and every jump variable stays at its steady-state value for all $k \geq 0$. At t, entry $e_t = \theta_t u_t - v_{pre,t}$ is unchanged because u_t and $v_{pre,t}$ are, so $\Delta e_t = 0$. At $t+1$, unemployment and pre-committed vacancies both rise by d, so entry $e_{t+1} = \theta_{t+1}u_{t+1} - v_{pre,t+1}$ changes by $(\bar{\theta}-1)d = 0$ and $\partial v_{t+1}/\partial s_t = d$. The state deviations then propagate, at fixed tightness, as

$$\Delta u_{t+2} = \big[1-(1-\bar{\delta})\bar{f}-\bar{\tau}\big]d = (1-\bar{\delta})(1-\bar{f}-\bar{s})\,d, \qquad \Delta v_{pre,t+2} = (1-\bar{\delta})\big[(1-\bar{q})d-\bar{s}d\big] = (1-\bar{\delta})(1-\bar{q}-\bar{s})\,d,$$

using $\bar{\tau} = \bar{\delta}+\bar{s}(1-\bar{\delta})$. These are equal because $\bar{\theta} = 1$ implies $\bar{f} = \bar{q}$. By induction the deviations in u and v_{pre} stay equal at every horizon, entry is unchanged at every horizon, and the guess is verified.

Outside these cases conditions (i) and (ii) are quantitative.

Part 2: δ shock. Consider a positive δ_t shock with s_t at its steady-state value $\bar{s}$. Under $p_0 = 0$, $\Lambda_t = 1$ at every date.

Unemployment. From (5) dated $t+1$, with $\tau_{t+1} = \delta_{e,t+1} + \bar{s}(1 - \delta_{e,t+1})$,

$$\frac{\partial u_{t+1}}{\partial \delta_t} = (1 - \bar{s})(1 - u_t) + M_t - (1 - \delta_t)(1 - \eta_L) q(\theta_t) \frac{\partial e_t}{\partial \delta_t}.$$

The first two terms count the separated incumbents and the new matches lost before producing. The third is the extra matching from entry at t. Unemployment rises if and only if the matching condition in Part 2 of the lemma holds.

Vacancies. By Lemma 3 and (I.1),

$$\frac{\partial v_{t+1}}{\partial \delta_t} = -\big[(1 - q(\theta_t))v_t + \bar{s}(1 - u_t)\big] + (1 - \delta_t)\big[1 - (1 - \eta_L)q(\theta_t)\big]\frac{\partial e_t}{\partial \delta_t} + \frac{\partial e_{t+1}}{\partial \delta_t},$$

which is negative if and only if the vacancy condition in Part 2 holds. Together with the rise in unemployment, this gives negative u–v comovement at $h = 1$. □

Proof of Proposition 6. With $p_0 > 0$, survival into $t+1$ is $1 - \delta_{e,t+1} = (1 - \delta_t)\Lambda_{t+1}$, and two exit responses appear. Next-period exit $\Lambda_{t+1} = F(\chi^c_{t+1})$ responds as in the old timing. Period-t exit Λ_t also responds, because s_t is known at the start of t: news of higher separations lowers the expected return on new product lines, which moves product-line entry N^e_t, the allocation of labor between production and entry, current retail profits, and hence χ^c_t. Every effect of exit on u and v passes through $\Delta\Lambda_t = F'(\chi^c_t)\Delta\chi^c_t$ or $\Delta\Lambda_{t+1} = F'(\chi^c_{t+1})\Delta\chi^c_{t+1}$. To first order in F', Steps 1–3 of the proof of Lemma 2 carry over with d replaced by the reposting inflow $(1 - \delta_t)\Lambda_{t+1}(1 - u_t)$, plus exit terms linear in $\Delta\Lambda_t$ and $\Delta\Lambda_{t+1}$. On the pre-committed vacancy stock the next-period term is

$$(1 - \delta_t)\frac{\partial \Lambda_{t+1}}{\partial s_t}\big[(1 - q(\theta_t))v_t + s_t(1 - u_t)\big],$$

and the period-t term has the same form, with $\partial\Lambda_t/\partial s_t$ multiplying the stocks it removes at t. These terms form the drain.

Joint system. The cutoff, product variety, and retail employment at $t+1$ are determined jointly. Let $B_t \equiv N_t + N^e_t$ be the product lines entering $t+1$ before exit and E^b_{t+1} pre-exit employment. Post-exit employment is $1 - u_{t+1} = F(\chi^c_{t+1})E^b_{t+1}$, and retail employment is $L^c_{t+1} = 1 - u_{t+1} - l^e_{t+1}$,

so that $Y^c_{t+1} = \rho(N_{t+1}) z_{t+1} L^c_{t+1}$. Then

$$\begin{aligned} \chi^c &= g(N,\, L^c), \\ N &= (1-\delta_t)\, F(\chi^c)\, B_t, \\ L^c &= F(\chi^c)\, E^b_{t+1} - l^e_{t+1}, \end{aligned}$$

with g as defined before Proposition 6 and $g_L = \rho(N) z (\mu - 1)/(\mu N) > 0$. The third equation is the feedback from exit to employment: more exit lowers retail employment, which lowers profits and raises exit further. Totally differentiating,

$$\det J\, \Delta\chi^c_{t+1} = g_N (1-\delta_t) \Lambda_{t+1} \Delta B_t + g_L \big(\Lambda_{t+1} \Delta E^b_{t+1} - \Delta l^e_{t+1} \big), \qquad \det J = 1 - (1-\delta_t) F'(\chi^c) B_t\, g_N - g_L F'(\chi^c) E^b_{t+1}.$$

Condition (55) is $\det J > 0$ at the steady state, so the implicit function theorem gives a finite response $\partial \chi^c_{t+1} / \partial s_t$. Its sign need not be determined: the dominance condition in Proposition 6 bounds the drain by its absolute value. The shocks enter through ΔE^b_{t+1}, which falls by the separations, through ΔB_t, which moves with product-line entry at t, and through the labor used in entry at $t+1$, Δl^e_{t+1}. All are finite, so the drain is of order $F'(\chi^c)$.

Conclusion. If the reposting inflow weakly dominates the drain, the pre-committed vacancy stock and unemployment respond as in Part 1 of Lemma 2 up to the entry terms, and Steps 1–3 deliver positive u–v comovement at $h = 1$ under conditions (i) and (ii) with the inflow in place of d. The first-order coefficients are continuous in ρ_s, so if dominance is strict at $\rho_s = 0$, it holds for all $\rho_s \in [0, \bar{\rho}_s)$ for some $\bar{\rho}_s > 0$. □

Appendix J. Steady-state equilibria: proof of Proposition 7

Throughout, restrict to DS-CES preferences, $\rho(N) = N^{1/(\varepsilon-1)}$ and $\mu = \varepsilon/(\varepsilon - 1)$ constant, and set $p_0 = 0$ so that $\delta_e \in (0, 1)$ is a parameter. Assumption 1 and condition (C) hold. Employment is $L(\theta) = (1 - \delta_e) f(\theta) / [\tau + (1 - \delta_e) f(\theta)]$, and steady-state entry is $e(\theta)$ in (59) with $\Lambda_r = 1$. Write $x \equiv \ln \theta$.

Step 1: $N_{JC}(\theta)$ is well defined and strictly increasing. Under DS-CES the job creation curve is $(z/\mu) N^{1/(\varepsilon-1)} = \Phi(\theta)$, with Φ defined in (62), which depends on θ alone. Since $1/(\varepsilon - 1) > 0$, it inverts to $N_{JC}(\theta) = (\mu \Phi(\theta)/z)^{\varepsilon - 1}$. Entry can be written $e(\theta) = \delta_e \{1 + \tau(\theta - 1)/[\tau + (1 - \delta_e) f(\theta)]\}$, whose derivative has the sign of $\tau + (1 - \delta_e)[f(\theta)(1 - \varepsilon_f(\theta)) + f'(\theta)] > 0$, because

$\varepsilon_f < 1$. So e, and with it $K(\theta) = (r + \delta_e)e(\theta)^{1/\xi}x_m/(1 + r)$, is strictly increasing. K/q rises because q falls, and f rises. Hence Φ, and with it N_{JC}, is strictly increasing.

Step 2: $N_{RC}(\theta)$ is strictly increasing and bounded. Under DS-CES (61) gives $N_{RC}(\theta) = (\mu-1)z(1-\delta_e)L(\theta)/[f_e(\delta_e\mu+r)]$. Since L is strictly increasing with $L(0^+) = 0$ and $L < 1$, N_{RC} is strictly increasing, with $N_{RC}(0^+) = 0$ and $N_{RC} < \bar{N}$.

Step 3: Boundary behavior. *As $\theta \to 0$.* By (M2), q is decreasing, and by (M1) it is bounded by one, so $q(0^+) \in (0, 1]$. Since $f(0^+) = 0$, $e(\theta) \to 0$ and $K(\theta) \to 0$, so $K/q \to 0$. Hence $\Phi(0^+) = b + \kappa(r + \tau)/[(1 - \delta_e)(1 - \phi)]$, which gives N^0_{JC} in (63). If $b > 0$ or $\kappa > 0$, then $N^0_{JC} > 0 = N_{RC}(0^+)$.

As $\theta \to \infty$. By (M1), $q(\theta) \leq 1/\theta$. For $\theta \geq \theta_0 > 0$, $K(\theta) \geq K(\theta_0) > 0$, so $K/q \geq K(\theta_0)\,\theta \to \infty$. Hence $\Phi \to \infty$ and $N_{JC} \to \infty$, while $N_{RC} < \bar{N}$. This uses only $q \to 0$, so it holds even where entry would reach its upper bound $e = 1$ and Q would be capped at x_m.

Step 4: At most two crossings. Steady states solve $N_{JC}(\theta) = N_{RC}(\theta)$. Both sides are positive for $\theta > 0$, so these are the zeros of

$$D(x) \equiv \ln N_{JC}(e^x) - \ln N_{RC}(e^x) = (\varepsilon - 1)\ln \Phi(e^x) - \ln L(e^x) + \text{const.}$$

By (C), $\ln \Phi$ is convex in x. The elasticity of L with respect to θ is $\varepsilon_f(\theta)\,\tau/[\tau + (1 - \delta_e)f(\theta)]$. By (M2) its first factor is nonincreasing, and its second falls because f rises. So $\ln L$ is concave in x, and D is convex. By Step 3, $D(x) \to +\infty$ as $x \to -\infty$, since $\ln N_{JC}$ stays finite while $\ln N_{RC} \to -\infty$, and as $x \to +\infty$. A convex function that tends to $+\infty$ at both ends has no zero if its minimum is positive, one zero (a tangency) if its minimum is zero, and exactly two zeros if its minimum is negative, apart from the nongeneric case in which it vanishes on an interval. A tangency imposes one equation on the parameters, so it is a codimension-one event. Hence, generically, there are either no steady states or exactly two, $\theta_L < \theta_H$. There are two if and only if $N_{JC}(\theta) < N_{RC}(\theta)$ for some θ. Since $N = N_{RC}(\theta)$ at a steady state and N_{RC} is strictly increasing, $N_L < N_H$.

Step 5: Uniqueness in the double limit. Let $\varepsilon \to \infty$ with $f_e = \bar{f}_e/\varepsilon$, as in Proposition 3. Then $(\mu - 1)/f_e \to 1/\bar{f}_e$, so $N_{RC}(\theta) \to z(1 - \delta_e)L(\theta)/[\bar{f}_e(r + \delta_e)]$, which is finite and positive for $\theta > 0$. On any compact subset of $N \in (0, \infty)$, $(z/\mu)N^{1/(\varepsilon-1)} \to z$, so the job creation curve becomes $\Phi(\theta) = z$. It is vertical in (θ, N) space, and Φ does not involve μ. Since Φ is continuous and strictly increasing from $\Phi(0^+)$ to ∞, a solution θ^* exists, and is unique, if and

only if $z > \Phi(0^+)$. The equilibrium is $(\theta^*, N_{RC}(\theta^*))$. With f_e fixed instead, $\mu - 1 \to 0$ drives N_{RC} to zero, which is the degenerate limit ruled out before Proposition 3.

Appendix J.1. Details of comparative statics

Comment out this section temporarily

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